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Measuring successful, failed and deceptive exploration of evolutionary algorithms based on attraction basins

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HIGHLIGHTS

- Clear identification of the exploration and exploitation phases.
- Novel measures for direct exploration and exploitation based on attraction basins, without basin precomputation.
- Measuring successful, failed and deceptive exploration on various optimization problems for DE/rand/1/bin, JADE, LSHADE, ABC, and TLBO.
- A detailed explanation of the limitations of indirect diversity measures for exploration and exploitation.

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ABSTRACT

This paper introduces a novel direct measure of exploration and exploitation in Evolutionary Algorithms, based on attraction basins—regions of the search space with an attractor toward which neighboring solutions tend to evolve. Each solution can be associated with a particular attraction basin. If a new solution belongs to the same attraction basin as its parent, then the search process is identified as exploitation; otherwise, it is identified as exploration. Four different exploration type definitions were used: successful, failed, deceptive, and successful rejection. These are measured directly at the individual level (without precomputing basins) on six single-objective continuous benchmark problems and a real-world problem using DE/rand/1/bin, JADE, LSHADE, ABC, and TLBO. The results revealed that EAs differentiate themselves in how they handle different types of exploration. The approach outperformed prior basin-based measures in efficiency, and exposed the limitations of diversity-based methods, enabling explicit quantification and control of the balance between exploration and exploitation.

1. Introduction

Exploration and exploitation have been considered core concepts of Evolutionary Algorithms (EAs) since their inception. The inventor of Genetic Algorithms (GAs) had already discovered a striking balance between exploration and exploitation in GAs [1]. However, exploration and exploitation have never been defined formally, and only a rough definition exists [2]:

- *"Exploration is a process of visiting entirely new regions of a search space."*
- *"Exploitation is a process of visiting regions of a search space based on the neighborhood of previously visited points."*

A common misconception in the early days of EAs was that EAs explore the search space through crossover and mutation operators, while exploitation is achieved by selection. An early discussion on exploration and exploitation in EAs [3] revealed that there was no common agreement on this issue among researchers, and 15 years after the work [3] was published, the survey paper on exploration and exploitation [2] still came to the astonishing conclusion: "Exploration and exploitation

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are fundamental concepts of any search algorithm, and surprisingly these concepts are not understood better in EAs." With the new developments and further investigation of exploration and exploitation, it was recognized that all the operators, (selection, crossover, mutation), contribute to exploration and exploitation in EAs. However, it is difficult to distinguish exploration from exploitation in these operators [2]. It is acknowledged nowadays that too much emphasis on exploitation will lead to premature convergence to a local optimum. In contrast, focusing on exploration will lead to searching in inferior regions, slowing down the convergence rate. It was also recognized that diversity plays an important role in exploration and exploitation in EAs. There is a rough correlation between diversity and exploration/exploitation. If diversity is high, then the individuals in the population are very different from each other. Hence, the EA should be in the exploration phase. Conversely, if diversity is low, then the individuals in the population are very similar, and the EA should be in the exploitation phase. However, diversity is a search-space independent measure. Measuring diversity and switching between exploration and exploitation [4] was one of the first approaches, aimed at balancing exploration and exploitation in EAs better. Another important factor for exploration and exploitation is the size of the population. To explore the search space thoroughly, a larger population size is needed, while, in a smaller population size, the diversity can be lost faster, and the solutions can get trapped in a local optimum. Researchers recognized that dividing a population into two sub-populations, where each sub-population is responsible either for exploration or exploitation, might be beneficial [5]. In each sub-population, the operators emphasize exploration or exploitation, and maintain a high diversity in the sub-population responsible for exploration [6].

Implicit balancing of exploration and exploitation in EAs can also be achieved by the proper setting of the control parameters of the EAs. Two opposite extremes are random search and the hill climbing algorithm. Namely, if the crossover and mutation rates are very high, much of the space will be explored, but there is a high probability of losing good solutions. In such a case, the behavior of the EA resembles random search. Conversely, if the crossover and mutation rates are low, the search space is not explored at all. In this scenario, the EA resembles a hill climbing algorithm more closely. A better balance of exploration and exploitation can be achieved with properly set control parameters, which can be divided further into parameter tuning and parameter control approaches [7], especially when adaptive parameter control is applied successfully (e.g., [8]), forcing more exploration at the beginning of an evolutionary run or when stagnation is detected, and exploitation at the end of an evolutionary run. However, in all the aforementioned cases, the balance between exploration and exploitation is implicit, since the exploration and exploitation have not been measured. Obviously, if we want to control exploration and exploitation explicitly, then we need to know how to measure it. However, how to measure exploration and exploitation in EAs directly is an open question [9]. Mostly, researchers nowadays are using implicit and indirect measures of exploration and exploitation, such as diversity-based measures, entropy-based measures, and fitness-based measures, which are imprecise, search-space independent, and very rough measures for exploration and exploitation. Only a few direct measures of exploration and exploitation exist. In [10,11], the authors modeled a neighborhood as a hypercube with equal diameters in all dimensions. If the Similarity to the Closest Neighbor (SCN) between a new solution and the previous solution is smaller (larger) than a threshold value X, the exploitation (exploration) phase is declared, where various distance measures can be applied to measure similarity. This approach [10,11] has a major shortcoming, in that the threshold value X must be known or approximated (guessed) in advance, as well as that the hypercube's diameter is the same in all dimensions. In [12], the authors improved their previous approach, and eliminated both shortcomings by introducing an explicit measure of exploration and exploitation based on attraction basins [13]. An attraction basin is a part of a search region with a point called an attractor to which a system tends to evolve [14]. As such, a multi-modal search space can be decomposed into several attraction basins. Each attraction basin has its own single local optimum. Hence, each point in the attraction basin has a monotonic path to the attractor—the local optimum in this attraction basin. A unimodal search space has only one attraction basin and only one attractor. Note that attraction basins for continuous optimization problems are operator independent, unlike for combinatorial (discrete) problems. They are a feature of the search space (fitness landscape). By using the concept of attraction basins, a direct measure of exploration and exploitation is defined in a natural way using the assumption that a neighborhood consists of all solutions that can be reached with a local search (e.g., the gradient method). If a newly created solution belongs to the same attraction basin as its parent solution, then the search process is in the exploitation phase. On the other hand, if a newly created solution belongs to a different attraction basin than its parent, then the search process is in the exploration phase. The first attempt to compute the amount of exploration and exploitation using attraction basins [12] had a major shortcoming—computing attraction basins of a fitness landscape in advance. This is a very expensive process computationally and unrealistic for real-world problems [15–17]. In this paper, we eliminate this shortcoming by simply checking if a newly created solution and its previous solution are attracted by the same attractor (local optimum). In that case, both solutions are in the same attraction basin, denoting the exploitation phase. Otherwise, the exploration phase is declared. Identifying the proper phases, exploration or exploitation, is more efficient and feasible for real-world problems. Recently, the authors in [18] proposed a new pbest selection mechanism based on attraction basins, which are approximated by a hypersphere with radius r, increasing the likelihood that the new trial vector would be generated in a different attraction basin. However, the amount of exploration has not been measured and quantified. The authors in [13] further identified four types of exploration: successful exploration, successful rejection, failed exploration, and deceptive exploration. However, these types were not evaluated across different optimization problems. Among them, failed and deceptive exploration are particularly problematic for EA search. To summarize, in this work we investigated the following Research Question RQ-1: *Is it possible to measure different types of exploration (successful, failed and deceptive exploration) directly based on attraction basins without computing attraction basins in advance?*

The main contributions of this paper are:

- Clear identification of the exploration and exploitation phases.
- Novel measures for direct exploration and exploitation based on attraction basins, where there is no need to compute the attraction basins in advance.
- For the first time, successful, failed and deceptive exploration are measured on various optimization problems for EAs: DE/rand/1/bin [19], JADE [20], LSHADE [21], ABC [22,23], TLBO [24,25].
- A detailed explanation of the limitations of indirect diversity measures for exploration and exploitation.

In this paper, the focus is on single-objective continuous optimization problems to narrow our research sufficiently. The landscapes and attraction basins for combinatorial (discrete) problems have many specifics (e.g., operator dependency, plateaus, and the local optima are not distributed randomly) discussed in [26]. Furthermore, local search algorithms for continuous and combinatorial problems exhibit essential differences due to the distinct types of search spaces. While combinatorial search spaces are finite for finite-size problems, continuous search spaces are infinite and not enumerable [27]. In combinatorial problems, the number of candidate solutions in the neighborhood N of a current candidate solution s is enumerable, and a local optimum (for minimization problems) can be defined as a candidate solution s' for which it holds that $\forall s \in N(s'), f(s') \leq f(s)$, where f is the fitness function $f : S \rightarrow \mathbb{R}$. For continuous optimization problems, the search space S consists of an infinite number of solutions. This makes enumeration

impossible for the search of the optimum. A local optimum in a continuous space is a solution $s' \in S$, such that $f(s') \leq f(s)$ for all points $s \in S$ that satisfy the relation $0 \leq \|s' - s\| \leq \varepsilon$. The set of solutions encircled in the region limited by the magnitude of ε is the neighborhood N of the local optimum s'. For the reasons above, combinatorial (discrete) problems are out of the scope of this paper.

The remainder of the paper is organized as follows: Related work is described in Section 2. The methodology for measuring successful exploration, successful rejection, failed exploration, and deceptive exploration is outlined in Section 3. The experimental results and analysis are presented in Section 4. Finally, Section 5 provides the conclusions of the study and future work.

2. Related work

In the survey paper on exploration and exploitation in EAs [2], the authors investigated and discussed several important issues, which are elaborated further in this section:

1. Which EA's components contribute to exploration and exploitation, and how?
2. How is the balance achieved between exploration and exploitation?

2.1. Which EA's components contribute to exploration and exploitation, and how?

Several important factors have been identified. Obviously, all evolutionary operators (selection, crossover, mutation) contribute to exploration and exploitation in EAs. By varying the selection pressure, the selection can control the level of exploration or exploitation. Namely, higher (lower) selection pressure pushes the search towards more exploitation (exploration). On the other hand, mutations can range widely, from minor to major changes in the genotype, covering exploitation as well as exploration. As such, the role of mutation varies in different EAs. In Evolutionary Strategies (ES), the mutation functions more as an exploration operator, whereas, in GAs, mutation tends to serve as an exploitation operator (particularly when only a few or no significant bits are changed). Similarly, a crossover operator combines two or more parents to generate a new solution. When the parents are diverse, it promotes exploration; when the parents are similar, it promotes exploitation. EAs employ different strategies for recombining solutions by mutation and crossover operators, where parents can be selected from different neighborhood topologies. For example, the DE/ran/1/bin strategy promotes more exploration, while DE/best/1/bin promotes more exploitation [28], and various neighborhood topologies in Particle Swarm Optimization (PSO) improve the balance between exploration and exploitation [6,29]. Nevertheless, the line between exploration and exploitation is blurred, and we should not claim that mutation and crossover operators are only responsible for either exploration or exploitation. Another factor influencing exploration and exploitation is the population size. The search space is explored more with a larger population size than with a smaller population size. Therefore, the balance of exploration and exploitation can be controlled indirectly through proper setting or controlling of the parameters [7] such as selection pressure, probability of mutation, probability of crossover, and population size. However, the parameters should not be set independently, as they are highly interdependent. For this reason, the multi-process-driven approach [30], where the balance between exploration and exploitation is coordinated between the different operators (selection, mutation, crossover), is preferred over the uni-process-driven approach, where each operator attempts to manage this balance independently. Furthermore, the control parameter setting is problem-dependent, which implies that different problems require different amounts of exploration and exploitation. It is important to notice that the control of exploration and exploitation is currently indirect, and a direct measure of exploration and exploitation is needed to quantify the effect of various EA components.

2.2. When and how are exploration and exploitation controlled?

When exploration and exploitation are controlled, online approaches, where different amounts of exploration and exploitation are emphasized during an evolutionary run, are preferred over offline approaches, where the proper settings of control parameters influence the EA's search capabilities.

Regarding the part on how exploration and exploitation are controlled it is more difficult. If both phases, exploration and exploitation, can be identified clearly, then direct measures can be used for this purpose. However, only recently have some direct measures of exploration and exploitation been proposed [10–12], and researchers have been using mainly indirect and imprecise measures, such as diversity-based, entropy-based, and fitness-based measures. Among the indirect measures of exploration and exploitation, the diversity-based measures prevail [4,28,31–38], due to a rough correlation between diversity and exploration/exploitation (already discussed in Section 1). Most often, the *dimension-wise diversity measure* [32] has been applied. After each generation/iteration, the diversity of the population is computed using Eqs. (1) and (2), where $mean(x^j)$ represents the mean of dimension j in the whole population, x_i^j is the dimension j of solution i, n is the population size, and m is the dimensionality of a problem.

$$\mathrm{Div}_j = \frac{1}{n}\sum_{i=1}^{n}\left|\mathrm{mean}(x^j) - x_i^j\right| \tag{1}$$

$$\mathrm{Div} = \frac{1}{m}\sum_{j=1}^{m}\mathrm{Div}_j \tag{2}$$

In Eqs. (1) and (2), instead of the $mean(x^j)$, the $median(x^j)$ has often been used (e.g., [34,39]). The amount of exploration and exploitation in each generation/iteration is then computed by Eqs. (3) and (4), where Div_{max} represents the maximum diversity over the whole evolutionary run. Since the amount of exploitation is the complementary percentage of exploration, it can be computed as $XPT\% = XPL\% - 1$.

$$\mathrm{XPL}\% = \left(\frac{\mathrm{Div}}{\mathrm{Div}_{\max}}\right) \times 100 \tag{3}$$

$$\mathrm{XPT}\% = \left(\frac{|\mathrm{Div} - \mathrm{Div}_{\max}|}{\mathrm{Div}_{\max}}\right) \times 100 \tag{4}$$

The authors in [34] applied Eqs. (3) and (4) to investigate what the optimal exploration and exploitation rates are in EAs. Since EAs apply different search strategies, the appropriate exploration/exploitation rates are expected to be different as well. Furthermore, optimization problems have different fitness landscapes [40,41], and, therefore, it is expected that the same EA will explore and exploit the search space differently on a smooth or rugged fitness landscape. The authors in [34] found that most EAs produce the best results when the balance between exploration and exploitation is approximately 10% to 90%, respectively. An interesting finding was also that the ratio between exploration and exploitation alone is not enough for good performance, and that the quality of the search mechanism matters. Namely, some EAs have the same ratio of exploration and exploitation as the best-performing EAs, but the results were significantly worse. Although the work [34] is interesting, measuring exploration and exploitation was again based on diversity. There are several problems with using diversity, Eqs. (3) and (4), for measuring exploration and exploitation:

- They measure diversity directly and exploration and exploitation only indirectly, where diversity is averaged across all dimensions (Eq. 2).
- They cannot measure exploration/exploitation at the micro level (e.g., how mutation/crossover contributes towards exploration/exploitation), but only at the macro level (population).

- The amount of exploration/exploitation depends on the maximum diversity over the whole evolutionary run. If the maximum diversity is an outlier, the computation is even more imprecise, and, more importantly, why should the amount of exploration depend on the maximum diversity?
- The diversity measure is not based directly on the properties of the search space (e.g., a given diversity could lead to exploration or exploitation, depending on the size of the attraction basins in the search space). It is a search-space independent measure.
- Diversity corresponds only roughly to exploration/exploitation, and such a measurement is imprecise and not accurate enough, as already shown in [12].

Entropy-based measures for exploration and exploitation are used less frequently. However, the rationale is similar to diversity-based measures, as well as their shortcomings. If the population is diverse, then the entropy is high, and the EA should be in the phase of exploration. On the contrary, when the population diversity is small, the entropy is low, and the EA should be in the phase of exploitation. Entropy can be regarded as a succinct measure of diversity. In [30], the whole population was grouped based on genotype similarity into k classes p_i using adaptive clustering and counting the size of each class. The entropy was then defined as Eq. (5):

$$Entropy = -\sum_{i}^{k} p_i \log_2 p_i \tag{5}$$

This measure was used in [30] as a trigger to switch between exploration and exploitation.

Fitness-based measures for exploration and exploitation use different rationales. Fitness improvements [36], stagnation [42], or survival time [43] are measured, and used as a trigger to switch between both phases. In the work [36], the population was sorted according to the fitness values, and the first half of the population (superior part) with better fitness values was responsible for exploitation, while the second half (inferior part) was responsible for exploration. Total fitness improvements were computed for the inferior and superior parts. Greater total fitness improvements of the inferior part indicate that, in the current stage, more exploration is needed, and exploitation is recommended if the total fitness improvements of the superior part are greater. In the work [42], the authors proposed a novel meta-level strategy that monitors EA's behavior externally to detect stagnation, which occurs when the current best solution has not been improved after a predefined number of function evaluations. Upon stagnation, the search is restarted with a reduced evaluation budget to enhance exploration. While the previous approaches (e.g., [44–47]) were algorithm-specific, the approach in [42] is algorithm-agnostic. In the work [43], another interesting measure, *survival time*, was proposed to determine the exploration and exploitation phase indirectly. Namely, when the average survival time is increasing (decreasing), it indicates that the search is in the exploitation (exploration) phase.

Only indirect measures (diversity-based, entropy-based, and fitness-based measures) for exploration and exploitation have been discussed so far. Although indirect measures are computationally less expensive than the proposed approach, there are several major shortcomings of indirect measures:

- The properties of the search space, such as the ruggedness of the fitness landscape and the number and size of attraction basins, are not reflected in these measures. They are search space independent measures.
- They measure diversity, entropy, or fitness improvements directly, but exploration only indirectly, and, as such, these measures are imprecise.
- They cannot identify the exploration or exploitation phase at the individual's level, only at the (sub-)population level.
- They cannot quantify the effect of selection, crossover, and mutation on exploration and exploitation.
- They cannot quantify the amount of successful exploration, successful rejection, failed exploration, and deceptive exploration.

One of the first attempts to measure exploration and exploitation directly was presented in [10,11], where the neighborhood was modeled as a hypercube with the same diameter over all the dimensions. When a new solution is created, it is checked if the Similarity to the Closest Neighbor (SCN) is bigger than threshold X. If the condition is met, then exploration is declared (Eq. 6); otherwise, exploitation (Eq. 7).

$$\mathrm{SCN}(\mathrm{ind}_{\mathrm{new}}, P) > X \quad \text{(exploration)}, \tag{6}$$

$$\mathrm{SCN}(\mathrm{ind}_{\mathrm{new}}, P) \leq X \quad \text{(exploitation)} \tag{7}$$

where similarity SCN can be computed using various distance measures at the genotype level, as follows (Eqs. 8 to 11), and $d(_, _)$ is a distance measure [2]:

- As a similarity to its parent(s), ind_{parent}.
- As a similarity to the most similar solution within the whole population P.
- As a similarity to the most similar solution in the sub-population $P' \wedge (P' \subset P)$.
- As a similarity to the most similar solution throughout the history of populations.

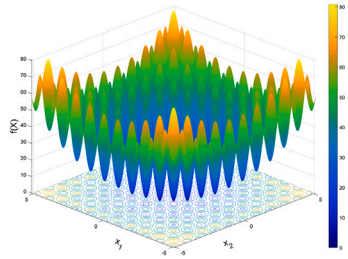
$$\mathrm{SCN}(\mathrm{ind}_{\mathrm{new}}, P) = d(\mathrm{ind}_{\mathrm{new}}, \mathrm{ind}_{\mathrm{parent}}), \qquad \mathrm{ind}_{\mathrm{parent}} \in P \tag{8}$$

$$\mathrm{SCN}(\mathrm{ind}_{\mathrm{new}}, P) = \min_{\substack{\mathrm{ind} \in P \\ \mathrm{ind}_{\mathrm{new}} \neq \mathrm{ind}}} d(\mathrm{ind}_{\mathrm{new}}, \mathrm{ind}) \tag{9}$$

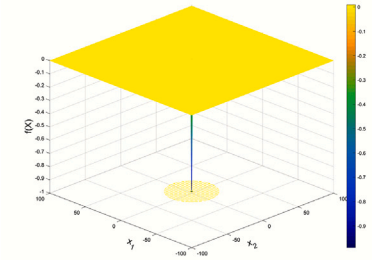
$$\mathrm{SCN}(\mathrm{ind}_{\mathrm{new}}, P') = \min_{\substack{\mathrm{ind} \in P' \\ P' \subset P \\ \mathrm{ind}_{\mathrm{new}} \neq \mathrm{ind}}} d(\mathrm{ind}_{\mathrm{new}}, \mathrm{ind}) \tag{10}$$

$$\mathrm{SCN}(\mathrm{ind}_{\mathrm{new}}, P^t) = \min_{\substack{\mathrm{ind} \in P^t \\ t\,=\,0,\ldots,\mathrm{current_gen} \\ \mathrm{ind}_{\mathrm{new}} \neq \mathrm{ind}}} d(\mathrm{ind}_{\mathrm{new}}, \mathrm{ind}) \tag{11}$$

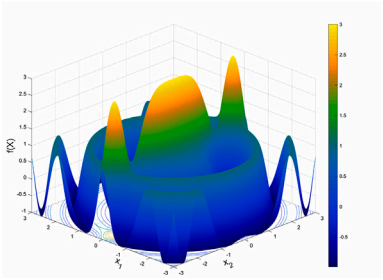
Using this approach, exploration and exploitation can be measured at the micro level after applying the operators (e.g., mutation, crossover). On the multi-objective 0/1 Knapsack problem, it has been shown that, indeed, various operators (mutation, crossover, repair) contribute to exploration, as well as to exploitation [11]. Hence, we cannot declare a particular operator as only an exploration/exploitation operator. The proposed approach has also been compared to parameter tuning and various parameter control approaches (deterministic, adaptive 1/5 rule, entropy-based control) on real-world problems [48], where it outperformed other approaches. However, measuring exploration and exploitation with the proposed approach has two major shortcomings: the proper threshold value X is problem-dependent and generally unknown, and the neighborhood has been modeled as a hypercube with the same diameter in all dimensions, which might be true only for special cases (e.g., for the Rastrigin function [12]). Clearly, a better approach should be developed without these shortcomings. A hint in the right direction was the work by [13], where the authors wrote: *"…exploration becomes the task of identifying the fittest attraction basin, and exploitation can be expressed as finding the local optimum within a given basin"*, and suggested that an attraction basin could be useful for computing the amount of exploration and exploitation. While in previous studies [26,49,50] the topology of attraction basins has been studied, such as the roundness of the attraction basins, the centrality of the local optima, and the interior and exterior of the attraction basins, which are related to the complexity of the fitness landscapes, or measuring escape rates as a probability of reaching a different attraction basin [51–53]. However, in the experiments [13] only the Rastrigin function was used, where attraction basins can be computed analytically. An algorithm for computing attraction basins was developed in [12], revealing that attraction basins are of various shapes, and only rarely hypercubes with the same diameter in all dimensions (Figs. 1 and 2).



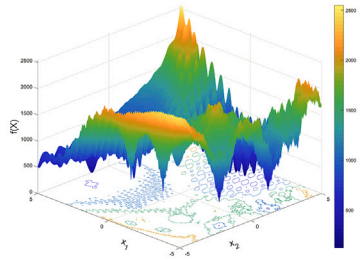
(a) Rastrigin function



(b) Easom function

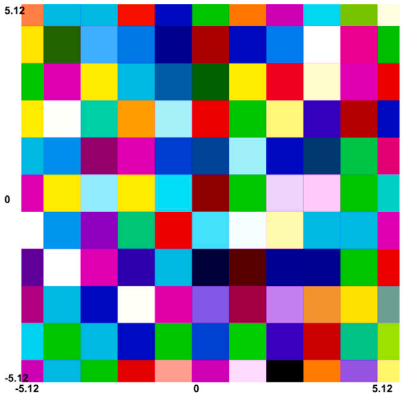


(c) Split Drop Wave function

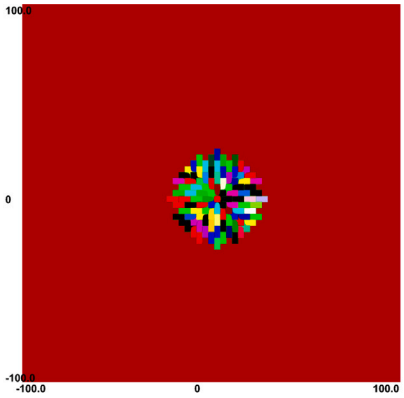


(d) Rotated Hybrid Composition Function 1

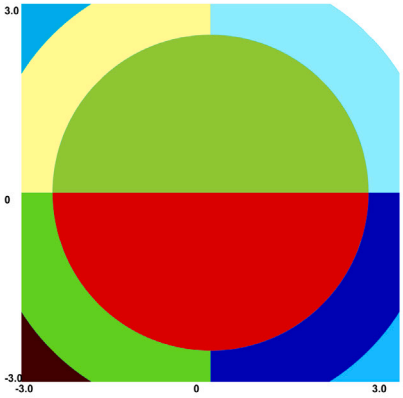
Fig. 1. Fitness landscapes of various benchmark functions.



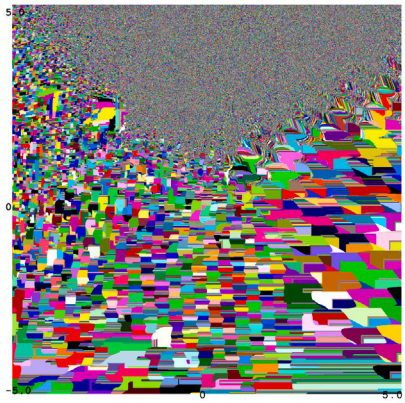
(a) Rastrigin function



(b) Easom function



(c) Split Drop Wave function



(d) Rotated Hybrid Composition Function 1

Fig. 2. Attraction basins of various benchmark functions.

After computing the attraction basins, which is a very CPU-intensive process, computing the amount of exploration and exploitation is straightforward. It is simply verified if the newly created solution is in the same attraction basin as its parent. If it holds, it is identified as exploitation; otherwise, it is identified as exploration. The main drawback of the approach in [12] is that attraction basins have to be computed for each optimization problem prior to the exploration/exploitation measurement, which is quite an unrealistic demand. In this paper, we eliminate this constraint by the following reasoning: two solutions are in the same attraction basin, and we can declare exploitation if the same local optimum can be reached from both solutions. Otherwise, the solutions are in different attraction basins, denoting exploration.

2.3. How is the balance achieved between exploration and exploitation?

Regarding the part on how the balance between exploration and exploitation is achieved, the work [2] identifies several approaches: achieving the balance between exploration and exploitation through diversity maintenance, through diversity control, diversity learning, and other more direct approaches.

In diversity maintenance, the population's diversity is not measured, but different techniques, such as varying the population size [54], duplicate elimination [55,56], restarts [42], migration between subpopulations [57], changing the selection pressure [58] or the probability of mutation/crossover [59], mating restrictions [60], and disruptive operators [61], are used to maintain diversity, and, hence, improve the balance between exploration and exploitation. The amount of exploration and exploitation is unknown. However, better results have been achieved which are credited to a better balance between exploration and exploitation.

In the diversity control approach, population diversity [4,37,62], entropy [11], or fitness improvements [36] are measured, and used as feedback to steer the evolution process towards exploration or exploitation.

In diversity learning, the long-term history is used, in combination with different machine learning techniques to learn (un)explored search areas [63,64], while, in the previous diversity control approach, the short-term history is used (e.g., current population). The authors of [64] correctly observed that random diversification of a population does not guarantee that the EA will explore *promising* regions. They proposed a space-partitioning-based (SPEA) reinforcement framework for learning promising regions, by estimating the explorative (exploitative) potential of subspaces. Promising regions were then clustered. Exploitative and explorative clusters approximate attraction basins, which may contain more than one optimum. By using the approach in [64], it is possible to detect promising regions for exploration and exploitation. However, the amount of exploration and exploitation has not been measured.

Examples of a direct approach are the previously mentioned approaches modeling neighborhoods with hypercubes [10,11] and with attraction basins [12].

With advancements in this field, once explicit measures of exploration and exploitation are developed, we anticipate the following taxonomy: achieving exploration and exploitation balance through exploration and exploitation maintenance, exploration and exploitation control, and exploration and exploitation learning. This work is a small, but important step in this direction.

Balancing exploration and exploitation for discrete optimization [65], multi-objective optimization [43], and constrained multimodal multi-objective optimization [66,67] have their own specifics which must be taken into account as well. However, all the mentioned works are based on various diversity measures. To keep our investigation focused on single-objective optimization, these issues are out of the current scope, and are considered as our future work.

3. Measuring successful, failed and deceptive exploration

In [12], the authors proposed a novel direct measure for exploration and exploitation based on attraction basins. An attraction basin $A(x^*)$ is a part of a search space S with a local optimum x^* called an attractor, which can be accessed from any other solution $y \in A(x^*)$ in this attraction basin simply by performing a local search. In our work we are currently using the Best-Improvement local search algorithm (Algorithm 1), while other approaches are also feasible (e.g., First-Improvement local search [68]). The computational complexity of the local search is $O(D \cdot k)$, where D is the number of dimensions and k is the number of iterations until convergence, as each iteration evaluates $2D$ neighboring solutions when a sparse neighborhood is used, and $3^D - 1$ neighboring solutions under a dense neighborhood. While this introduces computational overhead that scales linearly with dimensionality under a sparse neighborhood, making the approach more expensive for dense neighborhoods, large populations, expensive objective functions, or very high-dimensional problems (e.g., 100 + dimensions), it remains manageable for problems with moderate dimensionality. Importantly, many real-world optimization problems fall within this moderate range (typically $D < 50$) [69–71], and the insights gained about algorithm behavior, particularly regarding the exploration–exploitation dynamics and convergence patterns, provide a valuable understanding that justifies the computational cost. For problems where the evaluation budget is extremely limited or dimensionality is high, the measures can still be computed on a subset of solutions, or at selected generations rather than for every parent-offspring pair, providing a practical trade-off between analytical depth and computational feasibility. Another possibility to decrease the computational cost is the application of an approximate local search (e.g., fast local search or guided local search [72]). Furthermore, note that the computational complexity of the proposed approach is the same as for Memetic Algorithms, where every solution is improved by local search until a local optimum is found [27,73]. Each attraction basin can be seen as a neighborhood, since all solutions access the same attractor (local optimum) by local search. As such, a neighborhood is defined implicitly, and there is no need to approximate it with a hypercube. Hence, attraction basins and neighborhoods are presented independently of particular EA [14]. They are features of the search space (fitness landscape). Furthermore, there is no need to compute attraction basins in advance, as in [12], to find out if a newly created solution and parent are in the same attraction basin, denoting exploitation, or in different attraction basins, denoting exploration. The algorithm is straightforward (Algorithm 2).

Algorithm 1 Pseudocode of Local-Search using sparse neighborhood.

```
Input: Problem: contains the fitness function to optimize
Input: startingSolution: initial solution to begin search from
Input: stepSize: size of each step for generating neighbors
Initialize bestSolution ← startingSolution
Initialize D ← number of dimensions in Problem
Initialize improved ← true
while improved do
    improved ← false
    neighbors ← generate 2 * D neighbors by stepping ±stepSize in each dimension
    Evaluate each neighbor
    for each neighbor in neighbors do                ▷ Best-Improvement
        if neighbor is better than bestSolution then
            bestSolution ← neighbor
            improved ← true
        end if
    end for
end while
return bestSolution
```

Algorithm 2 Identifying exploration and exploitation.

for all pairs $(new_sol_i, parent_i)$ **do**
  $new_sol_{localopt}$ = Local-Search(new_sol_i)
  $parent_{localopt}$ = Local-Search($parent_i$)
  if $new_sol_{localopt} \neq parent_{localopt}$ **then**
    $mark_exploration()$
  else
    $mark_exploitation()$
  end if
end for

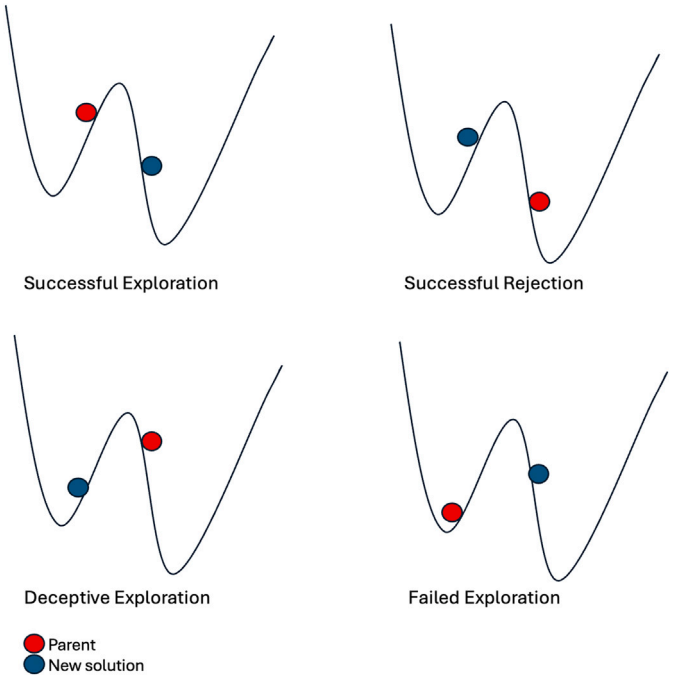


Fig. 3. Exploration types.

It is now important to consider the interplay between the creation of an offspring solution (which may be classified as either exploration or exploitation) and the effect of selection. Again, the term "selection" is used frequently in multiple contexts, so it is important to be precise. The concept of selection in (original) GAs is for a generational replacement of the entire population, so selection applies to the choice of picking parents for the mating pool [1]. It is referred to as parent selection [74]. In many current implementations of EAs (e.g., DE), a "parent" population of mating solutions is maintained, and the selection process determines if the created offspring solution is allowed to replace (one of) the parent solution(s). It is referred to as survival selection [74]. We are particularly interested in elitist fitness-based selection, which is used in many EAs for the admission of offspring solutions to the population of parent solutions, and it is used specifically in DE [19], JADE [20], LSHADE [21], ABC [22,23], and TLBO [24,25]. The goal of exploration is to move the search process towards more promising regions of the search space. However, this movement requires both the creation of an exploratory solution and its acceptance by selection. In the limit, a method that rejects every offspring solution will be even less effective than random search. The effect of survival selection on exploration is, therefore, a critical factor, that is often missing from discussions on exploration and exploitation. The application of elitist fitness-based selection to an offspring solution that is performing exploration can lead to four outcomes, as identified in [13]: successful exploration, successful rejection, failed exploration, and deceptive exploration (Fig. 3, Algorithm 3, and Table 1):

- Successful Exploration: A new solution (the blue dot in Fig. 3) is created from a parent (the red dot in Fig. 3) with a better fitness value than the parent. The new solution's attraction basin has a better attractor (local optimum) that is better than that of the parent's attraction basin. Hence, a better region is found and accepted.
- Successful Rejection: A new solution is created from a parent with a worse fitness value than the parent. The new solution's attraction basin has an attractor that is worse than that of the parent's attraction basin. Hence, a worse region is found and rejected.
- Deceptive Exploration: A new solution is created from a parent with a better fitness value than the parent. The new solution's attraction basin has an attractor that is worse than that of the parent's attraction basin. Hence, a worse region is found and accepted.
- Failed Exploration: A new solution is created from a parent with a worse fitness value than the parent. The new solution's attraction basin has an attractor that is better than that of the parent's attraction basin. Hence, a better region is found and rejected.

Algorithm 3 Identifying different types of exploration for minimization problems.

for all pairs $(new_sol_i, parent_i)$ **do**
  $new_sol_{localopt} = localsearch(new_sol_i)$
  $parent_{localopt} = localsearch(parent_i)$
  if $new_sol_{localopt} \neq parent_{localopt}$ **then**
    if $new_sol_i < parent_i \ \varsigma \ new_sol_{localopt} < parent_{localopt}$ **then**
      $mark_successful_exploration()$
    end if
    if $new_sol_i < parent_i \ \varsigma \ new_sol_{localopt} > parent_{localopt}$ **then**
      $mark_deceptive_exploration()$
    end if
    if $new_sol_i > parent_i \ \varsigma \ new_sol_{localopt} < parent_{localopt}$ **then**
      $mark_failed_exploration()$
    end if
    if $new_sol_i > parent_i \ \varsigma \ new_sol_{localopt} > parent_{localopt}$ **then**
      $mark_successful_rejection()$
    end if
  end if
end for

While successful exploration and successful rejection are desired outcomes (desired exploration), controversial for exploratory search are failed exploration and deceptive exploration (undesired exploration). In the failed exploration case, a better region has been found (attraction basin). However, this better region is rejected (if greedy selection is used), since the newly generated solution's fitness value is worse than its parent. It is expected that the amount of failed exploration will increase after a local optimum is found (new solutions will be worse than the local optimum, and rejected despite being in a better attraction basin), under the assumption that the attraction basin with a global optimum has not been found yet. Once the attraction basin with a global optimum is found, the possible outcome is successful rejection. In the deceptive exploration case, the search in a promising region (a better attraction basin) is replaced with a region that has a worse local optimum, since

Table 1
Four categories of solution exploration outcomes.

	Better parent attraction basin	Better new solution attraction basin
Better parent fitness	Successful rejection	Failed exploration
Better new solution fitness	Deceptive exploration	Successful exploration

Table 2
Control parameter settings for the selected algorithms.

Algorithm	Control parameters
DE/rand/1/bin	$F = 0.5$, $Cr = 0.9$
JADE	$p = 0.05$, $c = 0.1$
LSHADE	$r^{N^{init}} = 18$, $r^{arc} = 2.6$, $p = 0.11$, $H = 6$
ABC	$limit = 50$
TLBO	$keep = 0$

the fitness value of a newly created solution is better than that of its parent. We anticipate that the amount of deceptive exploration will be larger at the beginning of the evolutionary run rather than at its end. The interesting directions for future work are the following research questions:

- Could a better survival selection scheme (e.g., adaptive acceptance criteria, relaxed greedy selection, non-elitist selection [75]) minimize failed exploration by not rejecting (accepting) better (worse) attraction basins? A recent attempt to avoid failed exploration is presented in [76], where stalling in PSO was detected and the landscape was reshaped locally, so that particles could escape local minima more easily.
- Could archive-based memory schemes [77] minimize deceptive exploration by not accepting worse attraction basins?

If parent selection is used instead of survival selection, our categorization of exploration remains valid. However, failed exploration and successful rejection are not possible outcomes. Hence, there are only two types of exploration in parent selection: successful exploration and deceptive exploration. If the parent is in a better attraction basin than the newly created solution, which always survives, moving to a worse attraction basin will occur (the case for deceptive exploration). If a newly created solution is in a better attraction basin, it always survives, regardless of whether it is fitter than the parent solution or not, ultimately leading to successful exploration.

4. Experiments

The experiments were conducted on a set of classical optimization problems, on a set of more challenging problems from the CEC benchmark, and on a real-world problem (the Lennard-Jones problem). The classical problems include Rastrigin, Ackley, and Griewank. For the more challenging set, we selected three problems from the CEC2017 [78] and CEC2024 [79] benchmarks. The selected problems have different characteristics: F03 (multimodal function), F10 (hybrid function), and F20 (composition function). The experiments were run using problem dimensions of $m = 5$ and $m = 10$.

In the experiments, the successful, failed, deceptive exploration, and successful rejection of various optimization problems were measured for various EAs. For comparison purposes, several algorithms were included: DE/rand/1/bin [19], JADE [20], LSHADE [21], ABC [22,23], and TLBO [24,25]. The amount of *Exploration* was always distributed among successful, failed, deceptive exploration, and successful rejection in the amount of 100%. As such, although some EAs might have a very different % of *Exploration*, the distribution among different types of exploration might still be the same.

The control parameters used in the applied algorithms are presented in Table 2. For dimension $m = 5$, the maximum number of evaluations (maxFEs) was set to 50,000, and for dimension $m = 10$, it was set to 100,000. The population size was the same for both dimensions, set at $n = 50$. The reported results are based on 10 independent runs.

4.1. Rastrigin function

The results of the explicit measurement of exploration and exploitation on the Rastrigin function (dimension $m = 5$ and $m = 10$) are presented in Table 3, along with the diversity measure (Eq. 2) and best fitness value. All the results were averaged over 10 independent runs. On the five-dimensional problem, only TLBO didn't find the global minimum on every run (column *Best Fitness*), while, on the 10-dimensional problem, the same was true for DE and TLBO. An example showing that diversity is a doubtful measure for exploration can be observed by comparing *Exploration* and *Diversity* for JADE when $m = 5$ and $m = 10$. While an increase in diversity implies mostly an increase in exploration, this does not hold for JADE on the Rastrigin function, where bigger $Diversity = 1.1831$ can be observed, and smaller $Exploration = 0.0969$ when $m = 10$, compared to smaller $Diversity = 0.6652$ and bigger $Exploration = 0.1132$ when $m = 5$ (Table 3). The amount of exploration, when $m = 5$ was the lowest for JADE (*Exploration* = 11%), LSHADE (*Exploration* = 17%) and TLBO (*Exploration* = 22%) in the middle, and the highest for ABC (*Exploration* = 33%) and DE (*Exploration* = 34%). Deceptive exploration for all the algorithms, when $m = 5$, was below or around 5%, and it was not observable as problematic. Hence, the compared EAs are rarely switching to worse attraction basins. Successful rejection, when $m = 5$, is high (over 60%), except for TLBO (around 40%). Failed exploration for TLBO is higher than 50%, when $m = 5$, but lower for DE and JADE (around 30%) and ABC (18%). While LSHADE exhibits the smallest failed exploration (17%). TLBO, during the search, discovered promising regions, but they were rejected with the outcome of poor results. From Table 3, it can also be observed, when $m = 5$, that successful exploration is the highest for JADE and LSHADE (around 9%) and the smallest for TLBO and DE (around 3%). Switching to a higher dimension, $m = 10$, the results for JADE, LSHADE and ABC were very similar to the lower dimension ($m = 5$). It seems that the mechanisms of JADE, LSHADE and ABC in this case remained the same when performing the search. On the other hand, DE, when $m = 10$, increased the exploration rate drastically from 34% to 73%, while the ratio between successful exploration, failed exploration, deceptive exploration and successful rejection remained the same. While TLBO, when $m = 10$, didn't increase the exploration rate, there was an increase in failed exploration, from 54% to 70%, and a decrease in successful rejection, from 41% to 26%.

To better understand how the compared EAs performed exploration on the Rastrigin function, we visualized the process in Figs. 4–13. The

Table 3
Exploration and exploitation metrics on the Rastrigin function for dimensions $m = 5$ and $m = 10$.

Algorithm	Successful exploration	Failed exploration	Deceptive exploration	Successful rejection	Exploration	Diversity	Best Fitness
DE Rastrigin ($m = 5$)	0.0272 ± 0.0049	0.3133 ± 0.0055	0.0125 ± 0.0027	0.6470 ± 0.0077	0.3395 ± 0.0732	1.7397 ± 0.3682	0.0000 ± 0.0000
DE Rastrigin ($m = 10$)	0.0100 ± 0.0028	0.2850 ± 0.0114	0.0052 ± 0.0016	0.6997 ± 0.0144	0.7259 ± 0.1891	7.2605 ± 1.7015	1.2976 ± 1.4128
JADE Rastrigin ($m = 5$)	0.0889 ± 0.0085	0.2748 ± 0.0290	0.0309 ± 0.0043	0.6054 ± 0.0242	0.1132 ± 0.0101	0.6652 ± 0.0606	0.0000 ± 0.0000
JADE Rastrigin ($m = 10$)	0.0921 ± 0.0057	0.2692 ± 0.0301	0.0312 ± 0.0025	0.6075 ± 0.0260	0.0969 ± 0.0058	1.1831 ± 0.1122	0.0000 ± 0.0000
LSHADE Rastrigin ($m = 5$)	0.0892 ± 0.0045	0.1712 ± 0.0059	0.0561 ± 0.0026	0.6835 ± 0.0060	0.1750 ± 0.0039	1.7397 ± 0.3682	0.0000 ± 0.0000
LSHADE Rastrigin ($m = 10$)	0.0880 ± 0.0017	0.1577 ± 0.0031	0.0426 ± 0.0014	0.7116 ± 0.0038	0.1751 ± 0.0014	3.5136 ± 0.0981	0.0000 ± 0.0000
ABC Rastrigin ($m = 5$)	0.0568 ± 0.0029	0.1722 ± 0.0044	0.0149 ± 0.0012	0.7561 ± 0.0063	0.3284 ± 0.0105	2.2622 ± 0.0515	0.0000 ± 0.0000
ABC Rastrigin ($m = 10$)	0.0596 ± 0.0023	0.1782 ± 0.0012	0.0155 ± 0.0007	0.7467 ± 0.0031	0.3453 ± 0.0100	4.5328 ± 0.1340	0.0000 ± 0.0000
TLBO Rastrigin ($m = 5$)	0.0398 ± 0.0161	0.5360 ± 0.0574	0.0115 ± 0.0052	0.4126 ± 0.0500	0.2174 ± 0.1194	1.6558 ± 0.5970	0.2746 ± 0.8684
TLBO Rastrigin ($m = 10$)	0.0337 ± 0.0133	0.6934 ± 0.1148	0.0086 ± 0.0033	0.2642 ± 0.1046	0.2083 ± 0.1154	2.6812 ± 0.8233	0.0010 ± 0.0031

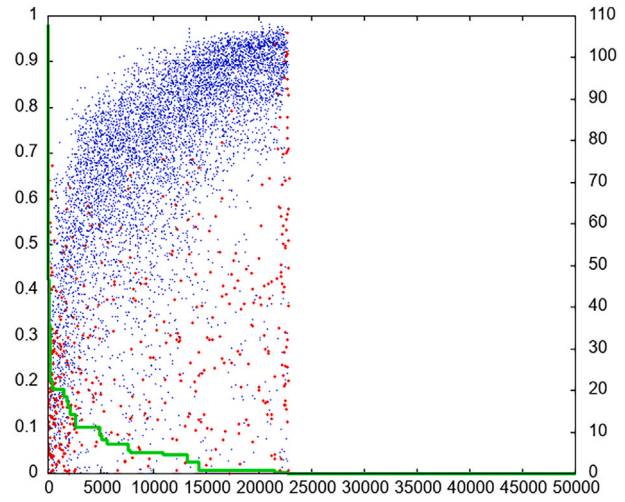


Fig. 4. Exploration behavior of DE on the Rastrigin function with $m = 5$.

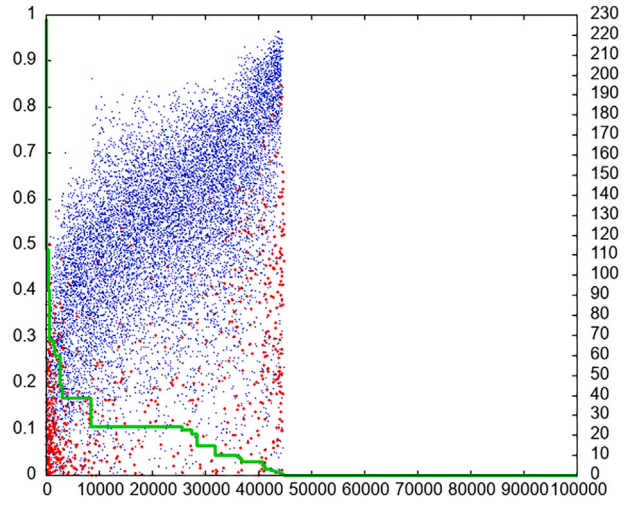


Fig. 5. Exploration behavior of DE on the Rastrigin function with $m = 10$.

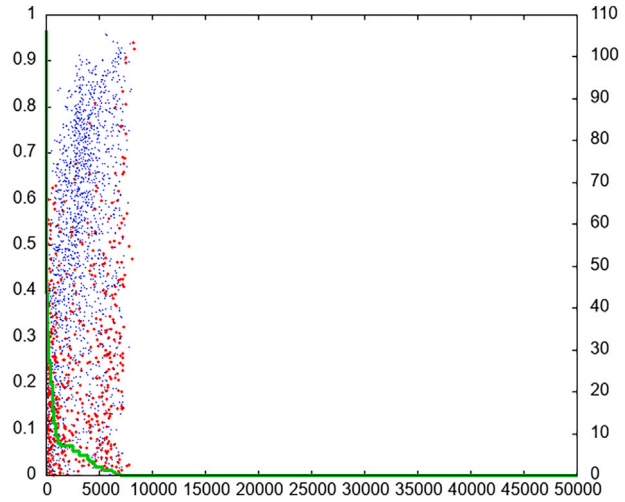


Fig. 6. Exploration behavior of JADE on the Rastrigin function with $m = 5$.

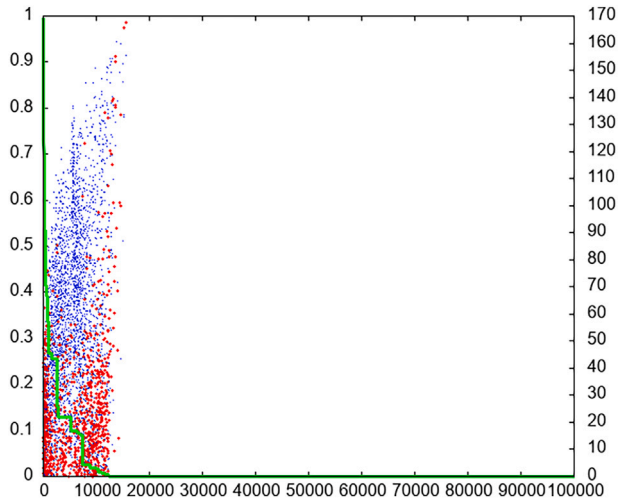


Fig. 7. Exploration behavior of JADE on the Rastrigin function with $m = 10$.

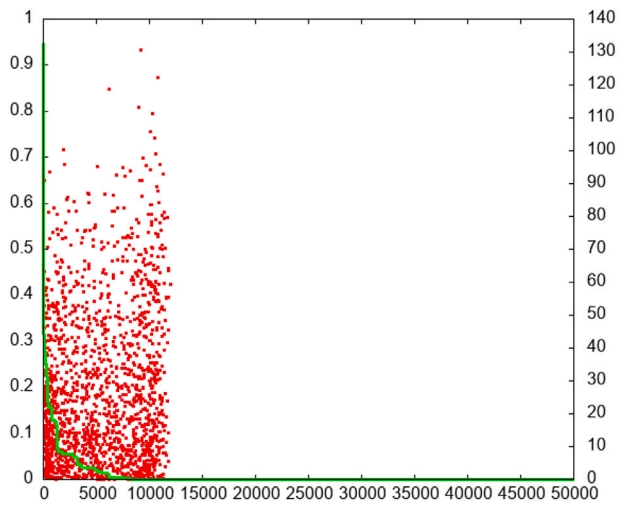


Fig. 8. Exploration behavior of LSHADE on the Rastrigin function with $m = 5$.

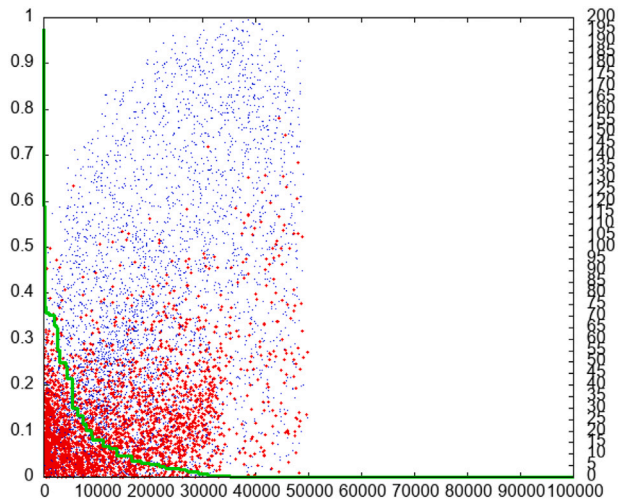


Fig. 9. Exploration behavior of LSHADE on the Rastrigin function with $m = 10$.

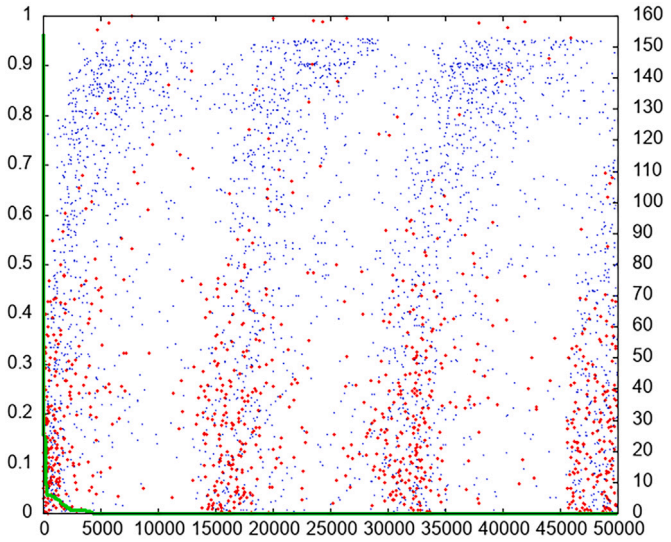


Fig. 10. Exploration behavior of ABC on the Rastrigin function with $m = 5$.

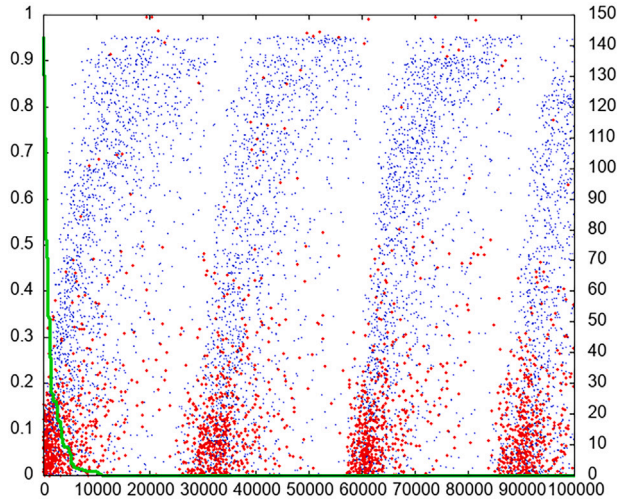


Fig. 11. Exploration behavior of ABC on the Rastrigin function with $m = 10$.

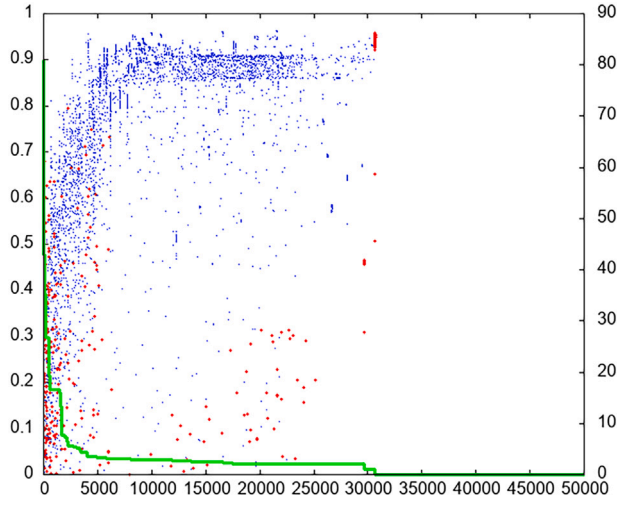


Fig. 12. Exploration behavior of TLBO on the Rastrigin function with $m = 5$.

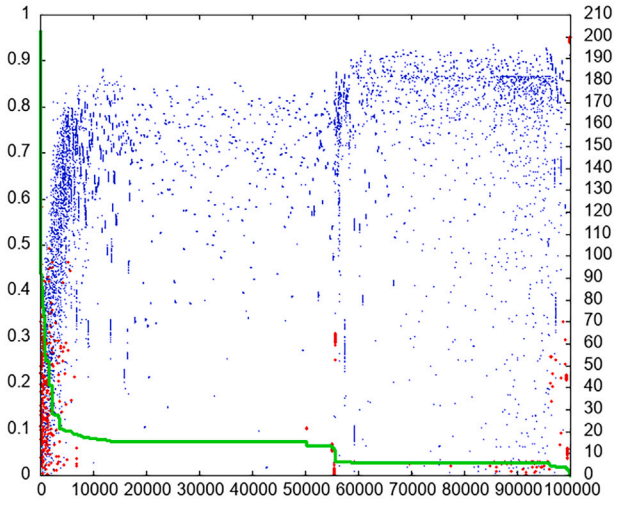


Fig. 13. Exploration behavior of TLBO on the Rastrigin function with $m = 10$.

red dots represent successful exploration, and the blue dots represent failed exploration. Since the amount of deceptive exploration was negligible, and to increase the readability of the figures, it was not included. The algorithm performance is also important, the convergence curve is represented with the green line (the y-coordinate represents the fitness values on the right-hand side). The number of fitness evaluations is represented on the x-coordinate, while the y-coordinate on the left-hand side represents relative improvements in fitness values. For example, the blue dot at $y = 0.9$ represents a solution with a 90% worse fitness than the parent. It was rejected, but it is in the promising region that has a fitter attractor than the attractor of the parent's attraction basin. Note that, an interval of blue dots with high values (e.g., Figs. 12 and 13), indicates an increasing risk of stagnation, likely caused by the parents getting closer to the local optimum. The red dot at $y = 0.5$ represents a solution that is 50% fitter than its parent, and was accepted, since its attraction basin represents a more promising region than the parent's attraction basin (has a better attractor). Note that the red dots mostly have smaller values than the blue dots, since much better solutions can be found quickly at the beginning of the search, but only small improvements are expected later and at the end of the evolutionary run (if the search converged properly). High blue dots also indicate the inefficiency of sampling, where most of the sampling solutions have worse fitness values than the parent solution despite better attraction basins having been found.

The behavior of DE, JADE, and LSHADE was similar (Figs. 4–7). They stopped with successful and failed exploration as soon as a global optimum had been found. In that case, successful exploration and failed exploration are not possible outcomes. It can be noticed that JADE found a global optimum much faster, and an increase in dimensions, from $m = 5$ to $m = 10$, caused bigger problems for DE, where it can be noticed (Fig. 5) that many fitter attraction basins were rejected (the blue dots). Comparing LSHADE (Figs. 8 and 9) with DE and JADE, it can be noticed that failed exploration is present in a smaller amount in LSHADE. An interesting behavior for ABC can be observed from Figs. 10 and 11. The main exploratory power of ABC is the scout bee phase [22,23], which depends on the control parameter *limit*. We can observe that an increase in exploration, successful and failed, came in waves, where scout bees found new promising regions, which were often rejected. From Figs. 12 and 13 it can be observed that TLBO had the slowest convergence and the biggest ratio between failed exploration (blue dots) and successful exploration (red dots). For $m = 10$, the failed exploration for TLBO was around 70% (Table 3).

While Table 3 shows the amount of different types of exploration at the end of the evolutionary run, it would be interesting to monitor

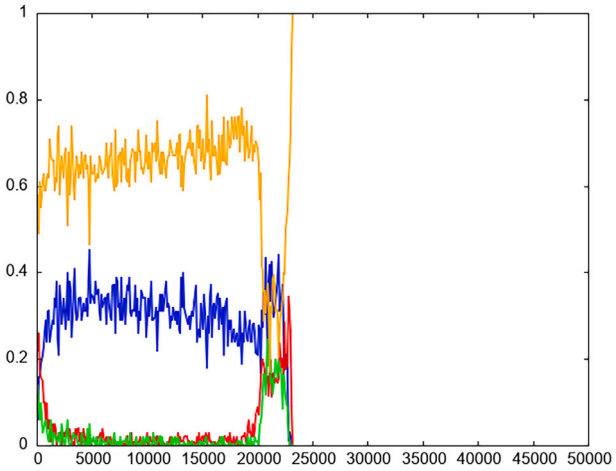


Fig. 14. Exploration types throughout the evolutionary run for DE on the Rastrigin function with $m = 5$.

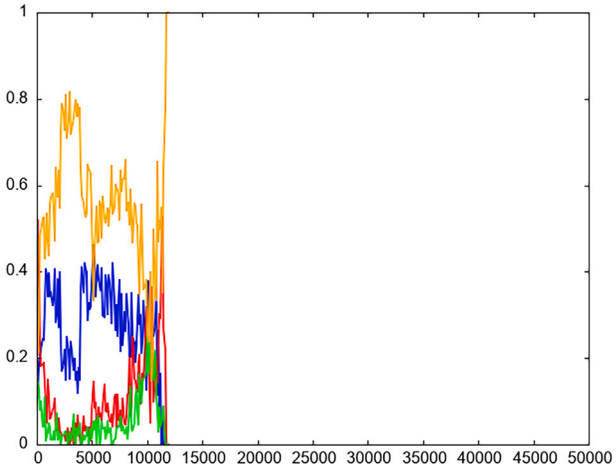


Fig. 15. Exploration types throughout the evolutionary run for JADE on the Rastrigin function with $m = 5$.

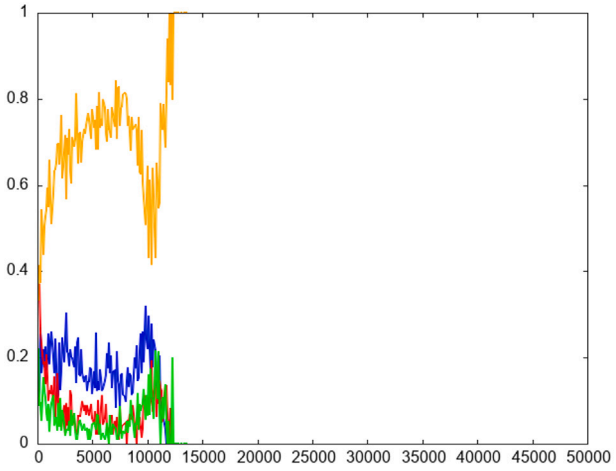


Fig. 16. Exploration types throughout the evolutionary run for LSHADE on the Rastrigin function with $m = 5$.

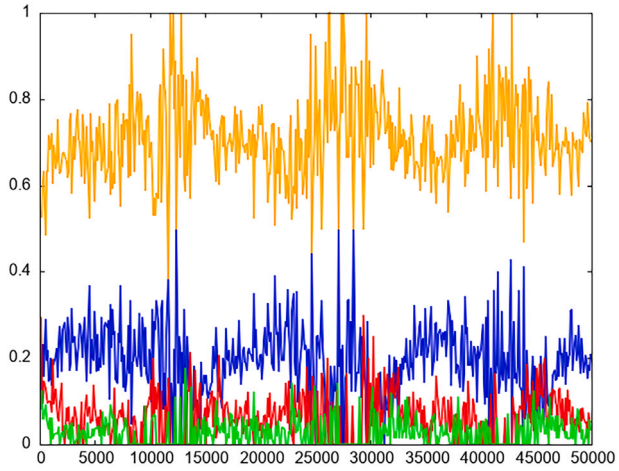


Fig. 17. Exploration types throughout the evolutionary run for ABC on the Rastrigin function with $m = 5$.

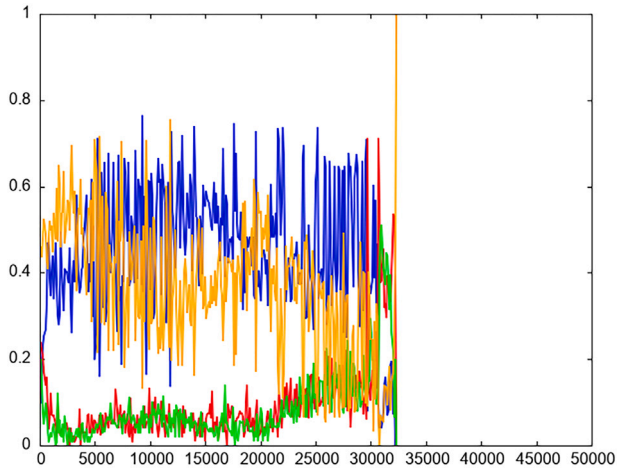


Fig. 18. Exploration types throughout the evolutionary run for TLBO on the Rastrigin function with $m = 5$.

different types of exploration during the evolutionary run (Figs. 14–18), where the red, blue, green and orange lines represent successful exploration, failed exploration, deceptive exploration, and successful rejection, respectively. It can be observed that successful rejection was consistently bigger than failed exploration for DE, JADE, LSHADE, and ABC on the Rastrigin function, except for TLBO, where, after 5000 FEs, the amount of failed exploration was bigger than successful rejection, culminating in worse fitness results among the compared metaheuristics. It can also be observed from Figs. 14–18 that the amount of deceptive exploration became bigger in the later stages of the evolutionary run for DE, JADE, LSHADE, and TLBO.

4.2. Ackley function

As the global optimum has always been found for the Ackley function, for $m = 5$ and $m = 10$, it didn't cause any problems for the selected EAs (Table 4). The exploration rate for all the algorithms was below 5%, except for ABC, where the exploration rate is around 18%, and for LSHADE with an exploration rate of 15% for $m = 5$ and 25% for $m = 10$. The rate of deceptive exploration is again low, and mostly around 5%. The amount of failed exploration was around 10% for all the selected EAs, and much lower than for the Rastrigin function. By summation

Table 4
Exploration and exploitation metrics on the Ackley function for the dimensions $m = 5$ and $m = 10$.

Algorithm	Successful exploration	Failed exploration	Deceptive exploration	Successful rejection	Exploration	Diversity	Best Fitness
DE Ackley ($m = 5$)	0.2602 ± 0.0094	0.0941 ± 0.0096	0.0518 ± 0.0047	0.5939 ± 0.0116	0.0468 ± 0.0024	1.2055 ± 0.1208	0.0000 ± 0.0000
DE Ackley ($m = 10$)	0.2098 ± 0.0090	0.0938 ± 0.0049	0.0441 ± 0.0039	0.6523 ± 0.0111	0.0473 ± 0.0026	2.1800 ± 0.1646	0.0000 ± 0.0000
JADE Ackley ($m = 5$)	0.3465 ± 0.0214	0.0909 ± 0.0077	0.0464 ± 0.0048	0.5162 ± 0.0188	0.0389 ± 0.0023	0.9158 ± 0.0721	0.0000 ± 0.0000
JADE Ackley ($m = 10$)	0.3625 ± 0.0139	0.0975 ± 0.0106	0.0481 ± 0.0047	0.4919 ± 0.0124	0.0300 ± 0.0017	1.2764 ± 0.1091	0.0000 ± 0.0000
LSHADE Ackley ($m = 5$)	0.1767 ± 0.0037	0.1312 ± 0.0046	0.0865 ± 0.0045	0.6057 ± 0.0062	0.1522 ± 0.0017	1.9551 ± 0.1522	0.0000 ± 0.0000
LSHADE Ackley ($m = 10$)	0.2220 ± 0.0057	0.0950 ± 0.0020	0.0596 ± 0.0020	0.6234 ± 0.0079	0.2503 ± 0.0020	6.0927 ± 0.2311	0.0000 ± 0.0000
ABC Ackley ($m = 5$)	0.1538 ± 0.0072	0.1096 ± 0.0057	0.0307 ± 0.0011	0.7059 ± 0.0057	0.1847 ± 0.0116	5.9984 ± 0.7811	0.0000 ± 0.0000
ABC Ackley ($m = 10$)	0.1554 ± 0.0062	0.1159 ± 0.0045	0.0307 ± 0.0017	0.6980 ± 0.0056	0.1876 ± 0.0079	10.2449 ± 0.7461	0.0000 ± 0.0000
TLBO Ackley ($m = 5$)	0.4089 ± 0.0225	0.1162 ± 0.0171	0.0742 ± 0.0060	0.4007 ± 0.0236	0.0304 ± 0.0021	0.9377 ± 0.0563	0.0000 ± 0.0000
TLBO Ackley ($m = 10$)	0.4352 ± 0.0174	0.0915 ± 0.0074	0.0623 ± 0.0071	0.4110 ± 0.0153	0.0196 ± 0.0009	1.1057 ± 0.1029	0.0000 ± 0.0000

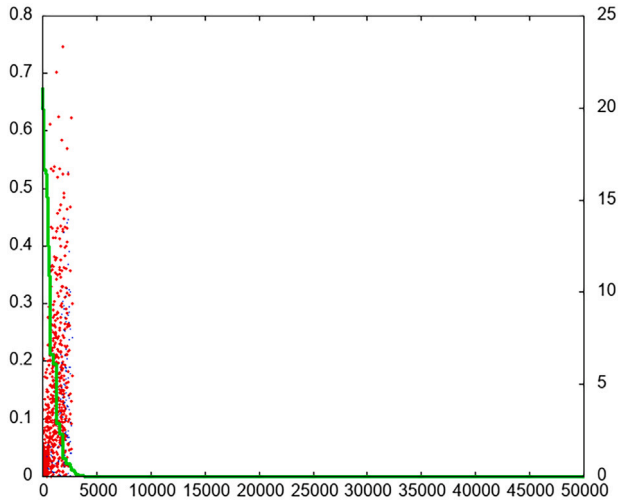


Fig. 19. Exploration behavior of DE on the Ackley function with $m = 5$.

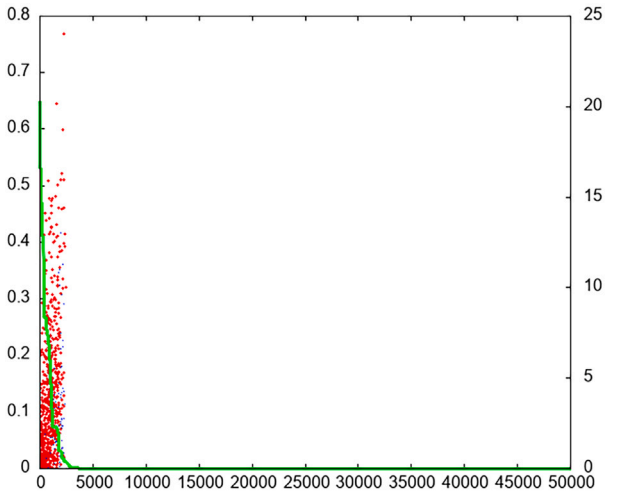


Fig. 21. Exploration behavior of JADE on the Ackley function with $m = 5$.

of the desired types of exploration (successful exploration and successful rejection), all the selected EAs had a ratio of 85% for the desired types of exploration to 15% for undesired types of exploration (failed and deceptive exploration). However, DE and JADE converge to the global optimum faster.

On Figs. 19–28, we can observe the exploration behavior of the selected EAs on the Ackley function visually. DE, JADE, and TLBO behaved similarly, where failed exploration was negligible. For ABC (Figs. 25 and 26), successful exploration came again in waves, due to the behavior of the scout bee phase. Scout bees, whose primary objective is to explore the search space, become active only after a predefined number of consecutive non-improvements (denoted by *limit*). When a better solution is found, this counter is reset to zero [22,23].

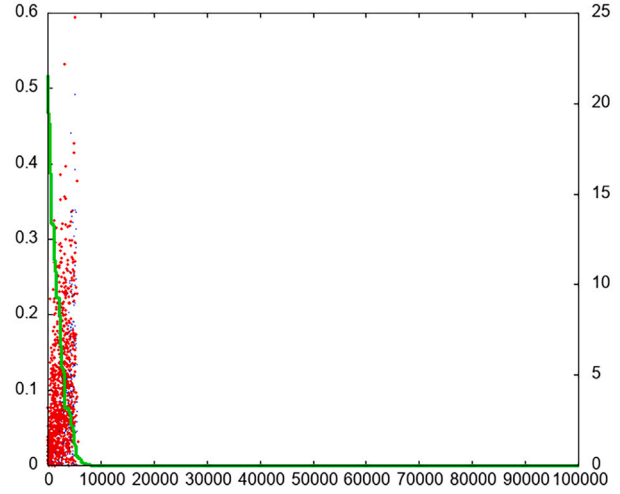


Fig. 20. Exploration behavior of DE on the Ackley function with $m = 10$.

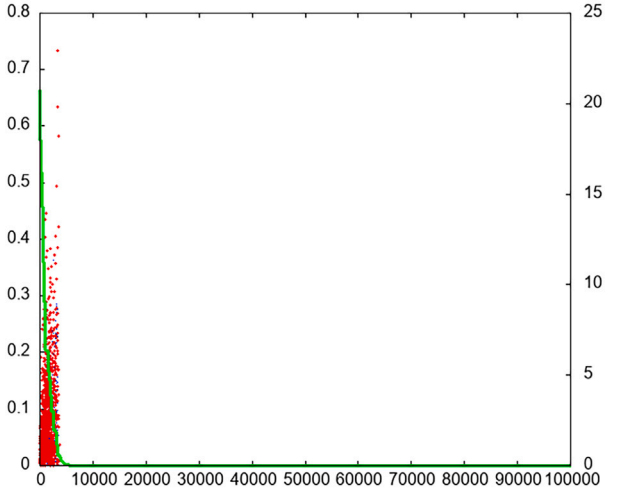


Fig. 22. Exploration behavior of JADE on the Ackley function with $m = 10$.

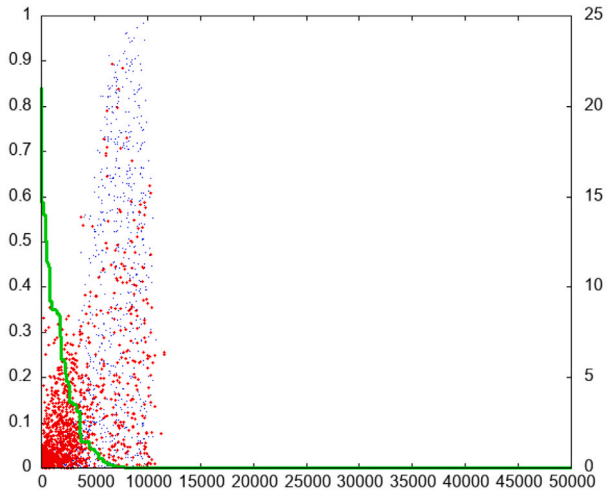


Fig. 23. Exploration behavior of LSHADE on the Ackley function with $m = 5$.

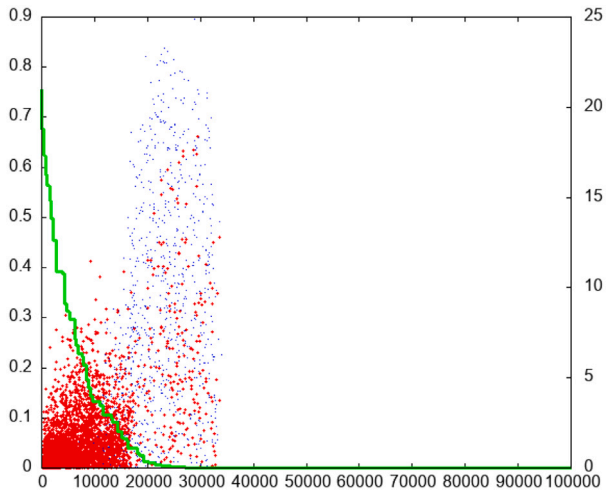


Fig. 24. Exploration behavior of LSHADE on the Ackley function with $m = 10$.

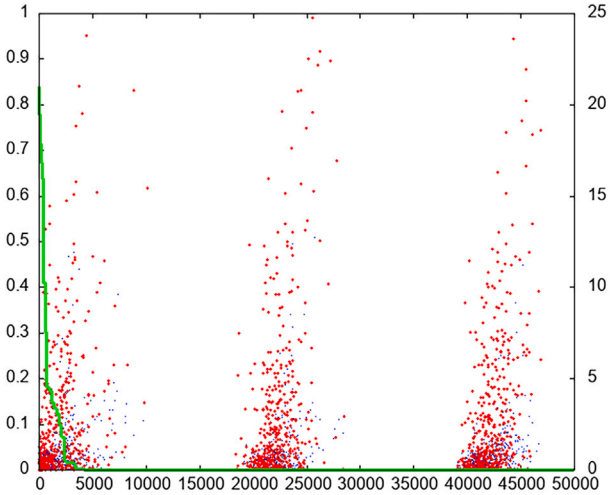


Fig. 25. Exploration behavior of ABC on the Ackley function with $m = 5$.

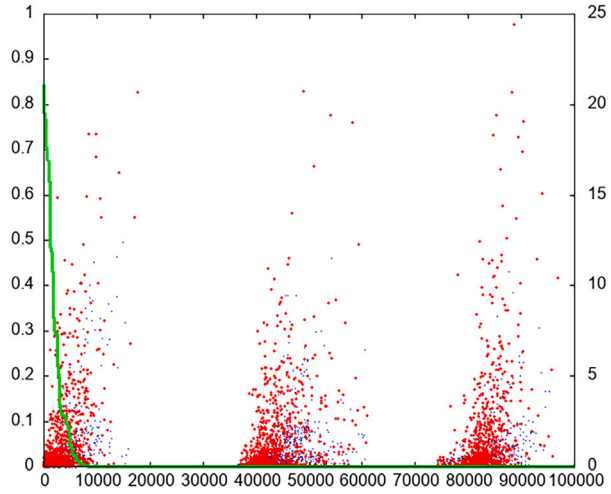


Fig. 26. Exploration behavior of ABC on the Ackley function with $m = 10$.

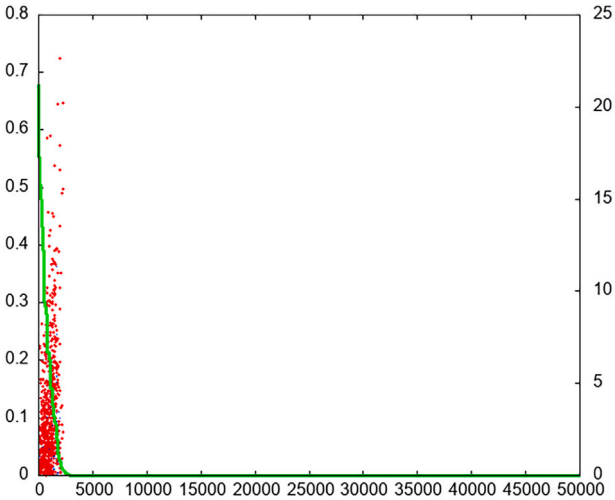


Fig. 27. Exploration behavior of TLBO on the Ackley function with $m = 5$.

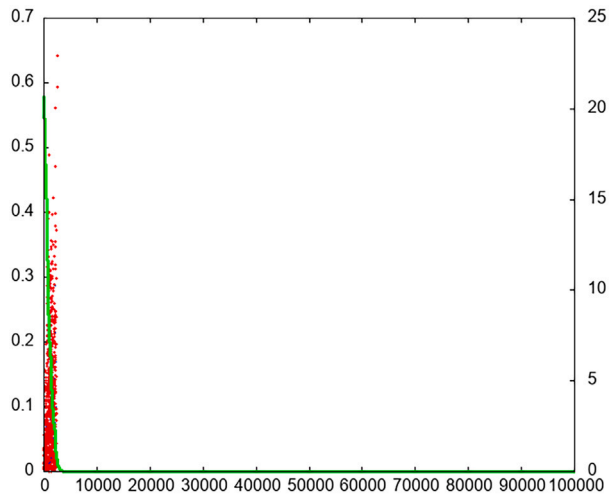


Fig. 28. Exploration behavior of TLBO on the Ackley function with $m = 10$.

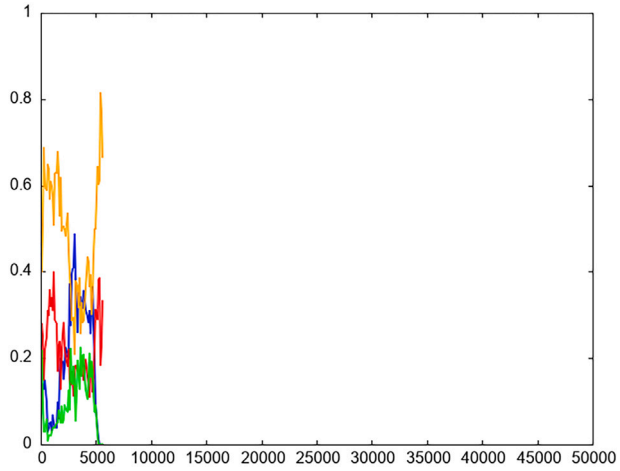


Fig. 29. Exploration types throughout the evolutionary run for DE on the Ackley function with $m = 5$.

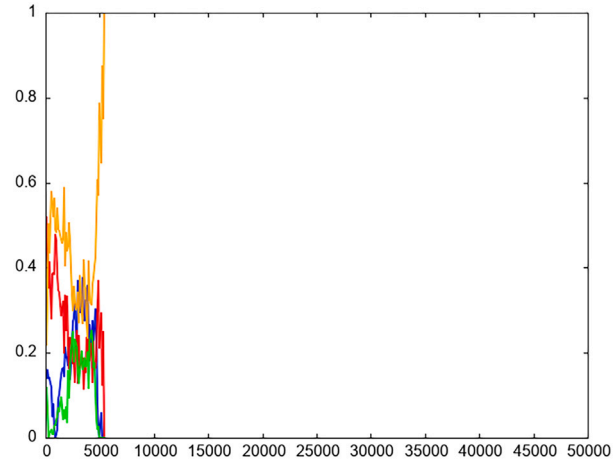


Fig. 30. Exploration types throughout the evolutionary run for JADE on the Ackley function with $m = 5$.

Different types of exploration during the evolutionary run are presented in (Figs. 29–33), where the red, blue, green and orange lines represent successful exploration, failed exploration, deceptive exploration, and successful rejection, respectively. Similar behavior for DE, JADE, and TLBO can be observed again. Exploration waves for ABC are clearly visible. LSHADE had prolonged exploration compared to DE and JADE.

4.3. Griewank function

Contrary to Ackley, the Griewank function is a harder problem for the selected EAs under the allowed number of fitness evaluations, and only JADE found the global optimum in every independent run (Table 5). The amount of exploration varied drastically among the selected EAs, from the lowest 10% for LSHADE, when $m = 5$, to the highest 75% for DE, when $m = 5$. Again, the amount of deceptive exploration was around 5% or lower, and not problematic. Failed exploration was extremely high for TLBO, more than 60%. Again, promising regions were found and rejected by TLBO's mechanics. The most successful was JADE, with successful exploration of 7%, failed exploration around 25%, and successful rejection around 65%. The amount of exploration was 33%, when $m = 5$, and 21%, when $m = 10$.

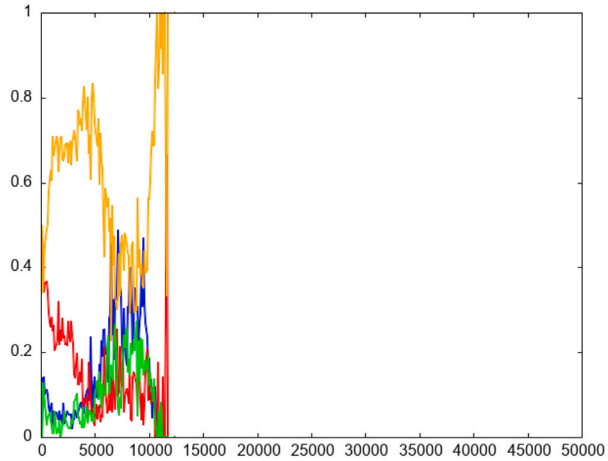


Fig. 31. Exploration types throughout the evolutionary run for LSHADE on the Ackley function with $m = 5$.

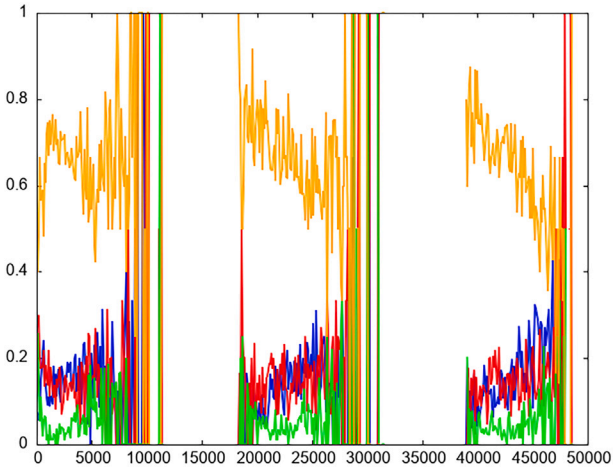


Fig. 32. Exploration types throughout the evolutionary run for ABC on the Ackley function with $m = 5$.

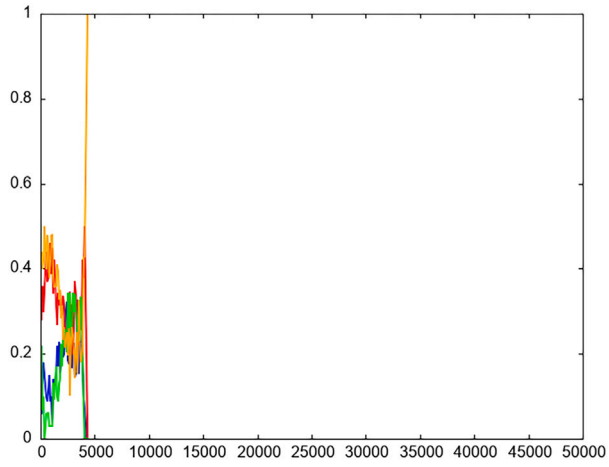


Fig. 33. Exploration types throughout the evolutionary run for TLBO on the Ackley function with $m = 5$.

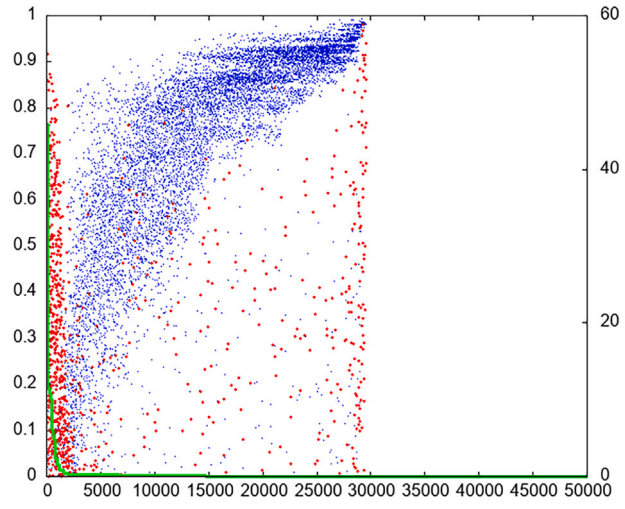


Fig. 34. Exploration behavior of DE on the Griewank function with $m = 5$.

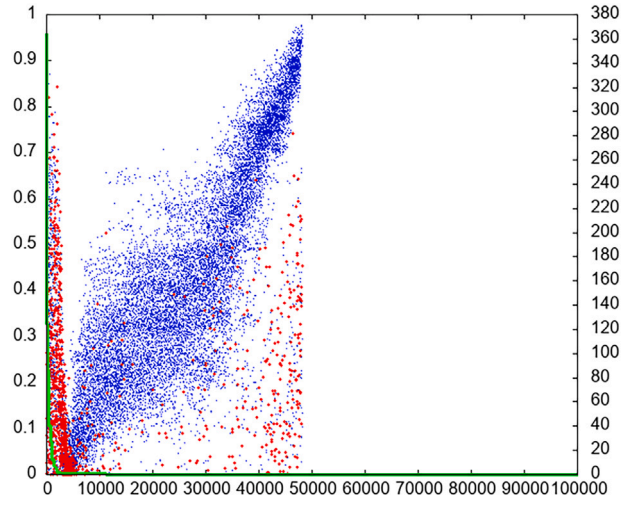


Fig. 35. Exploration behavior of DE on the Griewank function with $m = 10$.

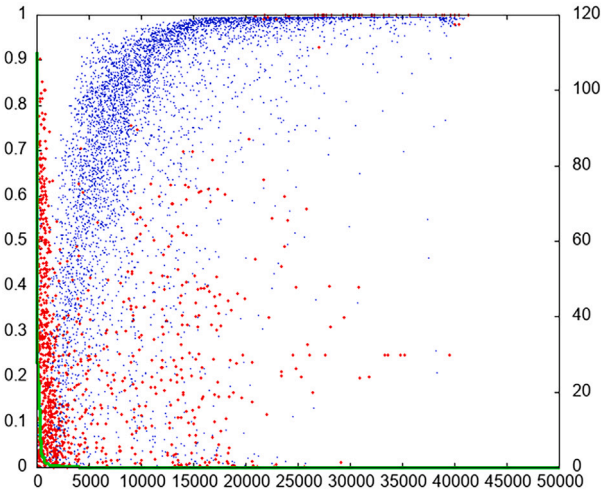


Fig. 36. Exploration behavior of JADE on the Griewank function with $m = 5$.

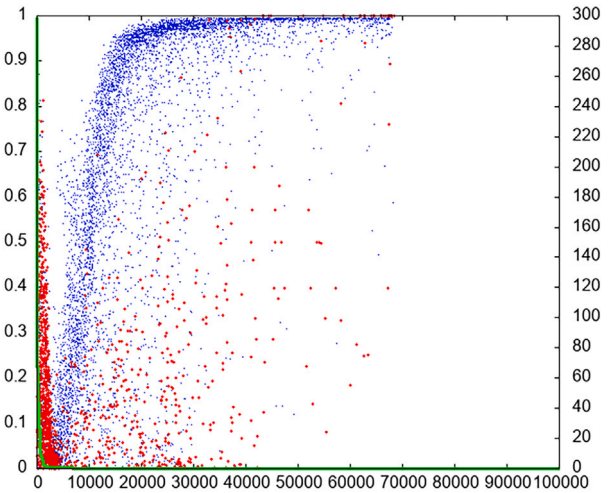


Fig. 37. Exploration behavior of JADE on the Griewank function with $m = 10$.

In Figs. 34–43, we can observe the exploration behavior of the selected EAs on the Griewank function visually. Again, DE, JADE and LSHADE behaved similarly. In the beginning, successful exploration prevailed quickly, and better promising regions were found. However, in the continuation, JADE excelled over DE and LSHADE, with far fewer failed explorations. LSHADE, compared to DE and JADE, prolonged the exploration phase. For ABC, successful exploration came again in waves, due to the scout bee phase. This was more observable when $m = 10$. The lack of a sufficient amount of exploration in TLBO is clearly visible, which was 12% when $m = 10$.

The different types of exploration during the evolutionary run are presented in (Figs. 44–48), where the red, blue, green and orange lines represent successful exploration, failed exploration, deceptive exploration, and successful rejection, respectively. The overperformance of JADE over DE and LSHADE can be attributed to a larger amount of successful exploration in the later stages of the evolutionary run. The amount of successful exploration in the second half of the evolutionary run is surprisingly very low for LSHADE. For TLBO, the number of failed explorations was again larger than the number of successful rejections, especially at the beginning of the evolutionary run.

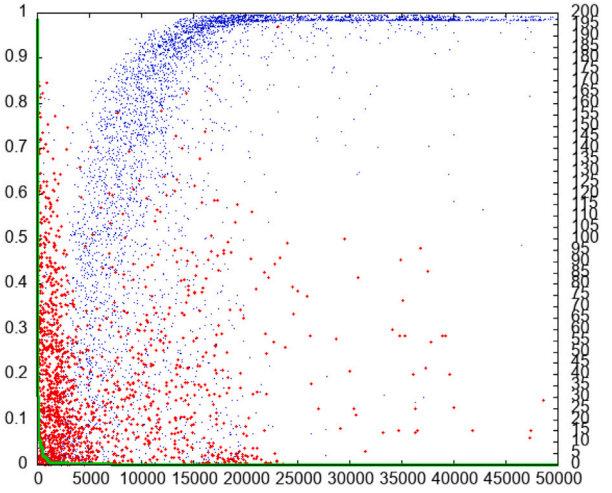


Fig. 38. Exploration behavior of LSHADE on the Griewank function with $m = 5$.

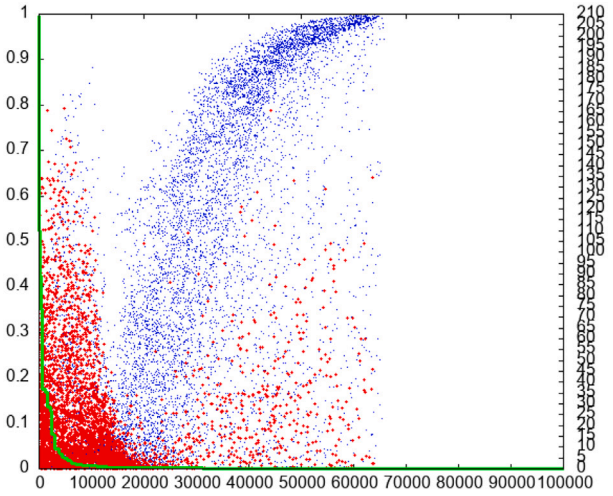


Fig. 39. Exploration behavior of LSHADE on the Griewank function with $m = 10$.

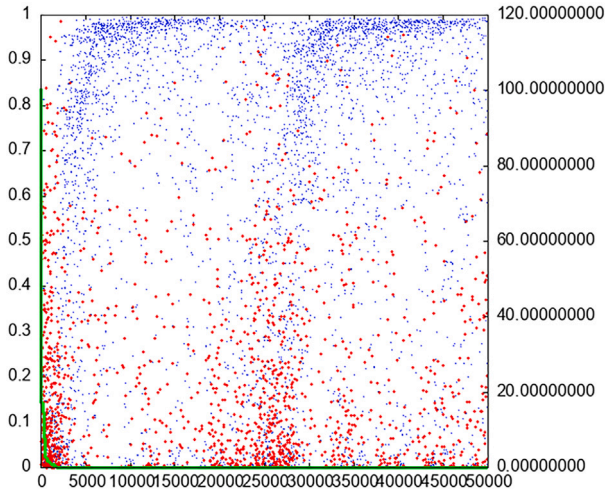


Fig. 40. Exploration behavior of ABC on the Griewank function with $m = 5$.

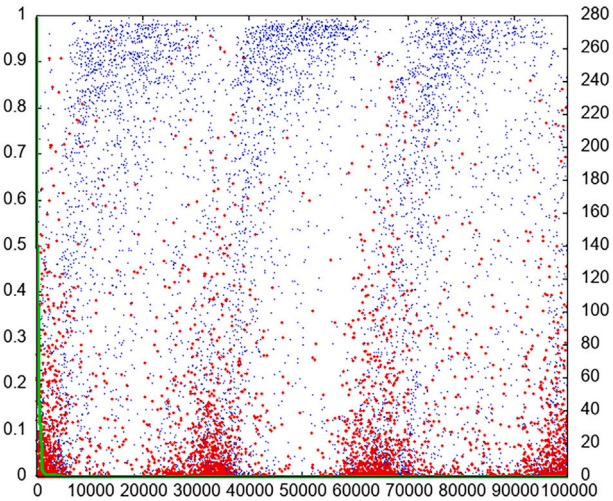


Fig. 41. Exploration behavior of ABC on the Griewank function with $m = 10$.

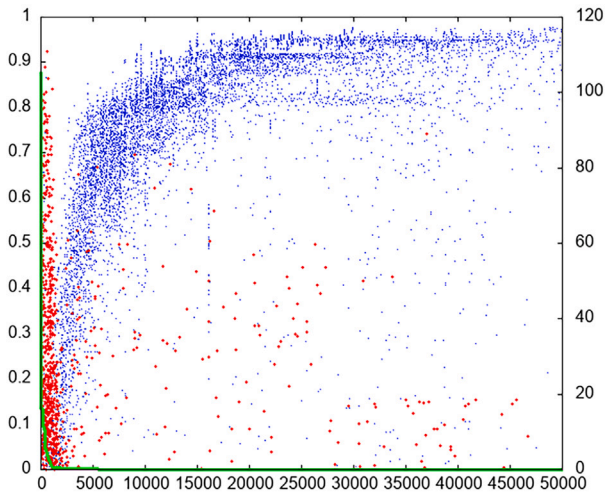


Fig. 42. Exploration behavior of TLBO on the Griewank function with $m = 5$.

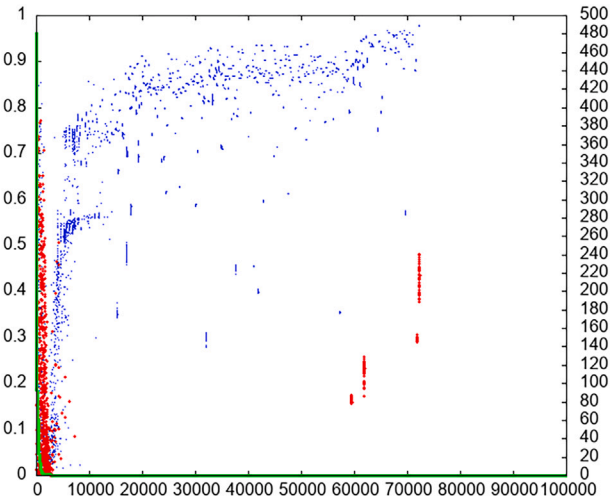


Fig. 43. Exploration behavior of TLBO on the Griewank function with $m = 10$.

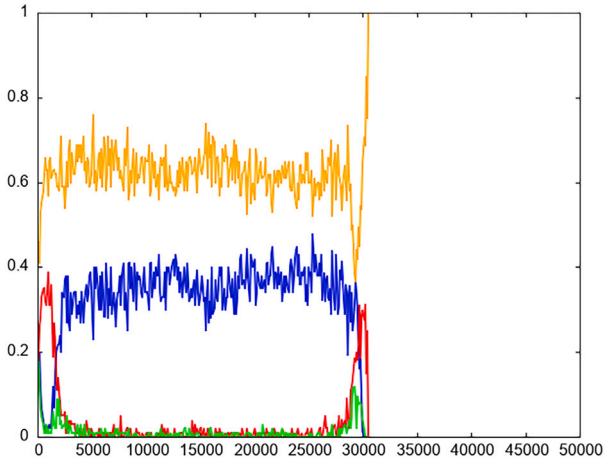


Fig. 44. Exploration types throughout the evolutionary run for DE on the Griewank function with $m = 5$.

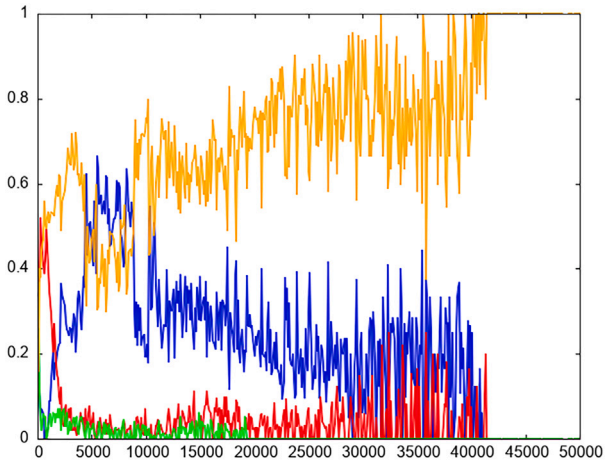


Fig. 45. Exploration types throughout the evolutionary run for JADE on the Griewank function with $m = 5$.

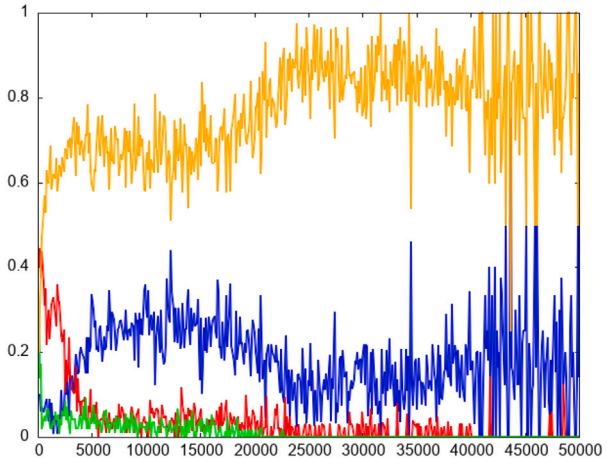


Fig. 46. Exploration types throughout the evolutionary run for LSHADE on the Griewank function with $m = 5$.

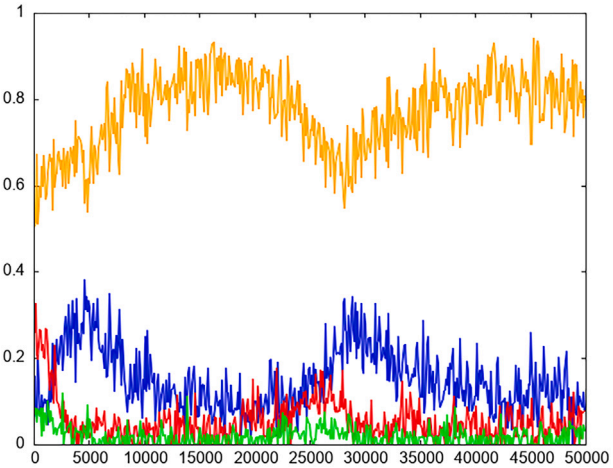


Fig. 47. Exploration types throughout the evolutionary run for ABC on the Griewank function with $m = 5$.

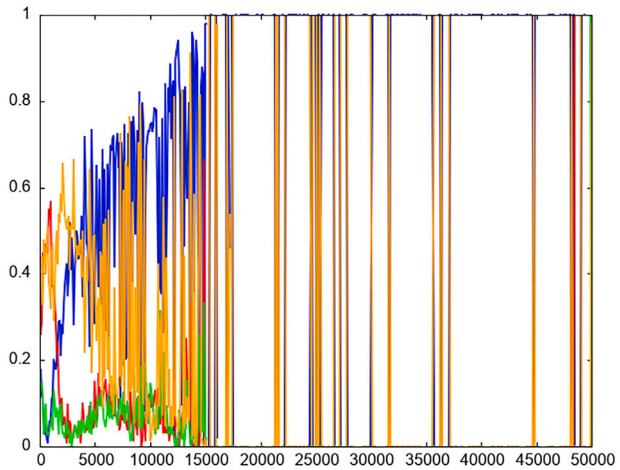


Fig. 48. Exploration types throughout the evolutionary run for TLBO on the Griewank function with $m = 5$.

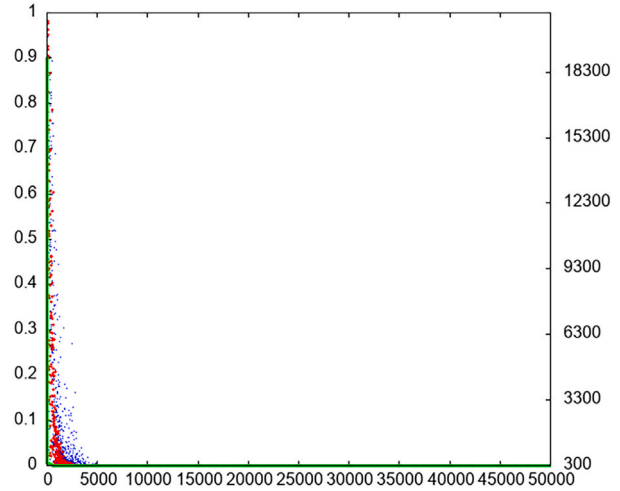


Fig. 49. Exploration behavior of DE on the F03 function with $m = 5$.

4.4. F03 function

For the multimodal function F03, only JADE and LSHADE found the global optimum in all the runs (Table 6), with the exploration ratio around 15% for JADE and lower for LSHADE, 5% ($m = 5$) and 7% ($m = 10$). All the other selected EAs explored the search too extensively (e.g., DE 88%, TLBO 78%, ABC 74%). DE and TLBO exhibited a high deceptive exploration rate of 15%, the highest observed so far. In addition, there was a low successful rejection rate (around 35%). For ABC, failed exploration (more than 40%) was a reason for concern. LSHADE again exhibits the lowest deception exploration rate. Figs. 49–58 additionally supported these observations.

The different types of exploration during the evolutionary run are presented in Figs. 59–63, where the red, blue, green and orange lines represent successful exploration, failed exploration, deceptive exploration, and successful rejection, respectively. There was an increased amount of deceptive exploration for DE, and failed exploration prevailed in the later stages of the evolutionary run. Better attraction basins were found in the later stages, but were rejected. While failed exploration prevailed in the early stages for JADE and LSHADE, the desired exploration (successful exploration and successful rejection) happened in the later stages.

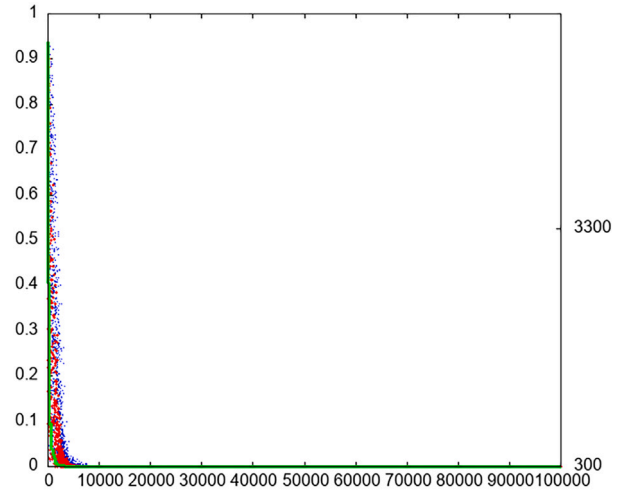


Fig. 50. Exploration behavior of DE on the F03 function with $m = 10$.

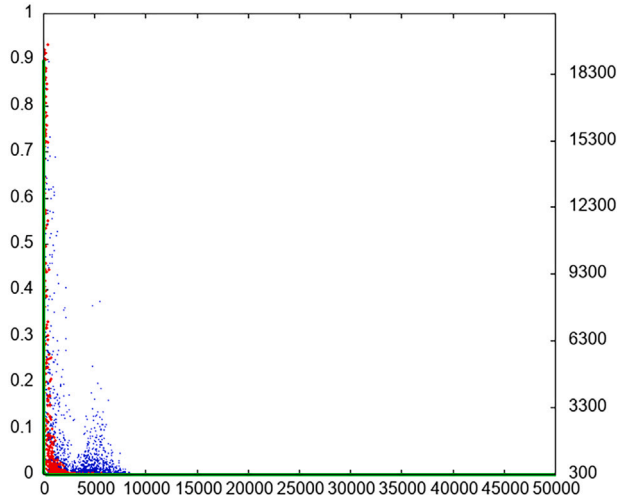


Fig. 51. Exploration behavior of JADE on the F03 function with $m = 5$.

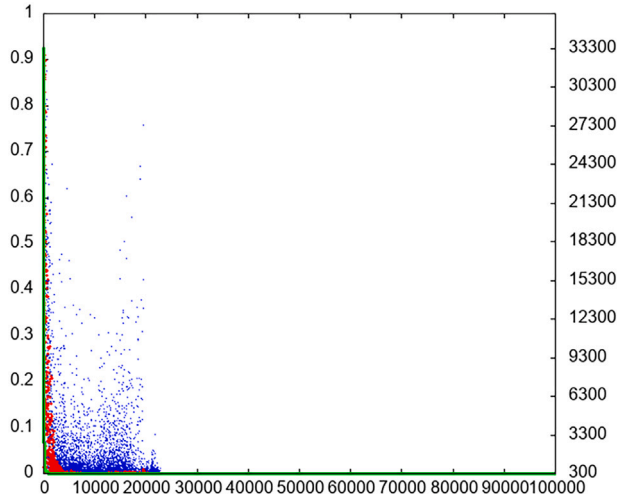


Fig. 52. Exploration behavior of JADE on the F03 function with $m = 10$.

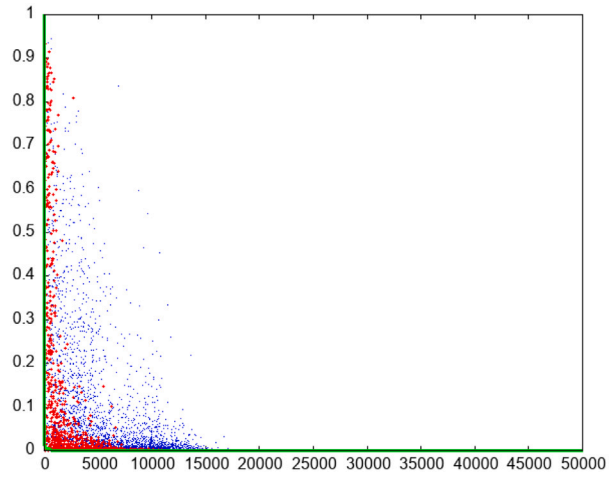


Fig. 53. Exploration behavior of LSHADE on the F03 function with $m = 5$.

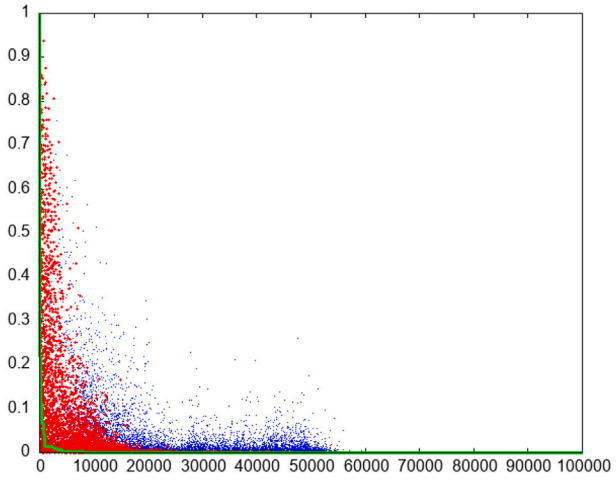


Fig. 54. Exploration behavior of LSHADE on the F03 function with $m = 10$.

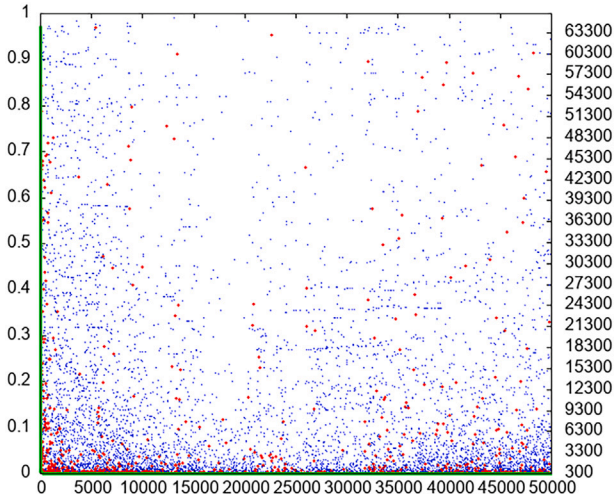


Fig. 55. Exploration behavior of ABC on the F03 function with $m = 5$.

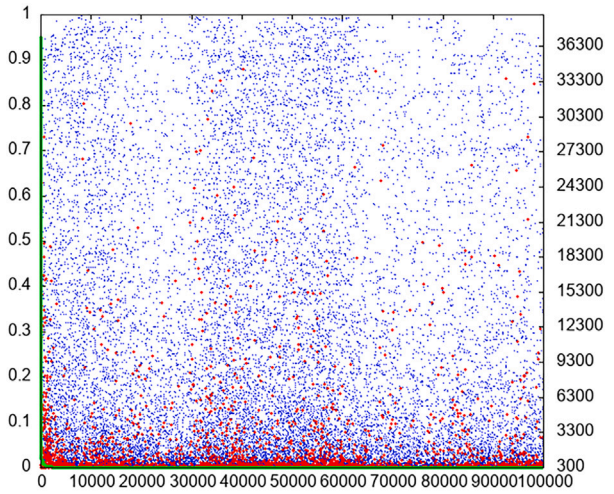


Fig. 56. Exploration behavior of ABC on the F03 function with $m = 10$.

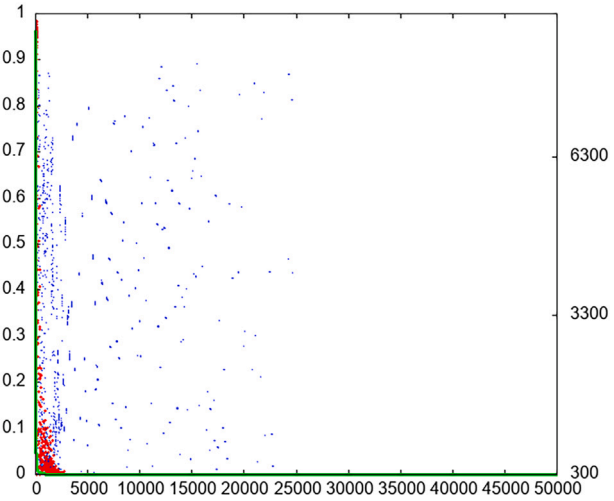


Fig. 57. Exploration behavior of TLBO on the F03 function with $m = 5$.

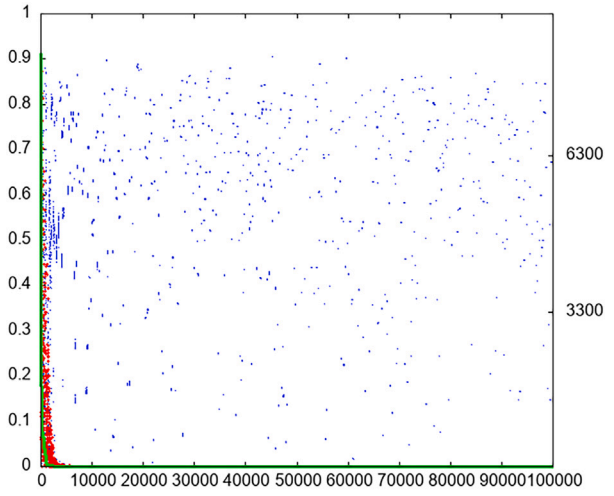


Fig. 58. Exploration behavior of TLBO on the F03 function with $m = 10$.

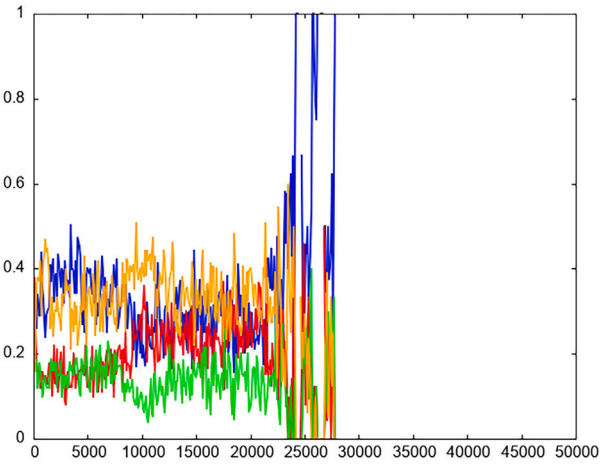


Fig. 59. Exploration types throughout the evolutionary run for DE on the F03 function with $m = 5$.

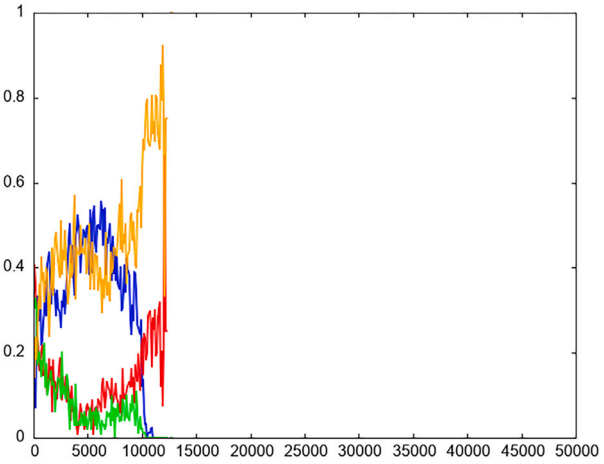


Fig. 60. Exploration types throughout the evolutionary run for JADE on the F03 function with $m = 5$.

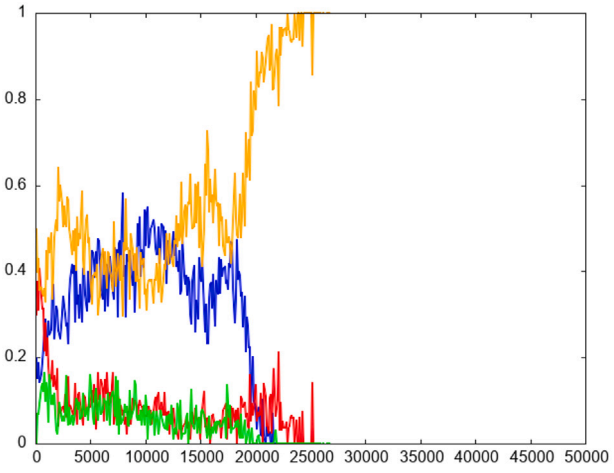


Fig. 61. Exploration types throughout the evolutionary run for LSHADE on the F03 function with $m = 5$.

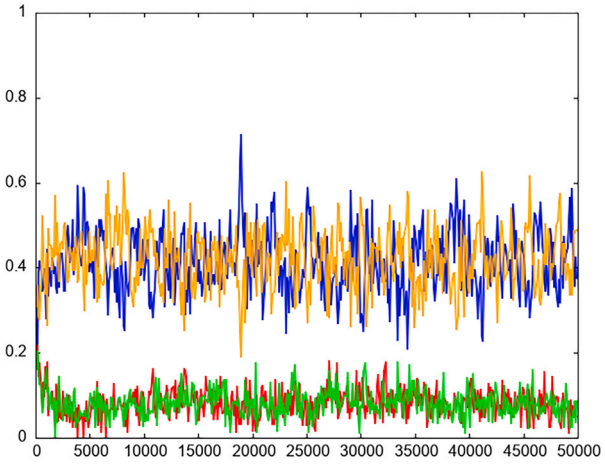


Fig. 62. Exploration types throughout the evolutionary run for ABC on the F03 function with $m = 5$.

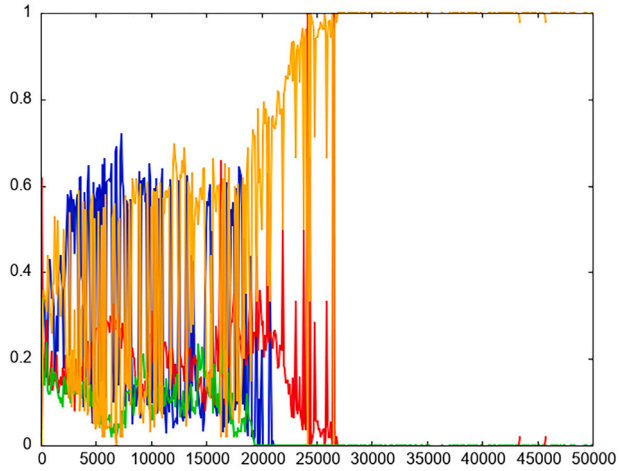


Fig. 63. Exploration types throughout the evolutionary run for TLBO on the F03 function with $m = 5$.

ABC struggled with failed exploration throughout the entire evolutionary run, whereas failed exploration can be observed in the early stages for TLBO and desired exploration in the later stages.

4.5. F10 function

The hybrid function F10 was solved in every run by JADE only when $m = 5$ (Table 7), although other selected EAs were not much behind. The exploration ratio varied among the selected EAs: DE (30%, when $m = 5$, and 72%, when $m = 10$), JADE (34%, when $m = 5$, and 50%, when $m = 10$), LSHADE (6%, when $m = 5$, and 18%, when $m = 10$), ABC (84%, when $m = 5$, and 59%, when $m = 10$), TLBO (82%, when $m = 5$, and 72%, when $m = 10$). Successful rejection was around 50% for all of the selected EAs. Deceptive exploration, around 10%, was problematic only for TLBO and DE when $m = 10$. On the other hand, failed exploration was very high, above 30%, even one of the highest for the most successful JADE (42%); successful exploration was also low for JADE, LSHADE and ABC.

Figs. 64–73 support the results from Table 7, additionally, where we can observe the high amount of failed exploration and a low amount of successful exploration, indicating a hard problem for the selected EAs. A high exploration rate for ABC is also clearly visible, where failed exploration prevails.

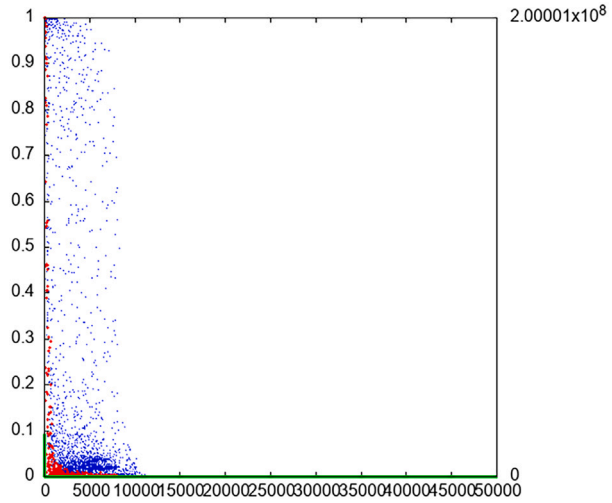


Fig. 64. Exploration behavior of DE on the F10 function with $m = 5$.

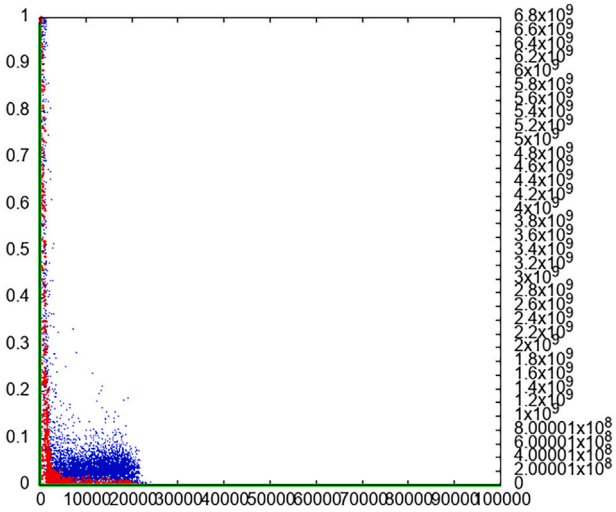


Fig. 65. Exploration behavior of DE on the F10 function with $m = 10$.

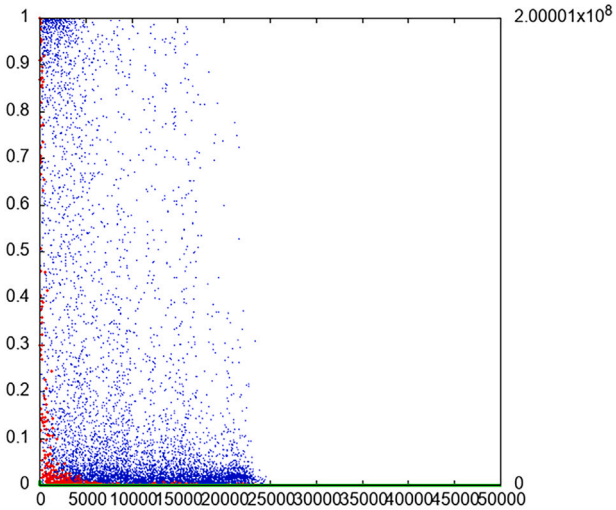


Fig. 66. Exploration behavior of JADE on the F10 function with $m = 5$.

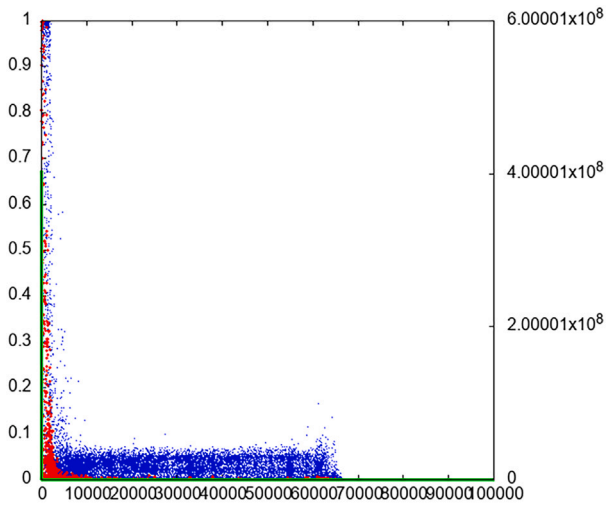


Fig. 67. Exploration behavior of JADE on the F10 function with $m = 10$.

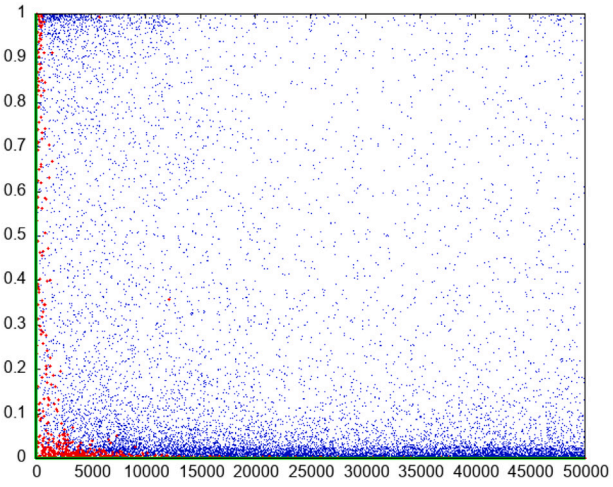


Fig. 68. Exploration behavior of LSHADE on the F10 function with $m = 5$.

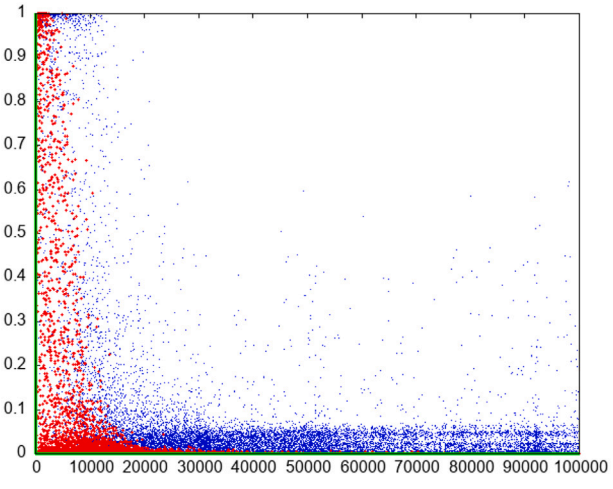


Fig. 69. Exploration behavior of LSHADE on the F10 function with $m = 10$.

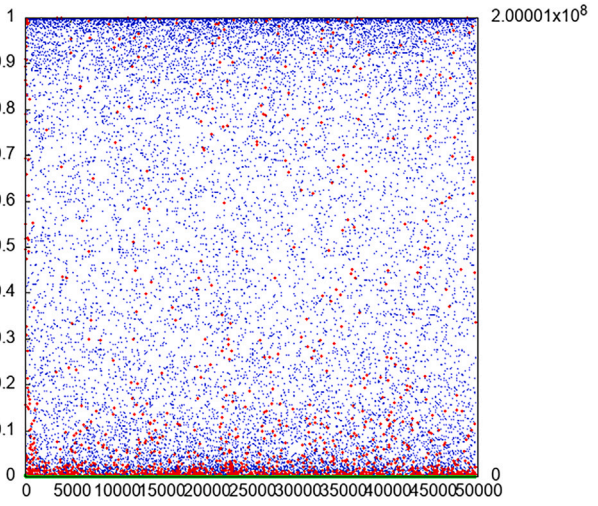


Fig. 70. Exploration behavior of ABC on the F10 function with $m = 5$.

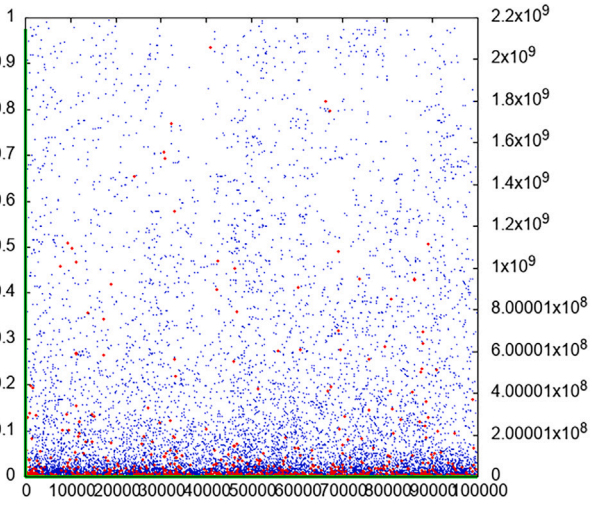


Fig. 71. Exploration behavior of ABC on the F10 function with $m = 10$.

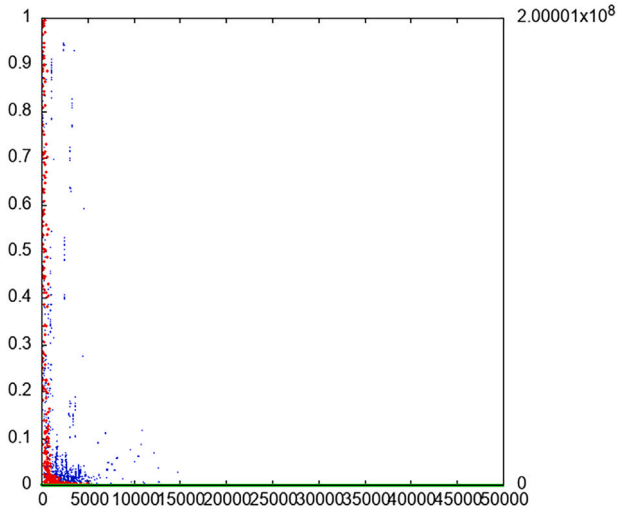


Fig. 72. Exploration behavior of TLBO on the F10 function with $m = 5$.

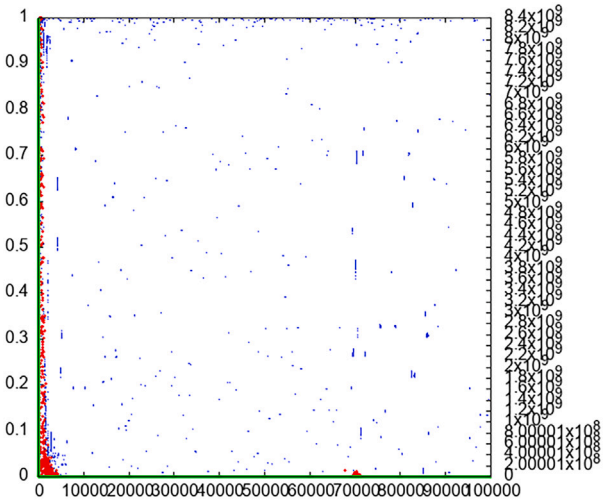


Fig. 73. Exploration behavior of TLBO on the F10 function with $m = 10$.

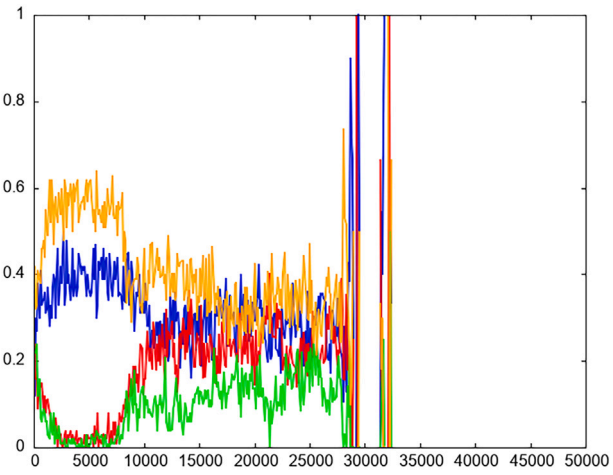


Fig. 74. Exploration types throughout the evolutionary run for DE on the F10 function with $m = 5$.

The different types of exploration during the evolutionary run are presented in Figs. 74–78, where the red, blue, green and orange lines represent successful exploration, failed exploration, deceptive exploration, and successful rejection, respectively. There was an increased amount of deceptive exploration for DE in the later stages, while failed exploration decreased gradually throughout the evolutionary run. JADE again increased the successful exploration in the later stages, despite a very large amount of failed exploration beforehand. The amount of successful exploration for LSHADE in the later stages is again problematic. LSHADE and ABC struggled with failed exploration throughout the whole evolutionary run. A significant amount of failed exploration can be observed in the early stages of TLBO, with an increase in desired exploration in the later stages.

4.6. F20 function

Among the selected optimization functions, the composition function F20, appeared to be the most difficult, since none of the selected EAs found the global optimum in all the runs (Table 8). The best average fitness was obtained by ABC, when $m = 5$. The exploration ratio varied among the selected EAs: DE (5%, when $m = 5$, and 42%, when $m = 10$), JADE (8%, when $m = 5$, and 23%, when $m = 10$), LSHADE (12%, when $m = 5$, and 14%, when $m = 10$), ABC (69%, when $m = 5$, and

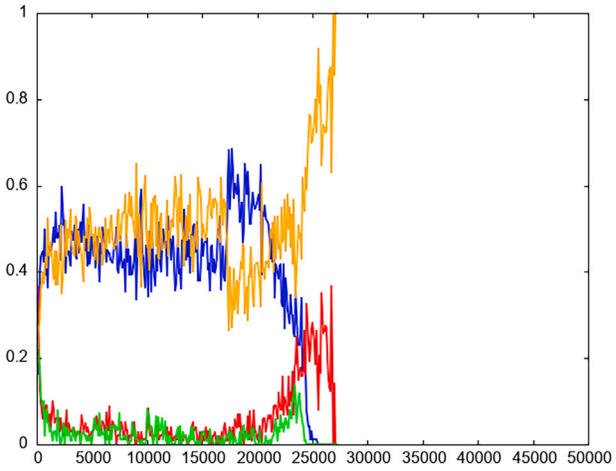


Fig. 75. Exploration types throughout the evolutionary run for JADE on the F10 function with $m = 5$.

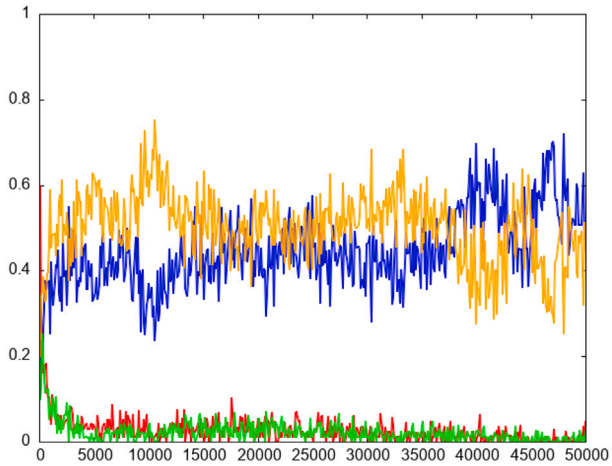


Fig. 76. Exploration types throughout the evolutionary run for LSHADE on the F10 function with $m = 5$.

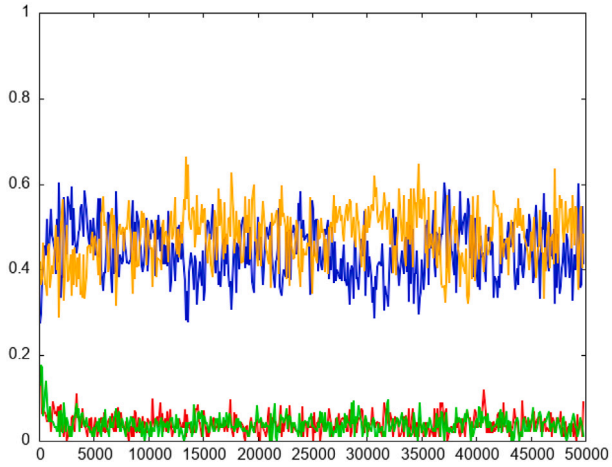


Fig. 77. Exploration types throughout the evolutionary run for ABC on the F10 function with $m = 5$.

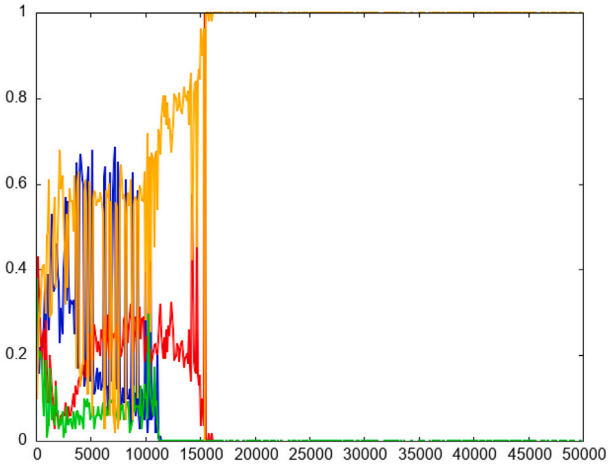


Fig. 78. Exploration types throughout the evolutionary run for TLBO on the F10 function with $m = 5$.

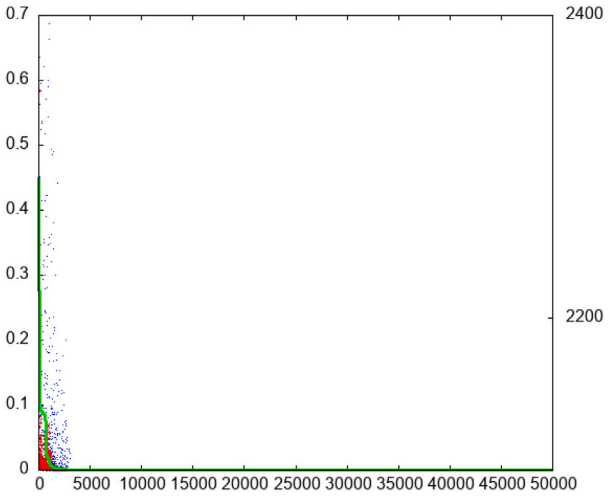


Fig. 79. Exploration behavior of DE on the F20 function with $m = 5$.

67%, when $m = 10$), TLBO (14%, when $m = 5$, and 29%, when $m = 10$). Successful rejection was around 50% or higher for all the selected EAs, while deceptive exploration was the lowest among all the selected optimization problems. However, the high amount of failed exploration for all the selected EAs, as well as the low amount of successful exploration, is problematic. The last observation is clearly visible in Figs. 79–88.

The different types of exploration during the evolutionary run are presented in Figs. 89–93, where the red, blue, green and orange lines represent successful exploration, failed exploration, deceptive exploration, and successful rejection, respectively. The DE and JADE behavior was similar, with an increase in deceptive exploration, a high amount of failed exploration, and decreasing successful rejection. LSHADE and ABC exhibited a low amount of successful exploration throughout the whole evolutionary run, with a very high amount of failed exploration. TLBO started with a high amount of deceptive exploration, followed by alternating between failed exploration and successful rejection.

4.7. Discussion

Although the number of problems (Rastrigin, Ackley, Griewank, F03, F10, F20) is small, a clear distinction in the exploration performance of the compared algorithms is evident from the graphs and tables presented

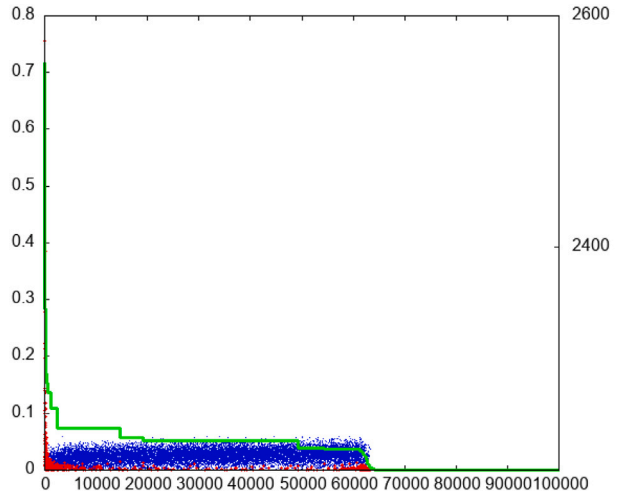


Fig. 80. Exploration behavior of DE on the F20 function with $m = 10$.

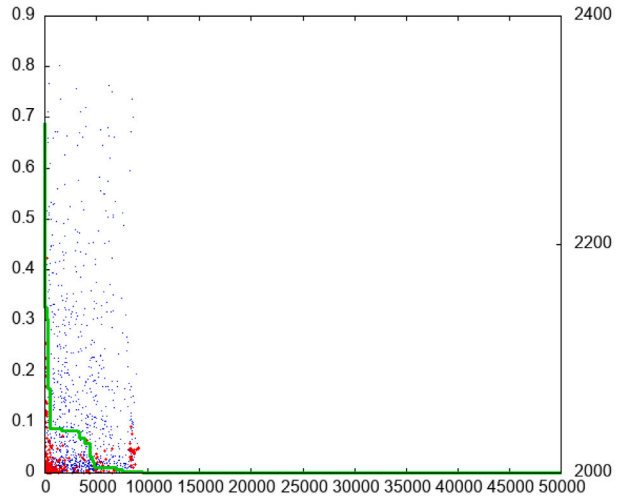


Fig. 81. Exploration behavior of JADE on the F20 function with $m = 5$.

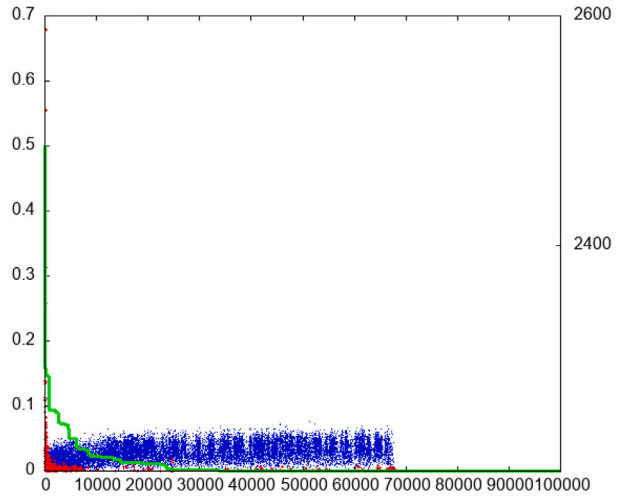


Fig. 82. Exploration behavior of JADE on the F20 function with $m = 10$.

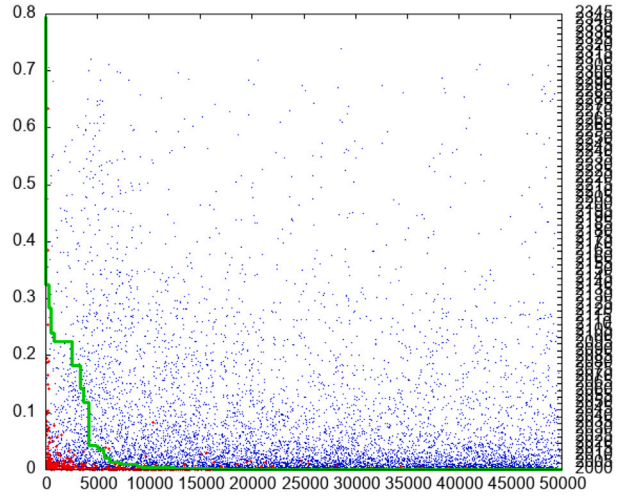


Fig. 83. Exploration behavior of LSHADE on the F20 function with $m = 5$.

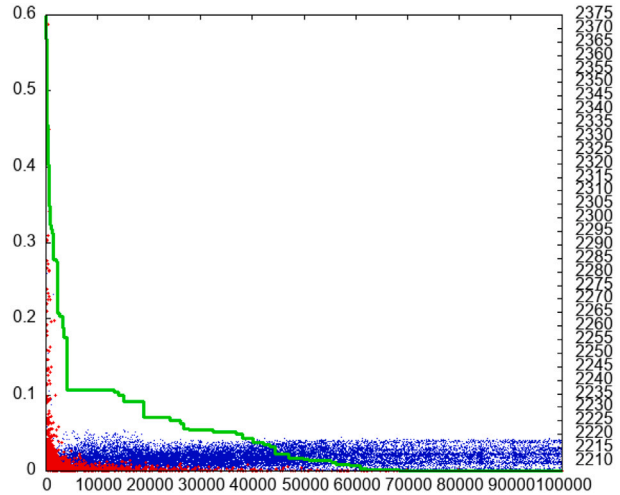


Fig. 84. Exploration behavior of LSHADE on the F20 function with $m = 10$.

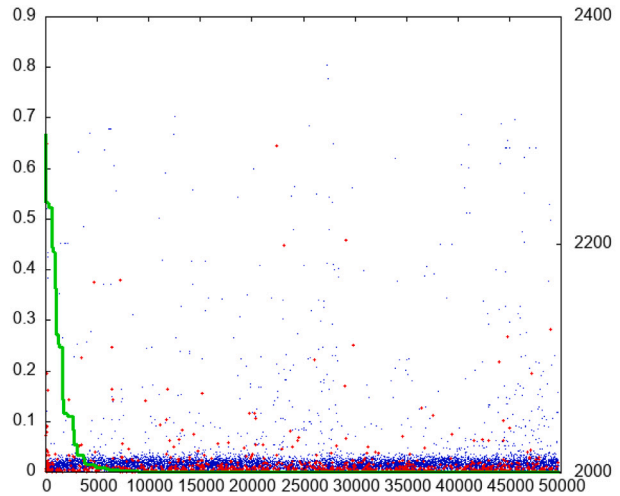


Fig. 85. Exploration behavior of ABC on the F20 function with $m = 5$.

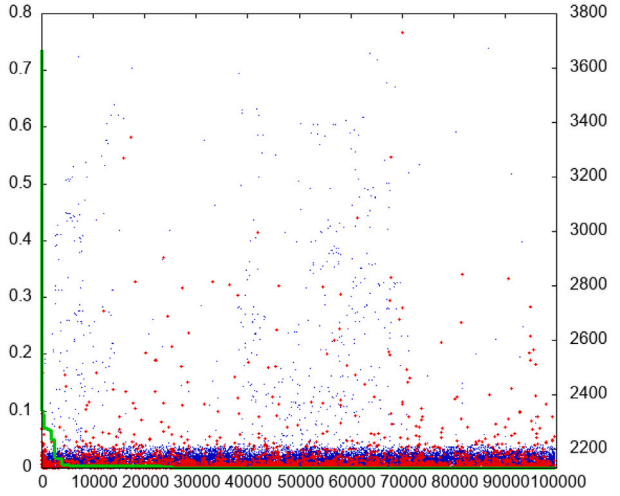


Fig. 86. Exploration behavior of ABC on the F20 function with $m = 10$.

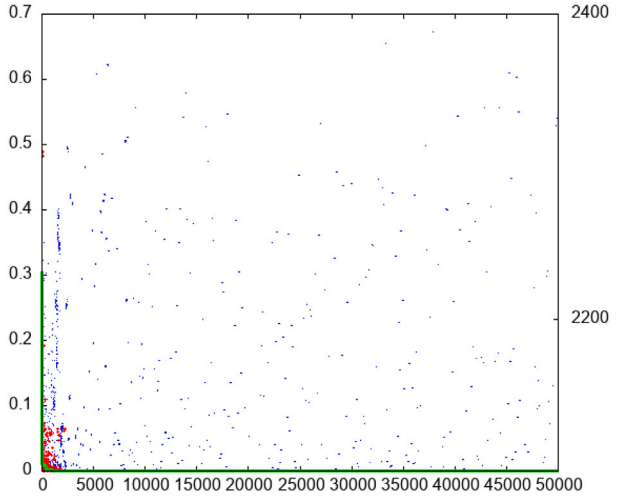


Fig. 87. Exploration behavior of TLBO on the F20 function with $m = 5$.

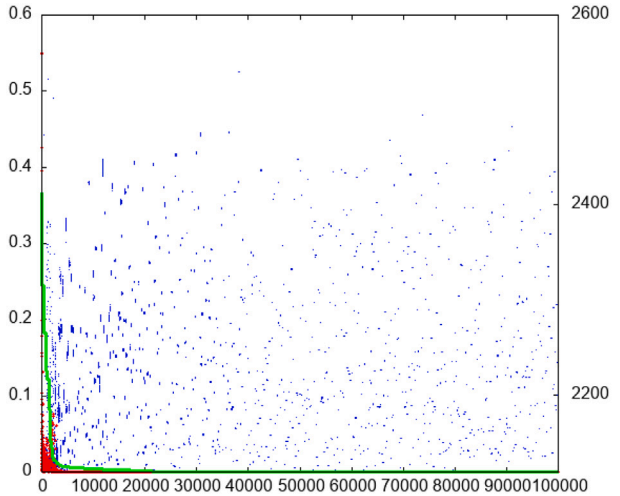


Fig. 88. Exploration behavior of TLBO on the F20 function with $m = 10$.

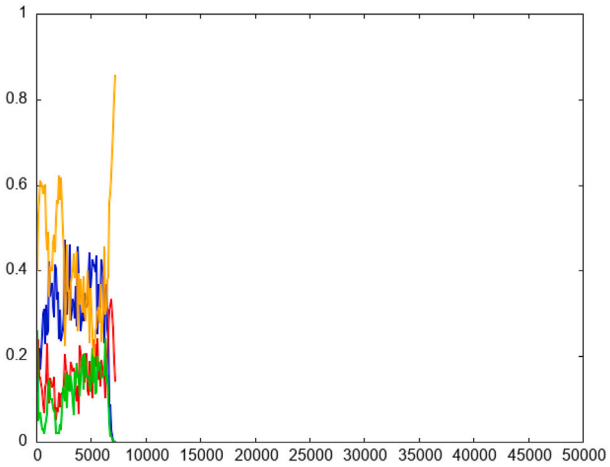


Fig. 89. Exploration types throughout the evolutionary run for DE on the F20 function with $m = 5$.

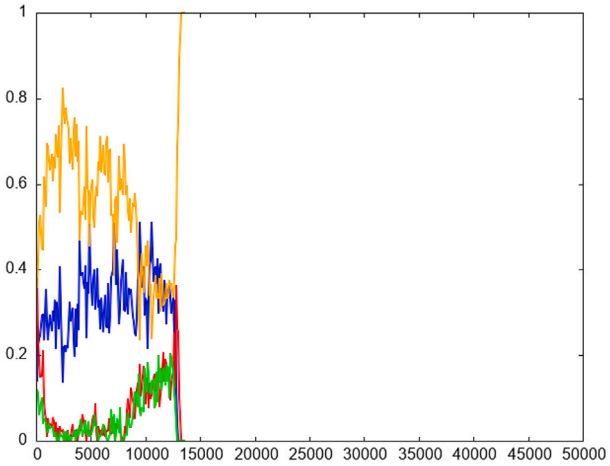


Fig. 90. Exploration types throughout the evolutionary run for JADE on the F20 function with $m = 5$.

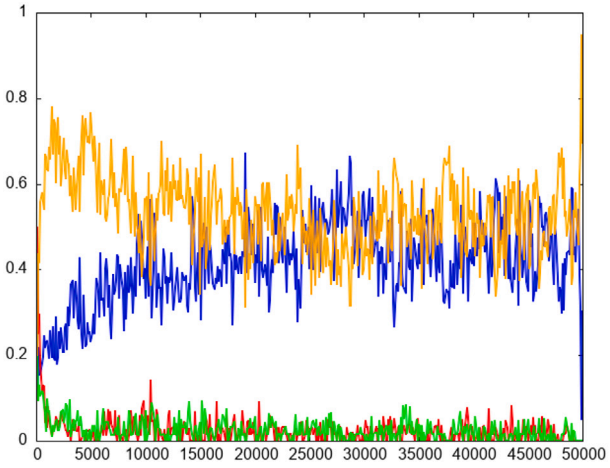


Fig. 91. Exploration types throughout the evolutionary run for LSHADE on the F20 function with $m = 5$.

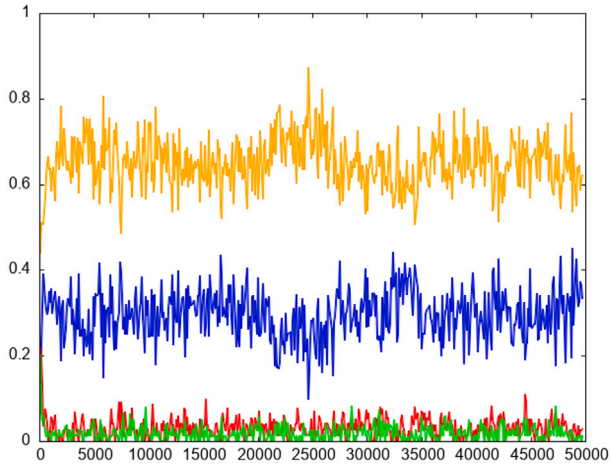


Fig. 92. Exploration types throughout the evolutionary run for ABC on the F20 function with $m = 5$.

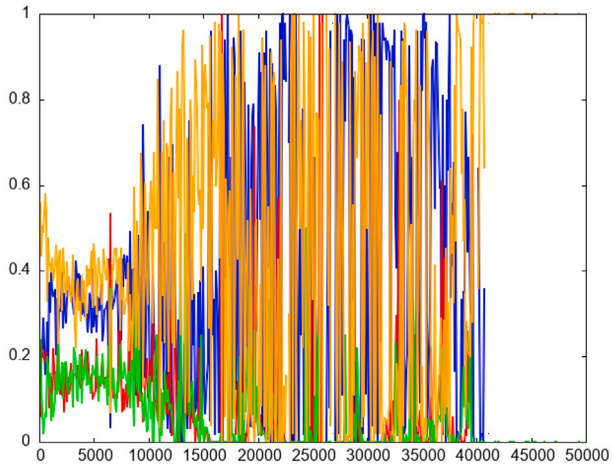


Fig. 93. Exploration types throughout the evolutionary run for TLBO on the F20 function with $m = 5$.

in the previous subsections (where the red, blue, green and orange lines represent successful exploration, failed exploration, deceptive exploration, and successful rejection, respectively). From the DE family, DE and JADE behave similarly, while LSHADE exhibits negligible deceptive exploration. On the other hand, TLBO has a larger amount of failed and deceptive exploration, and a smaller amount of successful rejection than other compared algorithms. The amount of exploration depends on the complexity of the fitness landscape, and a huge increase in exploration is observed for all algorithms on problems F03, F10, and F20. On these problems, the failed exploration is particularly high (above 30%) for DE, JADE, LSHADE, ABC, and TLBO. It can also be observed that deceptive exploration drastically increased in later stages of the evolutionary run for DE and JADE, but it is matched with increased successful exploration. In comparison, deceptive exploration is pretty stable in ABC, especially for problems F03, F10 and F20. The experiments show that deceptive exploration is not as problematic as failed exploration caused by premature convergence. This is why special focus is given in our analysis to successful and failed exploration.

4.8. Additional experiments

In this subsection additional experiments are presented, to briefly investigate the influence of ϵ on measuring the different types of exploration (successful exploration, successful rejection, failed exploration,

Table 5
Exploration and exploitation metrics on the Griewank function for the dimensions $m = 5$ and $m = 10$.

Algorithm	Successful exploration	Failed exploration	Deceptive exploration	Successful rejection	Exploration	Diversity	Best Fitness
DE Griewank ($m = 5$)	0.0258 ± 0.0055	0.3410 ± 0.0096	0.0083 ± 0.0018	0.6249 ± 0.0083	0.7534 ± 0.1265	45.7989 ± 4.8580	0.0012 ± 0.0039
DE Griewank ($m = 10$)	0.0378 ± 0.0105	0.3449 ± 0.0158	0.0227 ± 0.0053	0.5946 ± 0.0176	0.3329 ± 0.0884	49.8411 ± 5.4227	0.0121 ± 0.0095
JADE Griewank ($m = 5$)	0.0681 ± 0.0069	0.2854 ± 0.0194	0.0150 ± 0.0018	0.6315 ± 0.0167	0.3295 ± 0.0313	26.9000 ± 1.5743	0.0000 ± 0.0000
JADE Griewank ($m = 10$)	0.0684 ± 0.0064	0.2583 ± 0.0086	0.0322 ± 0.0029	0.6412 ± 0.0107	0.2064 ± 0.0285	31.2999 ± 2.5222	0.0000 ± 0.0000
LSHADE Griewank ($m = 5$)	0.0619 ± 0.0032	0.1768 ± 0.0124	0.0209 ± 0.0017	0.7404 ± 0.0105	0.1034 ± 0.0082	56.2859 ± 1.5279	0.0007 ± 0.0023
LSHADE Griewank ($m = 10$)	0.0829 ± 0.0033	0.1942 ± 0.0102	0.0383 ± 0.0027	0.6846 ± 0.0079	0.1727 ± 0.0140	141.9049 ± 4.4918	0.0045 ± 0.0062
ABC Griewank ($m = 5$)	0.0710 ± 0.0049	0.1568 ± 0.0062	0.0205 ± 0.0017	0.7517 ± 0.0118	0.5877 ± 0.0356	112.8317 ± 12.5740	0.0007 ± 0.0022
ABC Griewank ($m = 10$)	0.0687 ± 0.0029	0.1813 ± 0.0051	0.0433 ± 0.0019	0.7067 ± 0.0042	0.4514 ± 0.0149	193.3213 ± 9.4108	0.0007 ± 0.0023
TLBO Griewank ($m = 5$)	0.0464 ± 0.0167	0.6156 ± 0.1029	0.0101 ± 0.0030	0.3279 ± 0.0988	0.3422 ± 0.0878	26.7069 ± 3.0310	0.0266 ± 0.0157
TLBO Griewank ($m = 10$)	0.0980 ± 0.0712	0.6501 ± 0.1932	0.0515 ± 0.0343	0.2003 ± 0.1086	0.1191 ± 0.0655	27.4291 ± 5.8742	0.0022 ± 0.0036

Table 6
Exploration and exploitation metrics on the F03 function for the dimensions $m = 5$ and $m = 10$.

Algorithm	Successful exploration	Failed exploration	Deceptive exploration	Successful rejection	Exploration	Diversity	Best Fitness
DE F03 ($m = 5$)	0.1886 ± 0.0138	0.3111 ± 0.0111	0.1470 ± 0.0428	0.3532 ± 0.0495	0.3467 ± 0.2293	13.7643 ± 3.5670	300.1320 ± 0.2042
DE F03 ($m = 10$)	0.1665 ± 0.0024	0.3338 ± 0.0032	0.1661 ± 0.0027	0.3336 ± 0.0029	0.8770 ± 0.0009	39.5121 ± 7.3608	301.7620 ± 1.2535
JADE F03 ($m = 5$)	0.1297 ± 0.0103	0.3861 ± 0.0192	0.0911 ± 0.0093	0.3931 ± 0.0114	0.1648 ± 0.0142	17.4728 ± 1.4042	300.0000 ± 0.0000
JADE F03 ($m = 10$)	0.1600 ± 0.0186	0.3548 ± 0.0161	0.1388 ± 0.0170	0.3465 ± 0.0218	0.1457 ± 0.0218	54.3631 ± 8.5856	300.0000 ± 0.0000
LSHADE F03 ($m = 5$)	0.0899 ± 0.0029	0.3353 ± 0.0055	0.0643 ± 0.0037	0.5104 ± 0.0079	0.0471 ± 0.0016	30.8893 ± 2.8770	300.0000 ± 0.0000
LSHADE F03 ($m = 10$)	0.1333 ± 0.0049	0.2986 ± 0.0037	0.0578 ± 0.0015	0.5102 ± 0.0069	0.0741 ± 0.0010	126.8109 ± 3.1125	300.0000 ± 0.0000
ABC F03 ($m = 5$)	0.0724 ± 0.0073	0.4240 ± 0.0083	0.0717 ± 0.0078	0.4319 ± 0.0079	0.7359 ± 0.0228	65.1320 ± 20.5217	300.2619 ± 0.1208
ABC F03 ($m = 10$)	0.0659 ± 0.0015	0.4325 ± 0.0051	0.0662 ± 0.0017	0.4354 ± 0.0057	0.6891 ± 0.0047	206.0732 ± 20.6641	303.1569 ± 0.8715
TLBO F03 ($m = 5$)	0.1105 ± 0.0388	0.2300 ± 0.1084	0.0915 ± 0.0486	0.5679 ± 0.1893	0.5993 ± 0.2245	46.2635 ± 10.2184	300.1874 ± 0.3601
TLBO F03 ($m = 10$)	0.1414 ± 0.0340	0.3592 ± 0.0540	0.1435 ± 0.0338	0.3558 ± 0.0771	0.7780 ± 0.1061	87.5028 ± 12.4695	352.5173 ± 68.5422

Table 7
Exploration and exploitation metrics on the F10 function for the dimensions $m = 5$ and $m = 10$.

Algorithm	Successful exploration	Failed exploration	Deceptive exploration	Successful rejection	Exploration	Diversity	Best Fitness
DE F10 ($m = 5$)	0.1274 ± 0.0302	0.3177 ± 0.0083	0.0699 ± 0.0376	0.4849 ± 0.0665	0.2988 ± 0.1429	27.5617 ± 4.2541	1000.0327 ± 0.0705
DE F10 ($m = 10$)	0.1217 ± 0.0255	0.3350 ± 0.0081	0.1140 ± 0.0368	0.4293 ± 0.0570	0.7198 ± 0.2190	57.9857 ± 10.2569	1001.3149 ± 0.5185
JADE F10 ($m = 5$)	0.0699 ± 0.0134	0.4169 ± 0.0202	0.0358 ± 0.0049	0.4774 ± 0.0104	0.3388 ± 0.0677	43.6248 ± 10.7343	1000.0000 ± 0.0000
JADE F10 ($m = 10$)	0.0462 ± 0.0070	0.4285 ± 0.0127	0.0349 ± 0.0054	0.4904 ± 0.0060	0.4926 ± 0.0953	83.1350 ± 11.7413	1000.2289 ± 0.6694
LSHADE F10 ($m = 5$)	0.0274 ± 0.0012	0.4414 ± 0.0090	0.0250 ± 0.0020	0.5063 ± 0.0067	0.0615 ± 0.0042	92.6278 ± 11.3732	1000.0639 ± 0.0648
LSHADE F10 ($m = 10$)	0.0773 ± 0.0063	0.2483 ± 0.0323	0.0295 ± 0.0028	0.6449 ± 0.0290	0.1829 ± 0.0255	120.9346 ± 9.4372	1002.1009 ± 0.5116
ABC F10 ($m = 5$)	0.0377 ± 0.0011	0.4575 ± 0.0079	0.0378 ± 0.0007	0.4670 ± 0.0071	0.8414 ± 0.0068	255.6424 ± 4.5379	1002.6047 ± 1.0181
ABC F10 ($m = 10$)	0.0527 ± 0.0008	0.3548 ± 0.0049	0.0390 ± 0.0010	0.5535 ± 0.0059	0.5879 ± 0.0038	318.6602 ± 8.1153	1021.0021 ± 10.5243
TLBO F10 ($m = 5$)	0.1359 ± 0.0124	0.2337 ± 0.0544	0.1028 ± 0.0269	0.5277 ± 0.0642	0.8161 ± 0.0893	35.9667 ± 7.2467	1000.4019 ± 0.4800
TLBO F10 ($m = 10$)	0.1114 ± 0.0369	0.3621 ± 0.1463	0.1098 ± 0.0374	0.4167 ± 0.1106	0.7148 ± 0.1030	70.4177 ± 15.2291	1033.4052 ± 17.7073

and deceptive exploration), to measure the correlation between the desired exploration (successful exploration and successful rejection) and final fitness, to include higher dimensions ($m = 20$), and to demonstrate the applicability of the proposed approach in a real-world example.

We performed additional experiments varying ϵ for the Rastrigin, Ackley, and Griewank functions. In Tables 3–8 the ϵ was computed using the equation $\epsilon = (upperLimit + lowerLimit)/1000$, resulting in the following settings: $\epsilon = 0.001$ (Rastrigin), $\epsilon = 0.006$ (Ackley), $\epsilon = 0.12$ (Griewank). Meanwhile, in Table 9, the following settings were applied: $\epsilon = 0.01$ (Rastrigin), $\epsilon = 0.06$ (Ackley), $\epsilon = 0.01$ (Griewank). The results of measuring the different types of exploration (successful exploration, successful rejection, failed exploration, and deceptive exploration) for Rastrigin, Ackley, and Griewank under the new ϵ values were negligible (compare Tables 3–5 with Table 9). However, in very rugged landscapes, the differences due to the varying step sizes of ϵ might be more significant.

It would be interesting to find out if higher rates of desired exploration (successful exploration and successful rejection) yield better optima consistently. The Pearson correlation coefficient has been computed, to measure the strength and direction of the relationship between the desired exploration and final fitness. It can be observed (Table 10) that the negative correlation between the desired exploration and final fitness was strong for LSHADE ($r = -0.551$) and JADE ($r = -0.550$), medium for ABC ($r = -0.360$), and small for DE ($r = -0.219$). Since we are dealing with minimization problems, higher desired exploration contributes to a smaller and better fitness. The trend was the opposite for TLBO, where a medium positive correlation ($r = 0.341$) was observed, indicating a search behavior, that calls for additional investigation. Hence, we measured both the desired and undesired exploration throughout iterations. From Figs. 94 to 99, we can observe that desired exploration prevailed over undesired exploration in ABC, JADE, LSHADE and DE for most problems. However, in TLBO a higher amount of desired exploration did not culminate in better fitness values (smaller fitness values due to the minimization problem). In other words, successful exploration, where new solutions survived due to better fitness, was largely outweighed by successful rejection, which did not yield new solutions. Measuring correlation only between successful exploration and final fitness is presented in Table 11, where we can observe, that in all cases, a larger amount of successful exploration culminated in better final fitness. There was a strong negative correlation between successful exploration and final fitness for ABC ($r = -0.581$) and LSHADE ($r = -0.549$), medium for JADE ($r = -0.453$), and only a small one for TLBO ($r = -0.270$) and

Table 8
Exploration and exploitation metrics on the F20 function for the dimensions $m = 5$ and $m = 10$.

Algorithm	Successful exploration	Failed exploration	Deceptive exploration	Successful rejection	Exploration	Diversity	Best Fitness
DE F20 ($m = 5$)	0.1409 ± 0.0151	0.2940 ± 0.0228	0.0776 ± 0.0079	0.4875 ± 0.0183	0.0461 ± 0.0058	15.9594 ± 5.5065	2090.0000 ± 31.6228
DE F20 ($m = 10$)	0.0128 ± 0.0032	0.2973 ± 0.0114	0.0065 ± 0.0017	0.6833 ± 0.0122	0.4181 ± 0.1038	112.9467 ± 20.3273	2209.3507 ± 2.4246
JADE F20 ($m = 5$)	0.0771 ± 0.0145	0.4060 ± 0.0415	0.0421 ± 0.0096	0.4748 ± 0.0560	0.0827 ± 0.0182	28.3002 ± 22.4471	2080.0000 ± 42.1637
JADE F20 ($m = 10$)	0.0292 ± 0.0182	0.3973 ± 0.0884	0.0099 ± 0.0081	0.5636 ± 0.0653	0.2301 ± 0.1326	102.8534 ± 25.5837	2162.9284 ± 54.1735
LSHADE F20 ($m = 5$)	0.0570 ± 0.0121	0.3039 ± 0.0485	0.0700 ± 0.0164	0.5692 ± 0.0395	0.1189 ± 0.0221	67.8695 ± 41.2865	2050.0001 ± 52.7045
LSHADE F20 ($m = 10$)	0.0400 ± 0.0136	0.2840 ± 0.1037	0.0262 ± 0.0140	0.6498 ± 0.0788	0.1405 ± 0.0699	232.5142 ± 29.5292	2128.4361 ± 43.8438
ABC F20 ($m = 5$)	0.0262 ± 0.0006	0.2760 ± 0.0097	0.0125 ± 0.0006	0.6853 ± 0.0097	0.6875 ± 0.0085	220.9026 ± 6.9326	2002.0102 ± 0.7454
ABC F20 ($m = 10$)	0.0429 ± 0.0012	0.2979 ± 0.0035	0.0173 ± 0.0005	0.6418 ± 0.0033	0.6691 ± 0.0049	385.0443 ± 10.0938	2126.0887 ± 9.5461
TLBO F20 ($m = 5$)	0.1022 ± 0.0708	0.2446 ± 0.1717	0.0540 ± 0.0404	0.5991 ± 0.2300	0.1377 ± 0.1623	59.1879 ± 10.5092	2070.1117 ± 48.2881
TLBO F20 ($m = 10$)	0.0545 ± 0.0507	0.2932 ± 0.2983	0.0266 ± 0.0214	0.6257 ± 0.3043	0.2877 ± 0.2495	95.7618 ± 12.5040	2146.7479 ± 58.2920

Table 9
Results under the new ϵ settings; $\epsilon = 0.01$ (Rastrigin), $\epsilon = 0.06$ (Ackley), $\epsilon = 0.01$ (Griewank).

Algorithm	Successful exploration	Failed exploration	Deceptive exploration	Successful rejection	Exploration	Diversity	Best Fitness
DE Rastrigin ($m = 5$)	0.0272 ± 0.0044	0.3162 ± 0.0062	0.0127 ± 0.0014	0.6439 ± 0.0092	0.3346 ± 0.0494	1.7677 ± 0.2169	0.0000 ± 0.0000
JADE Rastrigin ($m = 5$)	0.0905 ± 0.0137	0.2657 ± 0.0180	0.0328 ± 0.0051	0.6110 ± 0.0130	0.1116 ± 0.0128	0.6823 ± 0.0647	0.0000 ± 0.0000
LSHADE Rastrigin ($m = 5$)	0.0919 ± 0.0040	0.1850 ± 0.0070	0.0692 ± 0.0028	0.6539 ± 0.0040	0.0920 ± 0.0018	0.9340 ± 0.0399	0.0000 ± 0.0000
ABC Rastrigin ($m = 5$)	0.0582 ± 0.0026	0.1645 ± 0.0052	0.0153 ± 0.0013	0.7620 ± 0.0067	0.3283 ± 0.0099	2.3184 ± 0.0827	0.0000 ± 0.0000
TLBO Rastrigin ($m = 5$)	0.0564 ± 0.0222	0.4908 ± 0.0989	0.0174 ± 0.0061	0.4354 ± 0.0760	0.1494 ± 0.0785	1.2037 ± 0.5150	0.0000 ± 0.0000
DE Rastrigin ($m = 10$)	0.0117 ± 0.0025	0.2940 ± 0.0113	0.0060 ± 0.0012	0.6882 ± 0.0121	0.6158 ± 0.1604	6.0464 ± 1.5358	0.5970 ± 0.6957
JADE Rastrigin ($m = 10$)	0.0952 ± 0.0117	0.2617 ± 0.0226	0.0335 ± 0.0036	0.6096 ± 0.0190	0.0948 ± 0.0097	1.1443 ± 0.1352	0.0000 ± 0.0000
LSHADE Rastrigin ($m = 10$)	0.0951 ± 0.0012	0.1581 ± 0.0043	0.0403 ± 0.0017	0.7066 ± 0.0052	0.2660 ± 0.0025	3.5136 ± 0.0981	0.0000 ± 0.0000
ABC Rastrigin ($m = 10$)	0.0596 ± 0.0014	0.1722 ± 0.0026	0.0162 ± 0.0007	0.7520 ± 0.0031	0.3450 ± 0.0111	4.5028 ± 0.1646	0.0000 ± 0.0000
TLBO Rastrigin ($m = 10$)	0.0511 ± 0.0140	0.6127 ± 0.1102	0.0144 ± 0.0046	0.3218 ± 0.1023	0.1072 ± 0.0409	1.5881 ± 0.5742	0.0000 ± 0.0000
DE Ackley ($m = 5$)	0.2657 ± 0.0108	0.0979 ± 0.0112	0.0520 ± 0.0044	0.5844 ± 0.0083	0.0457 ± 0.0020	1.2600 ± 0.1668	0.0000 ± 0.0000
JADE Ackley ($m = 5$)	0.3573 ± 0.0097	0.0904 ± 0.0100	0.0455 ± 0.0050	0.5068 ± 0.0154	0.0384 ± 0.0008	0.9304 ± 0.0568	0.0000 ± 0.0000
LSHADE Ackley ($m = 5$)	0.1886 ± 0.0070	0.1033 ± 0.0049	0.0697 ± 0.0028	0.6383 ± 0.0098	0.1520 ± 0.0016	1.9551 ± 0.1522	0.0000 ± 0.0000
ABC Ackley ($m = 5$)	0.1500 ± 0.0057	0.1107 ± 0.0058	0.0288 ± 0.0017	0.7105 ± 0.0067	0.1906 ± 0.0099	6.0572 ± 0.6271	0.0000 ± 0.0000
TLBO Ackley ($m = 5$)	0.4020 ± 0.0185	0.1157 ± 0.0092	0.0684 ± 0.0065	0.4139 ± 0.0192	0.0311 ± 0.0026	0.9788 ± 0.0905	0.0000 ± 0.0000
DE Ackley ($m = 10$)	0.2137 ± 0.0106	0.0966 ± 0.0072	0.0444 ± 0.0022	0.6453 ± 0.0099	0.0461 ± 0.0037	2.2463 ± 0.2357	0.0000 ± 0.0000
JADE Ackley ($m = 10$)	0.3694 ± 0.0070	0.0960 ± 0.0071	0.0504 ± 0.0044	0.4843 ± 0.0082	0.0295 ± 0.0009	1.2233 ± 0.0365	0.0000 ± 0.0000
LSHADE Ackley ($m = 10$)	0.2379 ± 0.0062	0.0824 ± 0.0024	0.0503 ± 0.0023	0.6293 ± 0.0065	0.3242 ± 0.0030	11.8390 ± 0.3134	0.0000 ± 0.0000
ABC Ackley ($m = 10$)	0.1531 ± 0.0054	0.1140 ± 0.0047	0.0307 ± 0.0016	0.7022 ± 0.0085	0.1879 ± 0.0067	10.1671 ± 0.8043	0.0000 ± 0.0000
TLBO Ackley ($m = 10$)	0.4283 ± 0.0226	0.0976 ± 0.0077	0.0670 ± 0.0068	0.4070 ± 0.0214	0.0203 ± 0.0011	1.1690 ± 0.0670	0.0000 ± 0.0000
DE Griewank ($m = 5$)	0.0265 ± 0.0031	0.3372 ± 0.0044	0.0086 ± 0.0011	0.6277 ± 0.0063	0.7182 ± 0.0999	45.0026 ± 3.0332	0.0010 ± 0.0031
JADE Griewank ($m = 5$)	0.0705 ± 0.0054	0.3194 ± 0.0209	0.0170 ± 0.0022	0.5930 ± 0.0214	0.3100 ± 0.0336	26.9737 ± 1.7506	0.0000 ± 0.0000
LSHADE Griewank ($m = 5$)	0.0667 ± 0.0037	0.1872 ± 0.0150	0.0224 ± 0.0015	0.7237 ± 0.0119	0.0967 ± 0.0079	56.2859 ± 1.5279	0.0007 ± 0.0023
ABC Griewank ($m = 5$)	0.0696 ± 0.0049	0.1688 ± 0.0060	0.0211 ± 0.0016	0.7405 ± 0.0116	0.5898 ± 0.0282	114.8913 ± 11.9587	0.0015 ± 0.0031
TLBO Griewank ($m = 5$)	0.0432 ± 0.0147	0.6045 ± 0.1018	0.0099 ± 0.0035	0.3424 ± 0.1003	0.3996 ± 0.1256	28.9203 ± 3.7901	0.0291 ± 0.0155
DE Griewank ($m = 10$)	0.0394 ± 0.0115	0.3427 ± 0.0172	0.0254 ± 0.0076	0.5926 ± 0.0185	0.3236 ± 0.1024	51.9603 ± 4.1240	0.0199 ± 0.0244
JADE Griewank ($m = 10$)	0.0682 ± 0.0058	0.2720 ± 0.0104	0.0335 ± 0.0033	0.6264 ± 0.0171	0.2101 ± 0.0258	32.8500 ± 2.5283	0.0000 ± 0.0000
LSHADE Griewank ($m = 10$)	0.1400 ± 0.0034	0.1747 ± 0.0061	0.0321 ± 0.0014	0.6532 ± 0.0073	0.1989 ± 0.0093	244.8503 ± 7.7220	0.0055 ± 0.0053
ABC Griewank ($m = 10$)	0.0679 ± 0.0031	0.1984 ± 0.0056	0.0429 ± 0.0023	0.6908 ± 0.0079	0.4489 ± 0.0213	194.5168 ± 12.9560	0.0012 ± 0.0026
TLBO Griewank ($m = 10$)	0.0634 ± 0.0356	0.7687 ± 0.1313	0.0350 ± 0.0204	0.1329 ± 0.0809	0.1860 ± 0.1028	26.3940 ± 3.9935	0.0256 ± 0.0218

Table 10
Pearson correlation between the desired exploration (successful exploration and successful rejection) and final fitness across functions for each algorithm.

Algorithm	Pearson r
DE	−0.219
JADE	−0.550
LSHADE	−0.551
ABC	−0.360
TLBO	0.341

DE ($r = -0.109$). The other improvements toward final fitness were due to exploitation.

A small additional experiment was performed using higher dimensions ($m = 20$) for the Rastrigin, Ackley and Griewank problems. The same control parameters were used as for $m = 10$ (Table 2, $maxFEs = 100{,}000$). Comparing Tables 3–5 (when $m = 10$) to Table 12, it can be observed for the Rastrigin function that the amount of total exploration for DE increased from 73% to 90%, while the portion of different types of exploration remained the same. However, DE struggled to find a good solution when $m = 20$. The exploration behavior of DE when $m = 10$ and $m = 20$ can be compared in Figs. 5 and 100(a). On the other hand, JADE increased the amount of exploration not so drastically, from 9% to 17%, still finding a global optimum. The exploration behavior of JADE when $m = 10$ and $m = 20$ can be compared in the Figs. 7 and 100(b). LSHADE increased the amount of exploration by double, from 18% when $m = 10$ to 35% when $m = 20$, while the ratio between successful exploration and successful rejection remained similar. The exploration behavior of LSHADE when $m = 10$ and $m = 20$ can be compared in the Figs. 9 and 100(c). The exploration behavior of ABC and TLBO when $m = 20$ was very similar to that for $m = 10$ (see also Figs. 11 and 100(d), Fig. 13 and 100(e)). For the Ackley function, the exploration behavior of all the compared metaheuristics (DE, JADE, ABC, TLBO) when $m = 20$ was very similar to that for $m = 10$ (see also Figs. 20 and 101(a), Fig. 22 and 101(b), Fig. 26 and 101(d), Fig. 28 and 101(e)), except for LSHADE which increased the amount of exploration by double again, from 22% when $m = 10$ to 40% when $m = 20$. The exploration behavior of LSHADE when $m = 10$ and $m = 20$ can be compared in the Figs. 24 and 101(c). For the Griewank function, the amount of total exploration for DE decreased

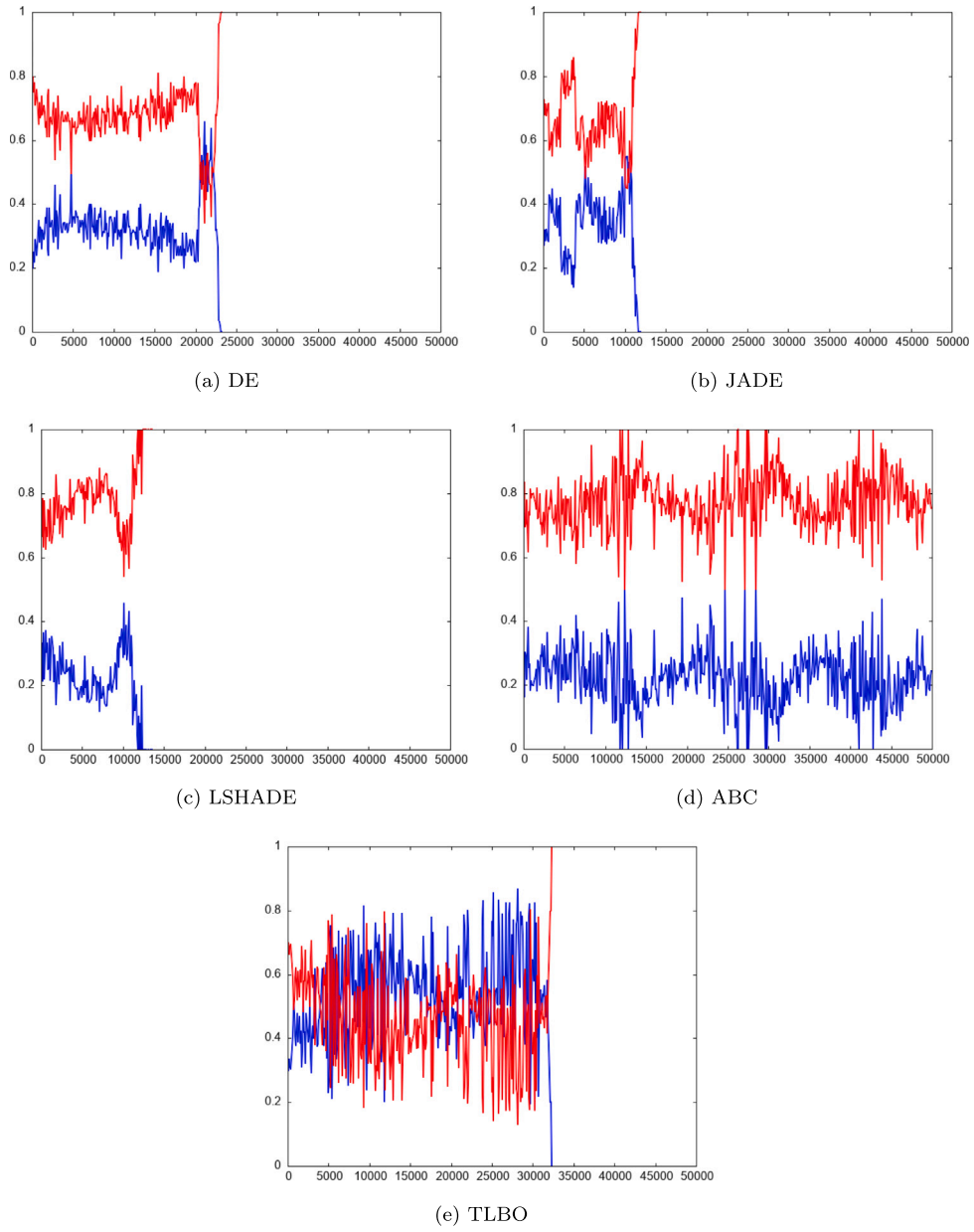


Fig. 94. Desirable exploration (red line) vs. undesirable exploration (blue line) on Rastrigin ($m = 5$).

from 33% to 13%, the amount of successful exploration increased from 4% to 10%, the amount of deceptive exploration increased from 2% to 9%, and the amount of successful rejection decreased from 59% to 46%. The exploration behavior of DE when $m = 10$ and $m = 20$ can be compared in Figs. 35 and 102(a). Similarly, the amount of total exploration for JADE decreased from 20% to 5%, the amount of successful exploration increased from 7% to 23%, the amount of deceptive exploration increased from 3% to 19%, and the amount of successful rejection decreased from 64% to 33%. The exploration behavior of JADE when $m = 10$ and $m = 20$ can be compared in Figs. 37 and 102(b). On the other hand, the amount of total exploration increased from 17% to 34% for LSHADE, while the amount of successful exploration increased from 8% to 26%. The exploration behavior of LSHADE when $m = 10$ and $m = 20$ can be compared in Figs. 39 and 102(c). The amount of total exploration for ABC decreased from 45% to 26%, while the portion of different types of exploration remained the same. The exploration behavior of ABC when $m = 10$ and $m = 20$ can be compared in Figs. 41 and 102(d). The amount of total exploration for TLBO decreased from 12% to 5%, the amount of successful exploration increased from 10% to 26%, the amount of failed exploration decreased from 65% to 32%, and the amount of deceptive exploration increased from 5% to 21%. The exploration behavior of TLBO when $m = 10$ and $m = 20$ can be compared in Figs. 43 and 102(e).

To demonstrate the applicability of the proposed method to a real-world optimization challenge, we considered the Lennard-Jones (LJ) cluster problem [80,81]. This problem is well-known for its energy landscape containing an exponentially large number of local minima [81]. It is widely regarded as NP-hard, and was included in the benchmark suite of the CEC 2011 competition on real-parameter optimization [82].

The objective is to determine the spatial arrangement of n_p atoms that minimizes the total potential energy of the molecular cluster. The pairwise LJ potential between two atoms separated by a distance r is defined as

$$LJ(r) = \frac{1}{r^{12}} - \frac{2}{r^6}. \tag{12}$$

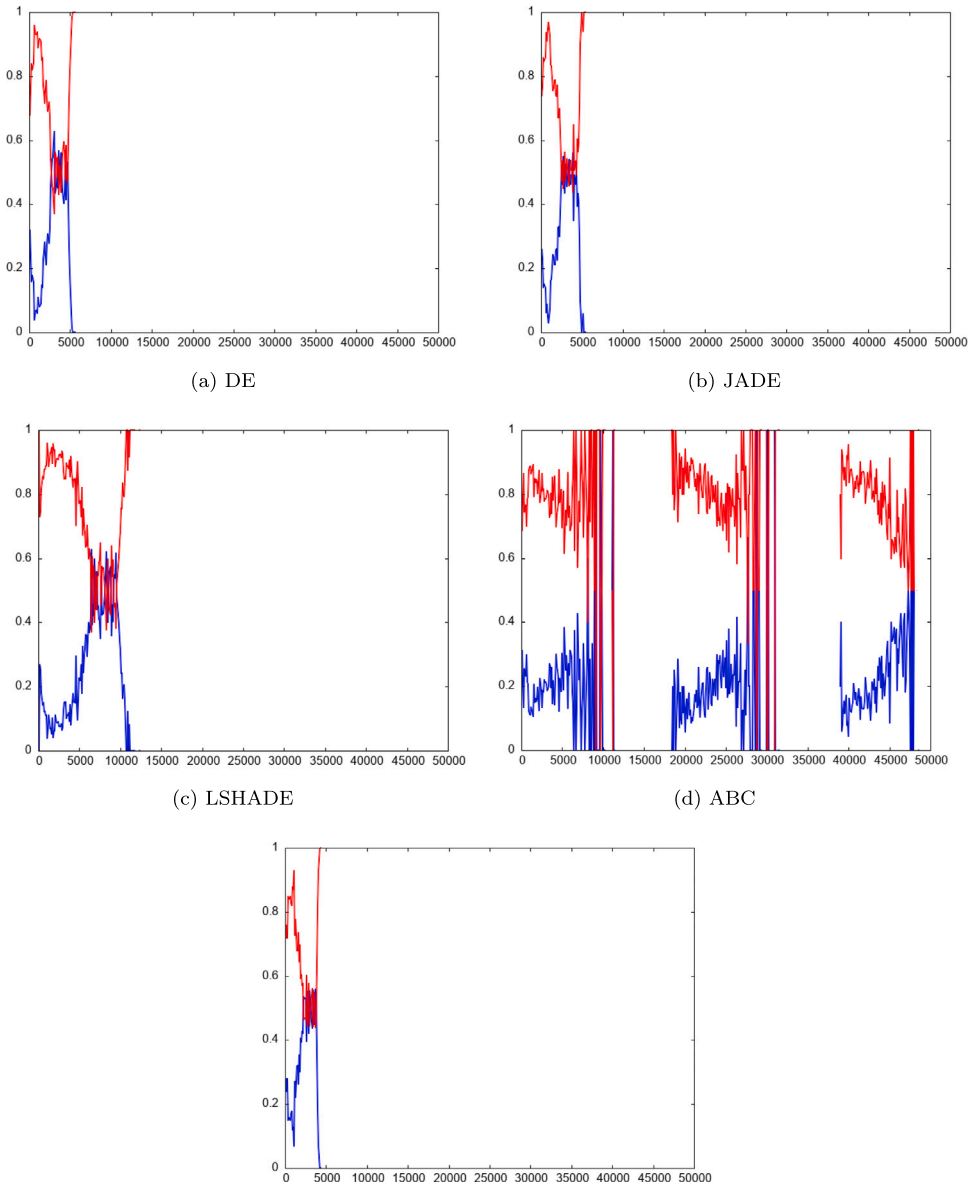


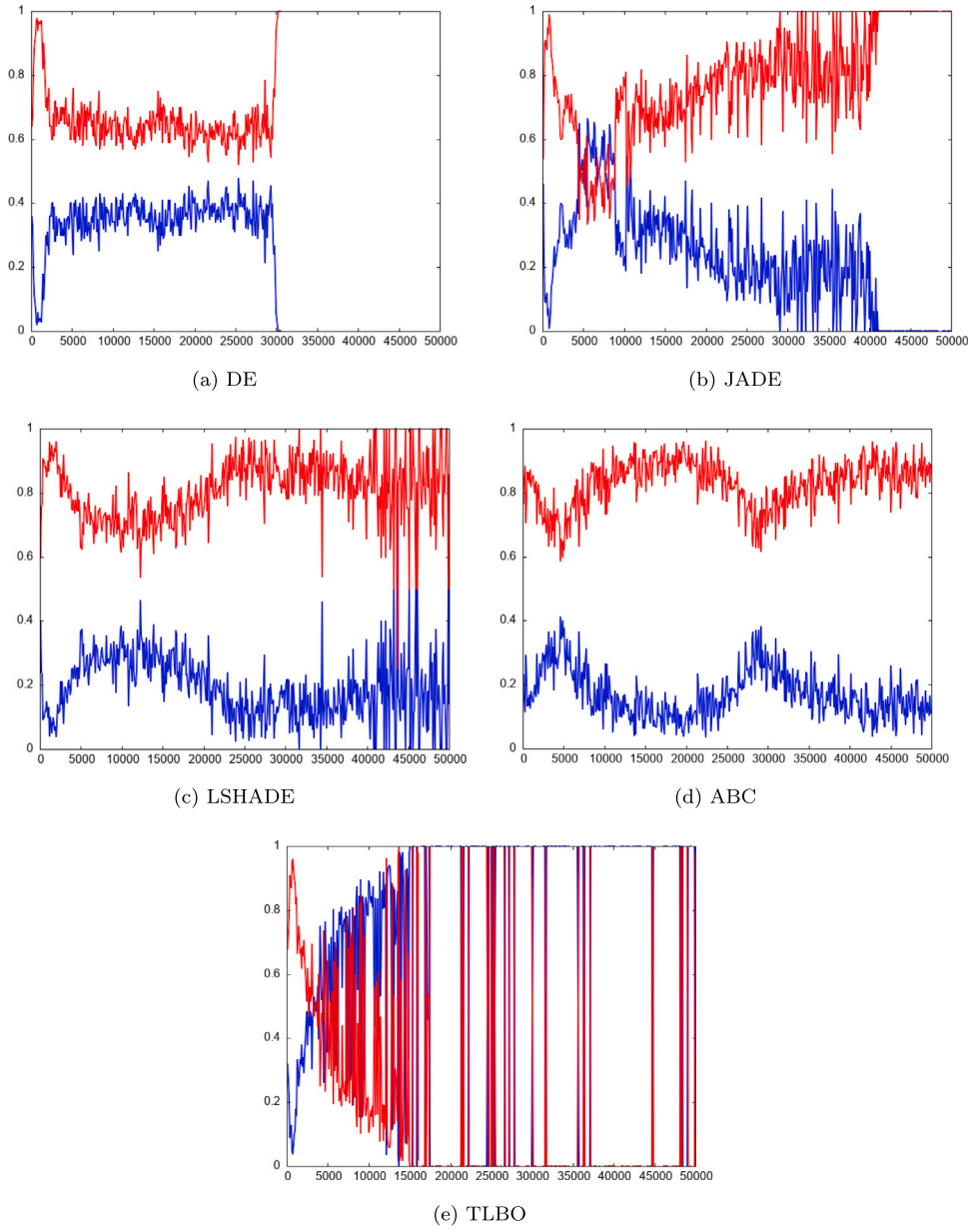
Fig. 95. Desirable exploration (red line) vs. undesirable exploration (blue line) on Ackley ($m = 5$).

The total potential energy f_{LJ} of a cluster of n_p atoms is then given by

$$f_{LJ}(X_1, \ldots, X_{n_p}) = \sum_{i<j} LJ(\|X_i - X_j\|), \tag{13}$$

where $X_i \in \mathbb{R}^3$ denotes the 3D coordinates of the i-th atom, and $\|\cdot\|$ is the Euclidean norm. We set the dimensionality to $m = 3$, corresponding to three spatial coordinates per atom.

The amount of exploration for all the compared metaheuristics, presented in Table 13, was very high, 99% for DE and JADE, 94% for ABC, and 79% for TLBO, except for LSHADE it was low (13%). The amount of successful exploration and deceptive exploration was very small. The amount of failed exploration is very high for JADE, LSHADE, and ABC (around 45%), as well as the amount of successful rejection, around 70% for DE and TLBO, and around 50% for JADE, LSHADE, and ABC. The exploration behavior is also shown in Figs. 103 and 104.

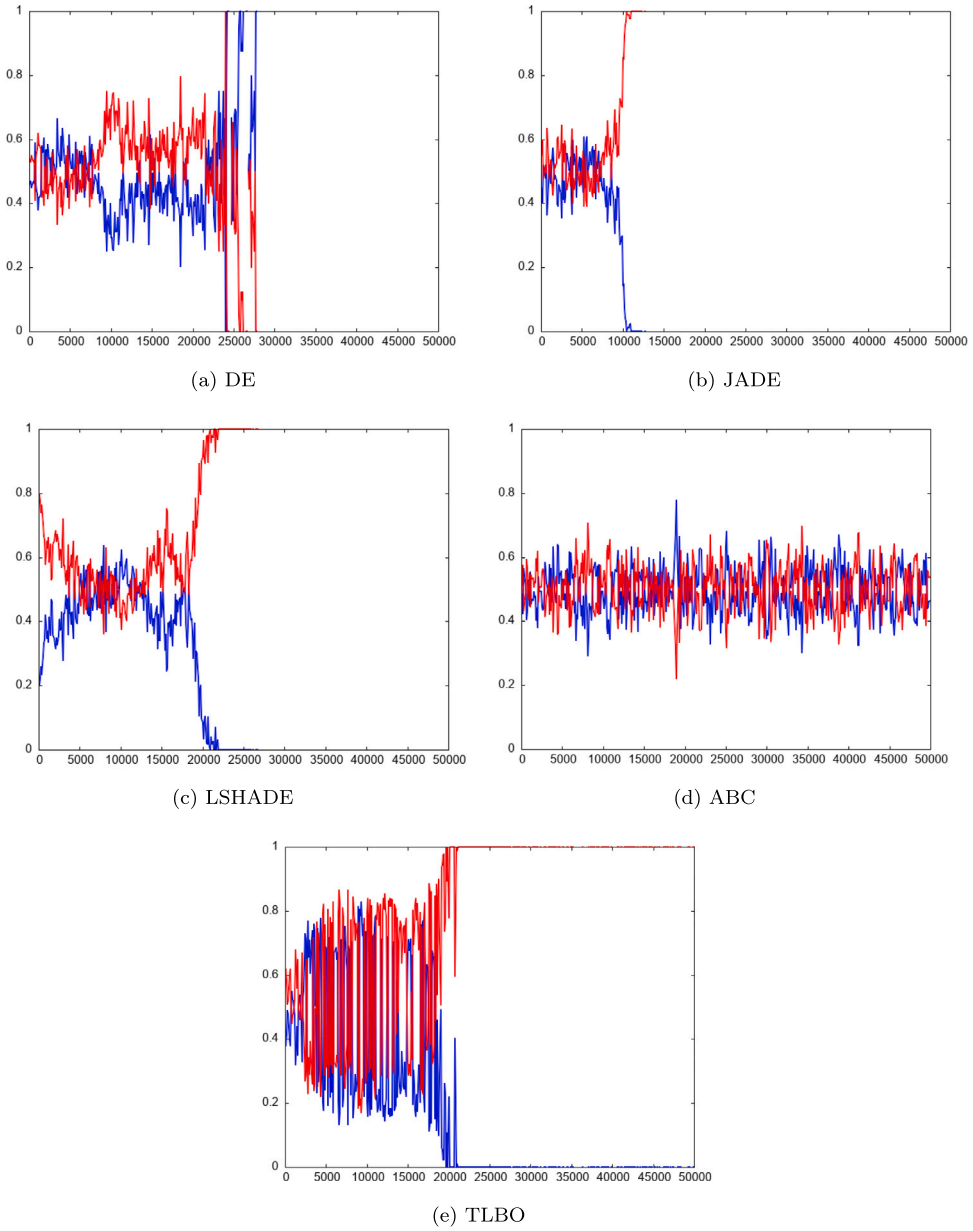


(a) DE (b) JADE

(c) LSHADE (d) ABC

(e) TLBO

Fig. 96. Desirable exploration (red line) vs. undesirable exploration (blue line) on Griewank ($m = 5$).



(a) DE (b) JADE

(c) LSHADE (d) ABC

(e) TLBO

Fig. 97. Desirable exploration (red line) vs. undesirable exploration (blue line) on F03 ($m = 5$).

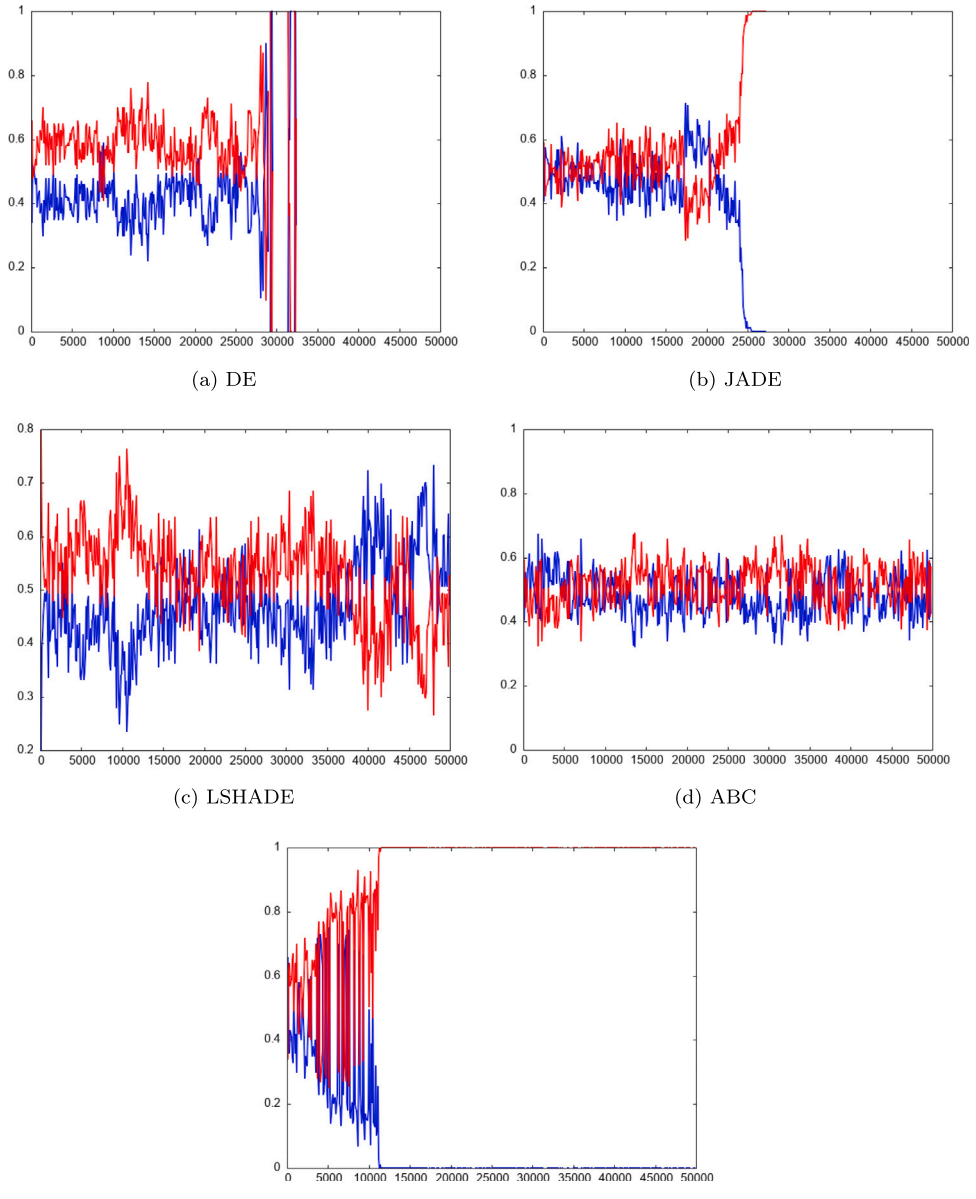
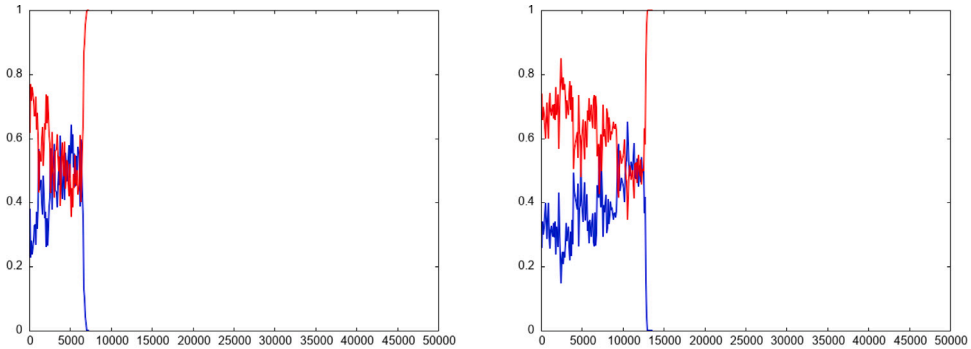
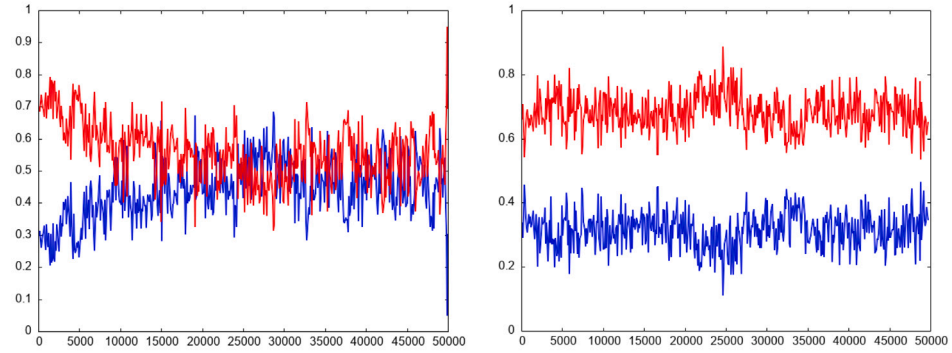


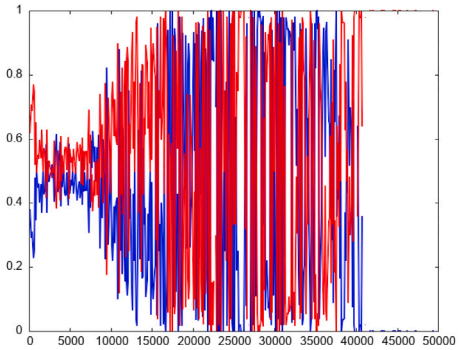
Fig. 98. Desirable exploration (red line) vs. undesirable exploration (blue line) on F10 ($m = 5$).



(a) DE (b) JADE



(c) LSHADE (d) ABC



(e) TLBO

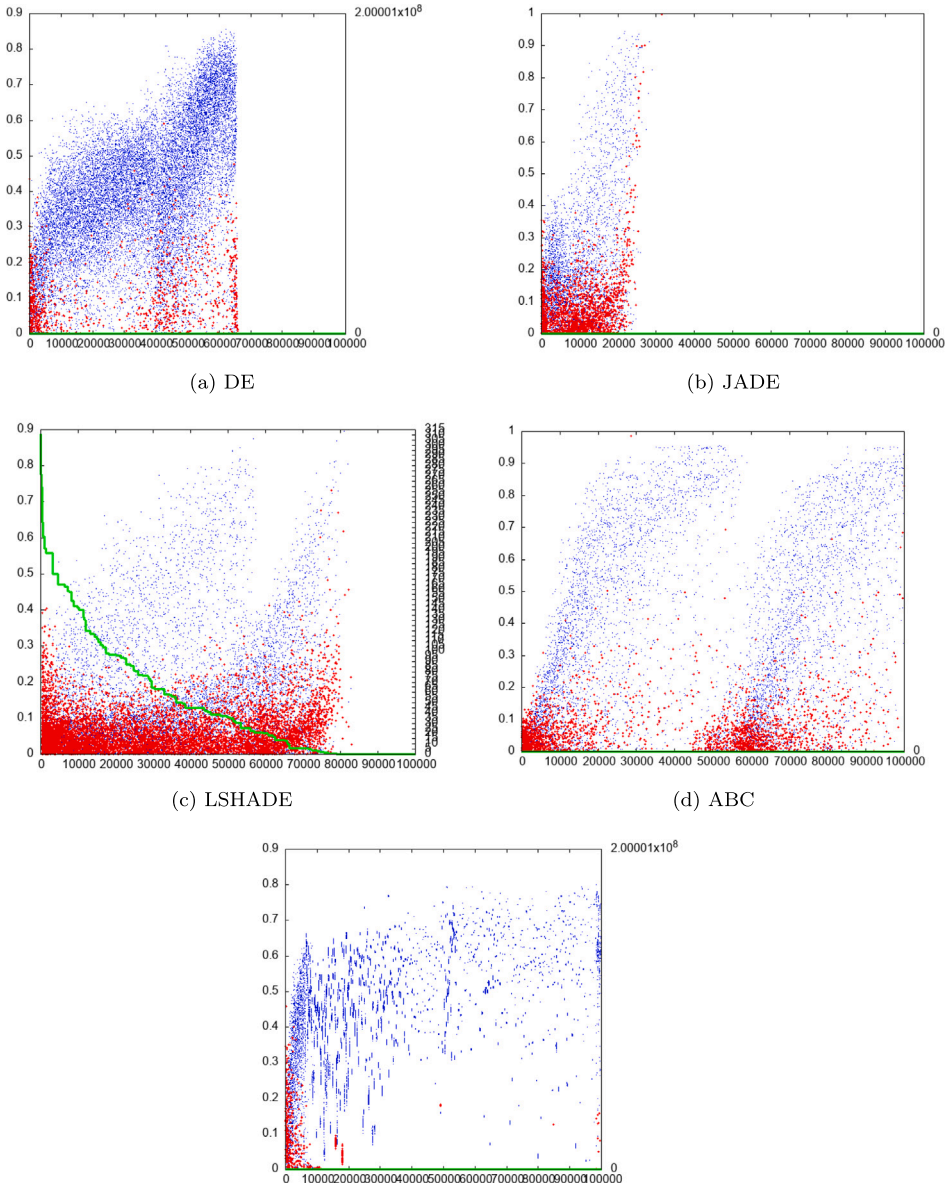
Fig. 99. Desirable exploration (red line) vs. undesirable exploration (blue line) on F20 ($m = 5$).

Table 11
Pearson correlation between successful exploration and final fitness across the functions for each algorithm.

Algorithm	Pearson r
DE	−0.109
JADE	−0.453
LSHADE	−0.549
ABC	−0.581
TLBO	−0.270

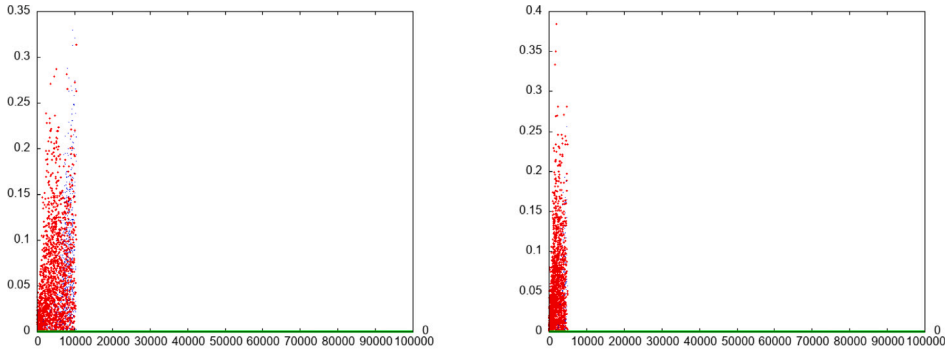
Table 12
Exploration and exploitation metrics on the classical problems for the dimension $m = 20$.

Algorithm	Successful exploration	Failed exploration	Deceptive exploration	Successful rejection	Exploration	Diversity	Best Fitness
DE Rastrigin	0.0091 ± 0.0033	0.2623 ± 0.0134	0.0052 ± 0.0019	0.7233 ± 0.0176	0.8954 ± 0.1335	17.5635 ± 3.5515	19.9275 ± 17.0156
JADE Rastrigin	0.0969 ± 0.0056	0.2451 ± 0.0223	0.0346 ± 0.0017	0.6234 ± 0.0204	0.1729 ± 0.0074	3.8784 ± 0.2367	0.0000 ± 0.0000
LSHADE Rastrigin	0.1111 ± 0.0049	0.1566 ± 0.0038	0.0441 ± 0.0011	0.6882 ± 0.0094	0.3469 ± 0.0034	18.9377 ± 0.4074	0.0000 ± 0.0000
ABC Rastrigin	0.0599 ± 0.0015	0.1814 ± 0.0041	0.0160 ± 0.0007	0.7427 ± 0.0040	0.3550 ± 0.0106	9.4812 ± 0.3666	0.0000 ± 0.0000
TLBO Rastrigin	0.0526 ± 0.0254	0.7084 ± 0.1396	0.0141 ± 0.0073	0.2249 ± 0.1106	0.1970 ± 0.1409	4.3619 ± 1.9467	2.0098 ± 6.3555
DE Ackley	0.1641 ± 0.0040	0.0869 ± 0.0072	0.0366 ± 0.0018	0.7124 ± 0.0097	0.0877 ± 0.0051	8.0317 ± 0.4561	0.0000 ± 0.0000
JADE Ackley	0.3831 ± 0.0166	0.0878 ± 0.0065	0.0510 ± 0.0031	0.4781 ± 0.0173	0.0410 ± 0.0018	3.7411 ± 0.2073	0.0000 ± 0.0000
LSHADE Ackley	0.3182 ± 0.0048	0.0640 ± 0.0019	0.0456 ± 0.0015	0.5722 ± 0.0066	0.3970 ± 0.0032	31.2614 ± 1.1273	0.0000 ± 0.0000
ABC Ackley	0.1595 ± 0.0042	0.0960 ± 0.0039	0.0309 ± 0.0013	0.7136 ± 0.0047	0.1731 ± 0.0048	23.5884 ± 0.7744	0.0000 ± 0.0000
TLBO Ackley	0.4804 ± 0.0206	0.0736 ± 0.0081	0.0586 ± 0.0065	0.3874 ± 0.0217	0.0218 ± 0.0012	2.6818 ± 0.2382	0.0000 ± 0.0000
DE Griewank	0.1009 ± 0.0135	0.3509 ± 0.0106	0.0895 ± 0.0122	0.4587 ± 0.0281	0.1346 ± 0.0255	155.7983 ± 10.7629	0.0042 ± 0.0060
JADE Griewank	0.2280 ± 0.0414	0.2541 ± 0.0111	0.1877 ± 0.0352	0.3302 ± 0.0721	0.0510 ± 0.0192	66.1417 ± 2.7108	0.0000 ± 0.0000
LSHADE Griewank	0.2612 ± 0.0171	0.0725 ± 0.0072	0.0511 ± 0.0031	0.6152 ± 0.0137	0.3373 ± 0.0025	617.7658 ± 19.1216	0.0000 ± 0.0000
ABC Griewank	0.0801 ± 0.0030	0.2363 ± 0.0061	0.0631 ± 0.0026	0.6205 ± 0.0091	0.2582 ± 0.0097	424.9595 ± 31.1132	0.0012 ± 0.0039
TLBO Griewank	0.2613 ± 0.1061	0.3201 ± 0.2611	0.2103 ± 0.0829	0.2083 ± 0.0737	0.0492 ± 0.0487	47.9880 ± 3.4609	0.0030 ± 0.0064

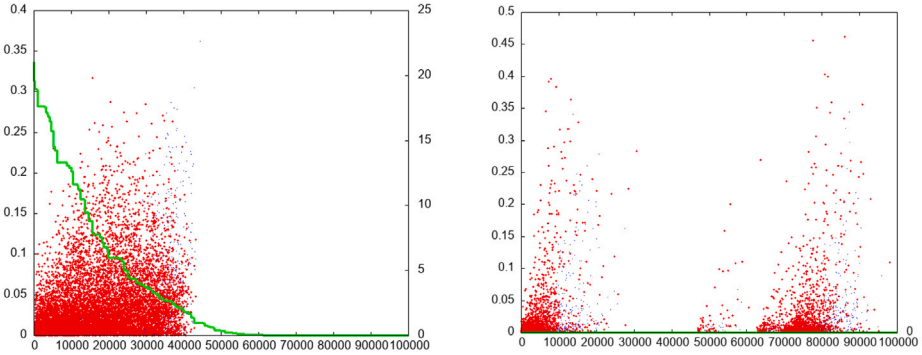


(a) DE (b) JADE (c) LSHADE (d) ABC (e) TLBO

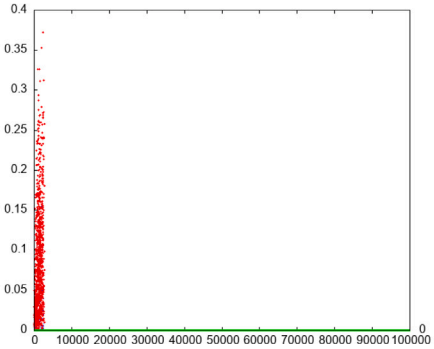
Fig. 100. Exploration on Rastrigin ($m = 20$).



(a) DE (b) JADE

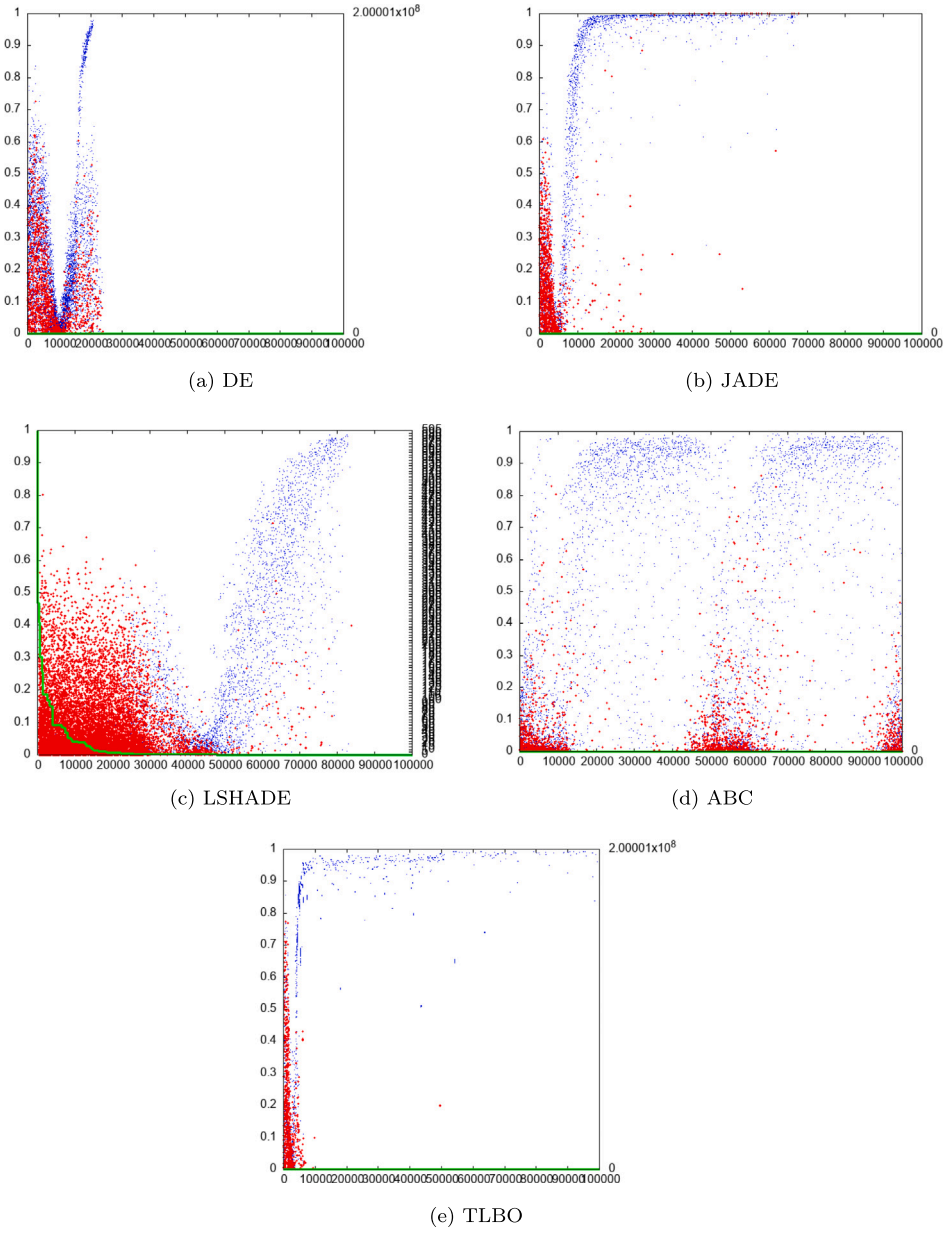


(c) LSHADE (d) ABC



(e) TLBO

Fig. 101. Exploration on Ackley ($m = 20$).



(a) DE (b) JADE

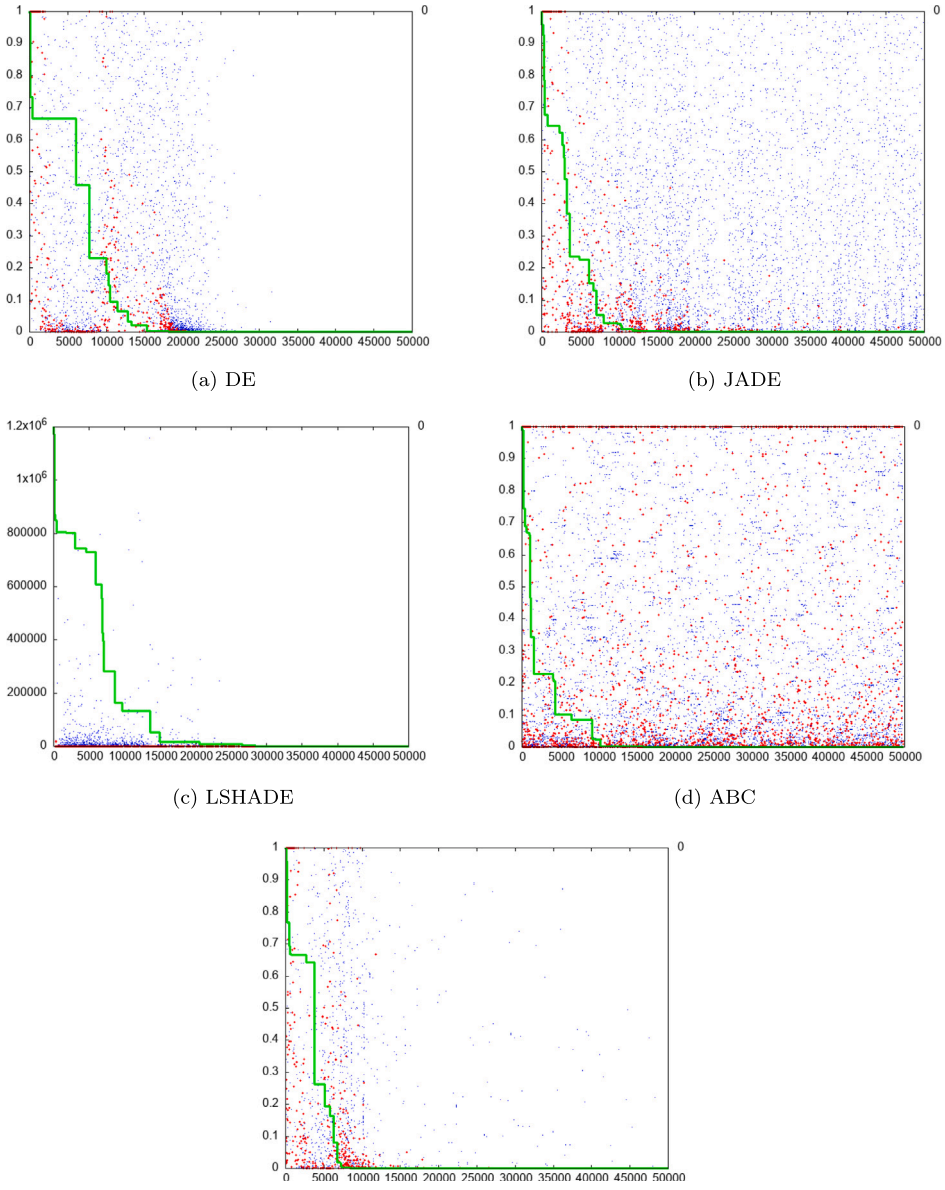
(c) LSHADE (d) ABC

(e) TLBO

Fig. 102. Exploration on Griewank ($m = 20$).

Table 13
Exploration and exploitation metrics on the Lennard-Jones problem for the dimension $m = 3$.

Algorithm	Successful exploration	Failed exploration	Deceptive exploration	Successful rejection	Exploration	Diversity	Best Fitness
DE	0.0313 ± 0.0054	0.2491 ± 0.0454	0.0103 ± 0.0008	0.7092 ± 0.0409	0.9996 ± 0.0007	28.8392 ± 7.5341	−3.0000 ± 0.0000
JADE	0.0138 ± 0.0004	0.4557 ± 0.0133	0.0133 ± 0.0004	0.5172 ± 0.0135	0.9909 ± 0.0015	20.1440 ± 2.3312	−2.9998 ± 0.0003
LSHADE	0.0306 ± 0.0012	0.4643 ± 0.0106	0.0293 ± 0.0007	0.4757 ± 0.0095	0.1307 ± 0.0154	33.1887 ± 1.8102	−2.9999 ± 0.0001
ABC	0.0490 ± 0.0010	0.4515 ± 0.0104	0.0482 ± 0.0008	0.4514 ± 0.0098	0.9443 ± 0.0027	46.4257 ± 1.1929	−2.9881 ± 0.0082
TLBO	0.0270 ± 0.0143	0.2180 ± 0.0926	0.0179 ± 0.0058	0.7371 ± 0.0894	0.7861 ± 0.0878	17.3821 ± 8.8414	−2.8000 ± 0.6325

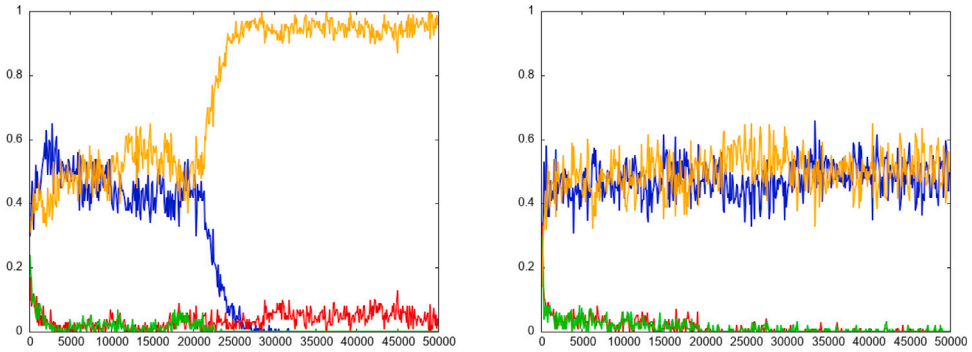


(a) DE (b) JADE

(c) LSHADE (d) ABC

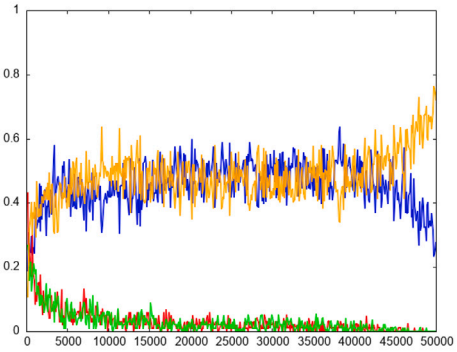
(e) TLBO

Fig. 103. Exploration behavior on the Lennard-Jones ($m = 3$).

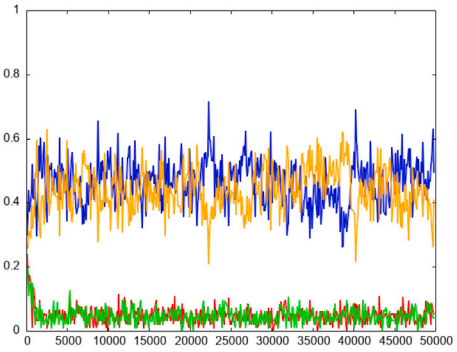


(a) DE

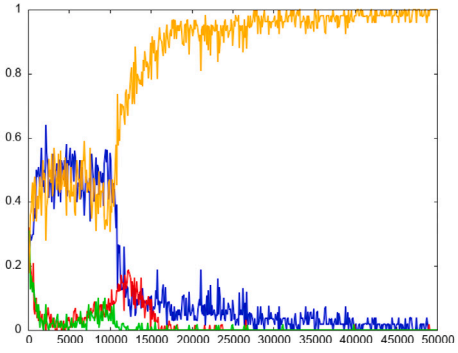
(b) JADE



(c) LSHADE



(d) ABC



(e) TLBO

Fig. 104. Exploration types throughout the evolutionary run on the Lennard-Jones problem ($m = 3$).

5. Conclusion

After 12 years, when the survey paper [2] was published, we can observe that the fundamental processes of any search algorithm, namely, exploration and exploitation, were understood much better by EA researchers. However, up until now, diversity-based measures, which are imprecise measures for exploration and exploitation, still prevail. In this paper, a novel direct measure for exploration and exploitation based on attraction basins has been proposed, which is computationally more efficient than previous similar measures [12], where attraction basins have to be computed in advance. As such, the answer to our research question RQ-1 is positive. Different types of exploration—successful, failed, deceptive exploration and successful rejection—have been investigated on a benchmark of six well-known artificial single-objective continuous optimization problems and on the real-world Lennard-Jones problem for DE/rand/1/bin, JADE, LSHADE, ABC, and TLBO. There was a clear distinction in EA's behavior with respect to exploration between ABC, TLBO and DE variants (DE/rand/1/bin, JADE, LSHADE). Furthermore, our work highlights the risks/limitations of elitist fitness-based selection, as the most problematic failed exploration was caused by greedy selection [83]. One of our key avenues of future research is to explore alternative forms of selection (e.g., [84]).

We are strongly convinced that, with the proposed approach, many open problems can be solved regarding exploration and exploitation, such as:

- Many EAs are the same in terms of updating operators, but with different mimicking scenarios and names. It would be possible to distinguish metaheuristics based on how they perform exploration and exploitation. The novelty of the proposed metaheuristics would be easier to identify.
- Many authors have tried to propose new genetic operators that influence exploration and exploitation. With the newly proposed direct exploration and exploitation measure, the authors can verify more easily whether their newly proposed genetic operators indeed work towards the improvement of exploration or exploitation.
- Until now, achieving a balance between exploration and exploitation has been controlled implicitly by proper parameter settings, off-the-fly or on-the-fly. With the newly proposed direct measure of exploration and exploitation, the effect of different parameter settings can be quantified more effectively.

This work lays down the foundations for explicit control of exploration and exploitation in Evolutionary Algorithms and Swarm Intelligence for continuous domains. We anticipate flourishing further research in many directions, such as investigating how different EAs' components (selection, mutation, crossover), factors (population size, strategies [85,86] and topologies), tuning and parameter control, influence exploration and exploitation by measuring them explicitly. Once explicit control of exploration and exploitation is well understood in single-objective optimization, further research is needed to apply explicit exploration measures to large-scale, constrained, multimodal, multi-objective, and dynamic optimization [35,66,67,87,88] and to analyse how much performance improvement can be achieved by controlling exploration relative to the computational cost.

CRediT authorship contribution statement

Miha Ravber: Validation, Software, Methodology, Investigation. **Jernej Jerebic:** Visualization, Validation, Software, Methodology, Investigation. **Matej Črepinšek:** Validation, Supervision, Methodology, Investigation. **Luka Mernik:** Validation, Methodology, Investigation. **Stephen Chen:** Validation, Supervision, Methodology, Investigation, Conceptualization. **Marjan Mernik:** Validation, Supervision, Methodology, Investigation, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The source code for the experiment is available at https://github.com/UM-LPM/exploration-and-exploitation.

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